NCR TVET COLLEGE NAMAKWALAND

Occupational Certificate: Qualification: Electrician

Phase: Knowledge Learning

NQF Level 4 - 360 Credits

DISCLAIMER

ETDP Courseware Release ETDP RSA PATENT 004/026785

(C) 2026 by Dr P.C. Wepener. This document is generated by the ETDP App under the authority and final approval of the authorised learning-material owner.

Neither Dr P.C. Wepener nor the ETDP App is accountable or liable for the correctness, completeness, factual, or academic correctness of this document. The accredited learning institution should be contacted for content inquiries, sources, references, or citations.

NOTICE OF RIGHTS

No part of this publication may be reproduced, transmitted, transcribed, stored in a retrieval system, or translated into any language or computer language, in any form or by any means, electronic, mechanical, magnetic, optical, chemical, manual, or otherwise, without prior written permission from the branded learning institution that owns the legal and intellectual property rights to the content of this document.

TRADEMARK NOTICE

Throughout this courseware title, trademark names may be used. Rather than placing a trademark symbol at every occurrence, names are used in an editorial manner for the benefit of the trademark owner, with no intention of infringement.

NOTICE OF LIABILITY

The information in this courseware title is distributed on an 'as is' basis, without warranty. While every precaution has been taken in preparation of this courseware, neither Dr P.C. Wepener nor the ETDP App shall have any liability to any person or entity for any loss or damage caused, or alleged to be caused, directly or indirectly by the instructions in this document or by the learning design and development processes described in it.

A sincere effort has been made to ensure typology accuracy of the material; however, no warranty, express or implied, is made regarding quality, correctness, reliability, accuracy, or freedom from error of this document or the products it describes. Data used in examples and sample files may be fictional. Any resemblance to real persons or companies is coincidental.

TERMS AND CONDITIONS

This document is developed for the learning institution holding a legal permit and may not be resold by the learning institution. Sample versions may be shared but may not be resold to a third party. For licensed users, this document may only be used under the terms of the license agreement between the learning institution and Dr P.C. Wepener.

Learning Institution: NCR TVET College Namaqualand

PC WEPENER (Ph.D.) BUSINESS MANAGEMENT UJ 2005

Pretoria, South Africa, 2026.

DOCUMENT REVISION AND QUALITY CONTROL

This document is a draft proposal. It is only valid for use once approved and signed off by the applicable Quality Council or Sector Education and Training Authority.

Document Title	Learner Guide	**Version**	2026/V1
SAQA Qualification Number	91761	**Qualification**	Occupational Certificate: Qualification: Electrician
QCTO / SETA Reference		**Learning Provider**	NCR TVET College Namaqualand
NQF Level	4	**Credits**	360
Document Type	Learning Material	**Phase**	Knowledge Learning
Document Developer	Kenny Saal	**Moderator**	
SETA			

PROVIDER APPROVAL			
Document Approved by:		**Name and Surname:**	
Appointment:		**Date Approved:**	dd/mm/yyyy
Signature:			
Remarks:			

QUALITY COUNCIL / SETA APPROVAL			
Document Approved by:		**Name and Surname:**	
Appointment:		**Date Approved:**	dd/mm/yyyy
Signature:			
Remarks:			

TABLE OF CONTENT

DISCLAIMER **2**
TABLE OF CONTENT **4**
CHAPTER 1 ..... **6**
CHAPTER 2 ..... **16**
CHAPTER 3 ..... **19**
CHAPTER 4 ..... **29**
CHAPTER 5 ..... **34**
CHAPTER 6 ..... **38**
CHAPTER 7 ..... **41**
CHAPTER 8 ..... **44**
CHAPTER 9 ..... **46**
CHAPTER 10 ..... **49**
CHAPTER 11 ..... **52**
CHAPTER 12 ..... **58**
CHAPTER 13 ..... **63**
CHAPTER 14 ..... **68**
CHAPTER 15 ..... **73**
CHAPTER 16 ..... **79**
CHAPTER 17 ..... **91**
CHAPTER 18 ..... **95**
CHAPTER 19 ..... **100**
CHAPTER 20 ..... **102**
CHAPTER 21 ..... **105**
CHAPTER 22 ..... **108**
CHAPTER 23 ..... **114**
CHAPTER 24 ..... **121**
CHAPTER 25 ..... **127**
CHAPTER 26 ..... **133**
CHAPTER 27 ..... **140**
CHAPTER 28 ..... **145**
CHAPTER 29 ..... **153**
CHAPTER 30 ..... **158**
CHAPTER 31 ..... **163**
CHAPTER 32 ..... **168**
CHAPTER 33 ..... **172**

CHAPTER 34.....177
CHAPTER 35.....181
CHAPTER 36.....185
CHAPTER 37.....191
CHAPTER 38.....195
CHAPTER 39.....204
CHAPTER 40.....208
CHAPTER 41.....215
CHAPTER 42.....219
CHAPTER 43.....227
CHAPTER 44.....233
CHAPTER 45.....242
CHAPTER 46.....249
CHAPTER 47.....256
CHAPTER 48.....261
CHAPTER 49.....266
CHAPTER 50.....270
CHAPTER 51.....275
CHAPTER 52.....286
CHAPTER 53.....290
CHAPTER 54.....294
CHAPTER 55.....298
CHAPTER 56.....306
CHAPTER 57.....313
CHAPTER 58.....320
CHAPTER 59.....326
CHAPTER 60.....334
CHAPTER 61.....343
CHAPTER 62.....348
CHAPTER 63.....353
CHAPTER 64.....358
CHAPTER 65.....363
CHAPTER 66.....368

CHAPTER 1

KM-01-KT01: SAFETY, HEALTH, ENVIRONMENT, RISK AND QUALITY PRINCIPLES IN THE WORKPLACE

In this Chapter we will discuss the following Topics:

KT0201 Legislation and regulations for workplace safety within electricity

KT0202 Occupation health and safety legislation

KT0203 Safety precautions and safe practices for working within industry

KT0204 Personal protective equipment

KT0205 Safety symbols and colour coding

KT0206 Fundamentals of isolating and locking out equipment and circuits

KT0207 Fundamentals of securing worksites

KT0208 Protection devices

KT0209 Causes, prevention and control of fires

KT0210 Hazard identification and risk principles

KT0211 Environmental, protection and pollution concepts

KT0212 Basic first Aid

KT0213 Incident reporting.

KT0214 Evacuation procedures.

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Describe and explain the application of legislation for working with electrical installation., equipment and appliances | Various types of personal protective equipment are identified and their uses are explained | Describe and explain the difference between acts and regulations, the responsibilities of the various role players and the application of the relevant health and safety regulations in the workplace. | Safety signs are identified and described in terms of associated risk and safe conduct | The inter-relationship between workplace safety and a productive work environment is explained | The types of hazardous waste is identified and the impact of incorrectly disposing of waste is described. | Environmental regulations for the disposal of relevant hazardous

waste is described | Regulations for the prevention and control of fires and the causes, effect and implication of fires are described | The attributes, characteristics, descriptions and properties of different types of fires are explained. | Basic first aid procedures are described for the attributes, characteristics and properties of various injuries. | The implication of injuries, their causes and effects are explained. | Basic risk assessment and hazard identification procedures are described (Weight 100%)

TOPIC KT0201: LEGISLATION AND REGULATIONS FOR WORKPLACE SAFETY WITHIN ELECTRICITY

Legislation and regulations for workplace safety within electricity

 10% of the declared nominal voltage (IEE Regulation, Appendix 2:14)

Occupation health and safety legislation

 10% of the declared nominal voltage (IEE Regulation, Appendix 2:14)

Safety precautions and safe practices for working within industry

 10% of the declared nominal voltage (IEE Regulation, Appendix 2:14)

Personal protective equipment

basic Electrical Installation Work The temporary nature of construction sites makes them one of the most dangerous places to work these regulations are made under the Health and Safety at Work Act 1974 and are designed specifically to promote safety at work in the construction industry basic Electrical Installation Work The temporary nature of construction sites makes them one of the most dangerous places to work these regulations are made under the Health and Safety at Work Act 1974 and are designed specifically to promote safety at work in the construction industry

Safety symbols and colour coding

describe, with the aid of a drawing, the application of the right-hand grip rule and show how a magnetic field is established when an electrical current flows through a conductor, by using formula, calculate the force on a current carrying conductor describe, with the aid of drawings, the effect on the magnetic field around a current carrying conductor when placed in a uniform magnetic field the legal responsibilities of both employers and employees Fire extinguishers are for dealing with small fi res, and different types of fi re must be attacked with a different type of extinguisher for example, water must not be used on a liquid or electrical fi re the normal procedure when dealing with electrical fi res is to cut off the electrical supply and use an extinguisher which is appropriate to whatever is burning

Fundamentals of isolating and locking out equipment and circuits

explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits

Protection devices

explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits

Hazard identification and risk principles

Protect against a risk to health and safety under the Health and Safety at Work Act, employers must provide free of charge any PPE and employees must make full and proper use of it safety signs such as those shown at Fig are useful reminders of the type of Safety signs showing type of PPE to be worn

PROTECT AGAINST A RISK TO HEALTH AND SAFETY

u under the Health and Safety at Work Act, employers must provide free of charge any PPE and employees must make full and proper use of it are useful reminders of the type of Safety signs showing type of PPE to be worn

Environmental, protection and pollution concepts

explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits

Incident reporting

occupational Health, Safety and Environment Students Learning Outcomes After completion of this chapter you will be able to: □ explain the importance of health and safety □ understand the safety symbols □ define various types of hazard (chemical, electrical) □ understand risk assessment process enlist any three hazards found at workplace Why safety laws are necessary? answer the following question in detail Write a note on importance of occupational health and safety Explain the personal protective equipment for protecting hands, face, ear, eyes and foot Explain health and safety precautions, regulations/guidelines explain the types of Fire Extinguisher and their uses occupational Health, Safety and Environment Students Learning Outcomes After completion of this chapter you will be able to: □ explain the importance of health and safety □ understand the safety symbols enlist any three hazards found at workplace Why safety laws are necessary? answer the following question in detail Write a note on importance of occupational health and safety Explain the personal protective equipment for protecting hands, face, ear, eyes and foot Explain health and safety precautions, regulations/guidelines

TOPIC KT0202: OCCUPATION HEALTH AND SAFETY LEGISLATION

671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

671101000-WM-03, Processes of testing and inspecting electrical installation. and control 671101000-WM-04, Processes of commissioning electrical installation. and control 671101000-WM-05, Maintenance processes for equipment, control systems and electrical Occupation or Specialization Addressed by this Curriculum 671101000: Occupational Certificate: Electrician Alternative Titles used by Industry Relation of this Curriculum to the Occupation and Qualification Progression o Occupational Certificate: Electrical Engine

TOPIC KT0203: SAFETY PRECAUTIONS AND SAFE PRACTICES FOR WORKING WITHIN INDUSTRY

671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

state why there is the need to leave the workplace clean and tidy state the procedure for disposing of waste materials To 3 phase supply Basic Electrical Installation Work Health and safety risks, precautions and procedures I n Chapter 1 of this book, we looked at some of the health and safety rules and regulations

Britain lose their lives at work. In addition, there are about 158,000 nonfatal injuries reported to the Health and Safety Executive (HSE) each year and an estimated 2.2 million people suffer ill health caused by, or made worse by, work. It is a mistake to believe that these things only happen in dangerous occupations such as deep-sea diving, mining and quarrying, fi shing industry, tunnelling and fire fighting or that it only happens in exceptional circumstances such as would never happen in your workplace

TOPIC KT0204: PERSONAL PROTECTIVE EQUIPMENT

671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

state why there is the need to leave the workplace clean and tidy state the procedure for disposing of waste materials To 3 phase supply Basic Electrical Installation Work Health and safety risks, precautions and procedures I n Chapter 1 of this book, we looked at some of the health and safety rules and regulations

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC KT0205: SAFETY SYMBOLS AND COLOUR CODING

671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

Describe, with the aid of a drawing, the application of the right-hand grip rule and show how a magnetic field is established when an electrical current flows through a conductor, by using formula, calculate the force on a current carrying conductor. Describe, with the aid of drawings, the effect on the magnetic field around a current carrying conductor when placed in a uniform magnetic field

TOPIC KT0206: FUNDAMENTALS OF ISOLATING AND LOCKING OUT EQUIPMENT AND CIRCUITS

671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

671101000-KM-02: Tools, Equipment and Materials, L4, Cr 8 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory required for the use of hand tools, power tools and measuring and testing instruments to practise as an Electrician

TOPIC KT0207: FUNDAMENTALS OF SECURING WORKSITES

671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

671101000-KM-02: Tools, Equipment and Materials, L4, Cr 8 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory required for the use of hand tools, power tools and measuring and testing instruments to practise as an Electrician

TOPIC KT0208: PROTECTION DEVICES

671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

Types and application of alternators and generators Testing principles of alternators and generators Protection of alternators and generators Calculate properties of alternators and generators Internal Assessment Criteria and Weight List the different types of alternators and generators and describe their construction Explain the principle of operation of alternators and generators Explain the fundamental differences between alternators and generators.

Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits

TOPIC KT0209: CAUSES, PREVENTION AND CONTROL OF FIRES

671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

Describe and explain the difference between acts and regulations, the responsibilities of the various role players and the application of the relevant health and safety regulations in Safety signs are identified and described in terms of associated risk and safe conduct The inter-relationship between workplace safety and a productive work environment is The types of hazardous waste is identified and the impact of incorrectly disposing of waste Environmental regulations for the disposal of relevant hazardous waste.

TOPIC KT0210: HAZARD IDENTIFICATION AND RISK PRINCIPLES

671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

Describe and explain the difference between acts and regulations, the responsibilities of the various role players and the application of the relevant health and safety regulations in Safety signs are identified and described in terms of associated risk and safe conduct The inter-relationship between workplace safety and a productive work environment is The types of hazardous waste is identified and the impact of incorrectly disposing of waste Environmental regulations for the disposal of relevant hazardous waste.

TOPIC KT0211: ENVIRONMENTAL, PROTECTION AND POLLUTION CONCEPTS

671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit. Internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose. Identify the basic components of various measuring and test instruments and describe the basic principle of operation

TOPIC KT0212: BASIC FIRST AID

671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

Describe and explain the difference between acts and regulations, the responsibilities of the various role players and the application of the relevant health and safety regulations in Safety signs are identified and described in terms of associated risk and safe conduct The inter-relationship between workplace safety and a productive work environment is The types of hazardous waste is identified and the impact of incorrectly disposing of waste Environmental regulations for the disposal of relevant hazardous waste.

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

TOPIC KT0213: INCIDENT REPORTING.

671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning

in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

 Provide regular eyesight tests; Provide health and safety information; Provide adjustable furniture (e.g. desk, chair, etc.); and Demonstrate that they have adequate procedures designed to reduce risks associated with DSE work, such as repetitive strain injury (RSI). The main provisions require employers to: Ensure that suitable personal protective equipment (PPE) is provided free of charge "wherever there are risks to health and safety that cannot be adequately controlled in other ways

TOPIC KT0214: EVACUATION PROCEDURES.

671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

Describe and explain the difference between acts and regulations, the responsibilities of the various role players and the application of the relevant health and safety regulations in Safety signs are identified and described in terms of associated risk and safe conduct The inter-relationship between workplace safety and a productive work environment is The types of hazardous waste is identified and the impact of incorrectly disposing of waste Environmental regulations for the disposal of relevant hazardous waste.

IN SUMMARY

In this unit we have focussed on:

KT0201 Legislation and regulations for workplace safety within electricity

KT0202 Occupation health and safety legislation

KT0203 Safety precautions and safe practices for working within industry

KT0204 Personal protective equipment

KT0205 Safety symbols and colour coding

KT0206 Fundamentals of isolating and locking out equipment and circuits

KT0207 Fundamentals of securing worksites

KT0208 Protection devices

KT0209 Causes, prevention and control of fires

KT0210 Hazard identification and risk principles

KT0211 Environmental, protection and pollution concepts

KT0212 Basic first Aid

KT0213 Incident reporting.

KT0214 Evacuation procedures.

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 2

KM-02-KT01: HAND TOOLS AND POWER TOOLS

In this Chapter we will discuss the following Topics:

KT0101 Types, uses and care of hand tools

KT0102 Types, uses and care of portable power tools

KT0103 Types, uses and care of fixed power tools

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Identify hand tools and describe their uses

Identify portable power tools and describe their uses

Identify fixed power tools and describe their uses (Weight 50%)

TOPIC KT0101: TYPES, USES AND CARE OF HAND TOOLS

671101000-KM-02: Tools, Equipment and Materials, L4, Cr 8 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory required for the use of hand tools, power tools and measuring and testing instruments to practise as an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

tools. (05 Hrs) 4-5 15. Operations of allied trade tools. Allied trades: Introduction to Hrs) fitting tools, safety appropriate 16. Workshop practice on filing and precautions. Description of hacksawing. (10Hrs) files, hammers, chisels accuracy as per drawing. 17. Prepare hand coil winding hacksaw frames, blades, their assembly. (5 Hrs) specification and grades. Practice on preparing T-joint, Marking tools description and straight joint and dovetail joint use. on wooden blocks

TOPIC KT0102: TYPES, USES AND CARE OF PORTABLE POWER TOOLS

671101000-KM-02: Tools, Equipment and Materials, L4, Cr 8 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory required for the use of hand tools, power tools and measuring and testing instruments to practise as an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

Scribers, dividers and centre punches were sharpened to ensure serviceability. PM-01-PS02: Select, use and care for power tools Scope of Practical Skill Given work instructions, work area, documents, templates, drilling machines, grinders, and saws standard operating procedures and, statutory requirements, you should be PA0201 Select a power tool pertaining to specific job requirements. PA0202 Use fixed power tools. PA0203 Use portable power tools

TOPIC KT0103: TYPES, USES AND CARE OF FIXED POWER TOOLS

671101000-KM-02: Tools, Equipment and Materials, L4, Cr 8 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory required for the use of hand tools, power tools and measuring and testing instruments to practise as an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

Scribers, dividers and centre punches were sharpened to ensure serviceability. PM-01-PS02: Select, use and care for power tools Scope of Practical Skill Given work instructions, work area, documents, templates, drilling machines, grinders, and saws standard operating procedures and, statutory requirements, you should be PA0201 Select a power tool pertaining to specific job requirements. PA0202 Use fixed power tools. PA0203 Use portable power tools

IN SUMMARY

In this unit we have focussed on:

KT0101 Types, uses and care of hand tools

KT0102 Types, uses and care of portable power tools

KT0103 Types, uses and care of fixed power tools

Complete the workbook activities linked to this chapter, which are activities 1 – 3.

CHAPTER 3

KM-02-KT02: MEASURING AND TESTING INSTRUMENTS

In this Chapter we will discuss the following Topics:

KT0201 Portable electrical measuring instruments

KT0202 Types of electrical measuring and testing instruments

KT0203 Safe use of measuring and testing instruments

KT0204 Construction and operating principles of measuring and testing instruments

KT0205 Methods of connecting measuring and testing instruments in circuits

KT0206 Applications and methods of using electrical measuring and testing instruments

KT0207 Care and maintenance of single and three phase measuring and testing instruments

KT0208 Instrument transformers

KT0209 Types and functions of panel mounted electrical measuring and testing instruments

KT0210 Safe use of panel mounted measuring and test instruments

KT0211 Protection of measuring instruments when connected in a circuit.

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Identify different types of electrical measuring instruments including fixed and portable and state the purpose. | Identify the basic components of various measuring and test instruments and describe the basic principle of operation. | Explain by means of circuit diagrams how a ammeter, voltmeter is directly connected and indirectly connected by using instrument transformers. | Describe the applications and methods of using electrical measuring and testing instruments in direct and alternating current circuits. | Describe safety and functionality checks to be performed on measuring and testing instruments before use | Describe correct methods of handling and storing measuring and testing instruments | Describe, with the aid of drawings, the connection of measuring instruments to single phase and three phase circuits | Explain the function of the wattmeter (electrodynamometer type), kilowatt-hour meter, frequency meter, power factor meter and maximum demand meter | Illustrate, by means of circuit diagrams, how the wattmeter (electrodynamometer type), kilowatt-hour meter, frequency meter, power factor meter and maximum demand meter are connected in single and three phase circuits: | Describe the various methods of protection when connecting measuring instruments in circuits (Weight 50%)

TOPIC KT0201: PORTABLE ELECTRICAL MEASURING INSTRUMENTS

Portable electrical measuring instruments

power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose identify the basic components of various measuring and test instruments and describe the basic principle of operation power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation

Types of electrical measuring and testing instruments

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose identify the basic components of various measuring and test instruments and describe the basic principle of operation Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation

Safe use of measuring and testing instruments

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose identify the basic components of various measuring and test instruments and describe the basic principle of operation Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation

Construction and operating principles of measuring and testing instruments

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose identify the basic components of various measuring and test instruments and describe the basic principle of operation Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation

Methods of connecting measuring and testing instruments in circuits

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose identify the basic components of various measuring and test instruments and describe the basic principle of operation Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation

Applications and methods of using electrical measuring and testing instruments

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose identify the basic components of various measuring and test instruments and describe the basic principle of operation Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation

Care and maintenance of single and three phase measuring and testing instruments

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose identify the basic components of various measuring and test instruments and describe the basic principle of operation Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection

of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation

Instrument transformers

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose identify the basic components of various measuring and test instruments and describe the basic principle of operation fluorescent lamps (where supply voltage is less than the rated voltage) reduced voltage motor starter series line boosters for fixed adjustment of line voltage have better voltage regulation servo-line voltage correctors voltage Regulation and Auto Transformer and regulation of single-phase instrument transformers (CT transformer at different loads and power factors perform series and parallel operation of two single phase transformers verify the terminals and accessories of three phase transformer HT and LT Perform 3 phase Method of connecting three operation (i) delta-delta, single phase transformers for three phase operation Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation fluorescent lamps (where supply voltage is less than the rated voltage) reduced voltage motor starter series line boosters for fixed adjustment of line voltage have better voltage regulation servo-line voltage correctors

Types and functions of panel mounted electrical measuring and testing instruments

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose identify the basic components of various measuring and test instruments and describe the basic principle of operation Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation

Safe use of panel mounted measuring and test instruments

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose identify the basic components of various measuring and test instruments and describe the basic principle of operation hrs) calibration of measuring Test single phase energy instruments Instrument transformers Types and functions of panel mounted electrical measuring and testing

Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation hrs) calibration of measuring Test single phase energy instruments

Protection of measuring instruments when connected in a circuit

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose identify the basic components of various measuring and test instruments and describe the basic principle of operation dynamometer instrument as a voltmeter: When this instrument is used as voltmeter, the fixed and moving coils are joined in series along with a high resistance (multiplier) (Fig 6) as the moving coil is made by winding small gauge (thin) wire, the above connection is unsuitable for measuring When the instrument is to be converted as an ammeter to measure large currents, the moving coil is connected across a shunt (Fig 5) Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation

TOPIC KT0202: TYPES OF ELECTRICAL MEASURING AND TESTING INSTRUMENTS

671101000-KM-02: Tools, Equipment and Materials, L4, Cr 8 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory required for the use of hand tools, power tools and measuring and testing instruments to practise as an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit. Internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose. Identify the basic components of various measuring and test instruments and describe the basic principle of operation

TOPIC KT0203: SAFE USE OF MEASURING AND TESTING INSTRUMENTS

671101000-KM-02: Tools, Equipment and Materials, L4, Cr 8 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory required for the use of hand tools, power tools and measuring and testing instruments to practise as an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit. Internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose. Identify the basic components of various measuring and test instruments and describe the basic principle of operation

TOPIC KT0204: CONSTRUCTION AND OPERATING PRINCIPLES OF MEASURING AND TESTING INSTRUMENTS

671101000-KM-02: Tools, Equipment and Materials, L4, Cr 8 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory required for the use of hand tools, power tools and measuring and testing instruments to practise as an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit. Internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose. Identify the basic components of various measuring and test instruments and describe the basic principle of operation

TOPIC KT0205: METHODS OF CONNECTING MEASURING AND TESTING INSTRUMENTS IN CIRCUITS

671101000-KM-02: Tools, Equipment and Materials, L4, Cr 8 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory required for the use of hand tools, power tools and measuring and testing instruments to practise as an Electrician

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit. Internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose. Identify the basic components of various measuring and test instruments and describe the basic principle of operation

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

TOPIC KT0206: APPLICATIONS AND METHODS OF USING ELECTRICAL MEASURING AND TESTING INSTRUMENTS

671101000-KM-02: Tools, Equipment and Materials, L4, Cr 8 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory required for the use of hand tools, power tools and measuring and testing instruments to practise as an Electrician

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit. Internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose. Identify the basic components of various measuring and test instruments and describe the basic principle of operation

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

TOPIC KT0207: CARE AND MAINTENANCE OF SINGLE AND THREE PHASE MEASURING AND TESTING INSTRUMENTS

671101000-KM-02: Tools, Equipment and Materials, L4, Cr 8 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory required for the use of hand tools, power tools and measuring and testing instruments to practise as an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit. Internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose. Identify the basic components of various measuring and test instruments and describe the basic principle of operation

TOPIC KT0208: INSTRUMENT TRANSFORMERS

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit. Internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose. Identify the basic components of various measuring and test instruments and describe the basic principle of operation

shock hazard, particularly if the step down ratio is high. Application: The common applications are: with a < 1 for step down. fluorescent lamps (where supply voltage is less than the rated voltage) reduced voltage motor starter series line boosters for fixed adjustment of line voltage have better voltage regulation servo-line voltage correctors. Power : Electrician (NSQF Revised - 2022) - Related Theory for Exercise 1.12

The following are the different types of current transformers. the cost of the meter will be high. Wound type current transformer: This is one in which The solution is to step-down the current and voltage with the primary winding is having more than one full turn wound instrument transformers, so that, they could be metered on the core (Fig 1) with instruments of moderate size

TOPIC KT0209: TYPES AND FUNCTIONS OF PANEL MOUNTED ELECTRICAL MEASURING AND TESTING INSTRUMENTS

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit. Internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose. Identify the basic components of various measuring and test instruments and describe the basic principle of operation

671101000-KM-02: Tools, Equipment and Materials, L4, Cr 8 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory required for the use of hand tools, power tools and measuring and testing instruments to practise as an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

TOPIC KT0210: SAFE USE OF PANEL MOUNTED MEASURING AND TEST INSTRUMENTS

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit. Internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose. Identify the basic components of various measuring and test instruments and describe the basic principle of operation

671101000-KM-02: Tools, Equipment and Materials, L4, Cr 8 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory required for the use of hand tools, power tools and measuring and testing instruments to practise as an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

TOPIC KT0211: PROTECTION OF MEASURING INSTRUMENTS WHEN CONNECTED IN A CIRCUIT.

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit. Internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose. Identify the basic components of various measuring and test instruments and describe the basic principle of operation

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

671101000-KM-02: Tools, Equipment and Materials, L4, Cr 8 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory required for the use of hand tools, power tools and measuring and testing instruments to practise as an Electrician

IN SUMMARY

In this unit we have focussed on:

KT0201 Portable electrical measuring instruments

KT0202 Types of electrical measuring and testing instruments

KT0203 Safe use of measuring and testing instruments

KT0204 Construction and operating principles of measuring and testing instruments

KT0205 Methods of connecting measuring and testing instruments in circuits

KT0206 Applications and methods of using electrical measuring and testing instruments

KT0207 Care and maintenance of single and three phase measuring and testing instruments

KT0208 Instrument transformers

KT0209 Types and functions of panel mounted electrical measuring and testing instruments

KT0210 Safe use of panel mounted measuring and test instruments

KT0211 Protection of measuring instruments when connected in a circuit.

Complete the workbook activity linked to this chapter, which is activity 2.

CHAPTER 4

KM-03-KT01: FUNDAMENTALS OF ELECTRICITY

In this Chapter we will discuss the following Topics:

KT0101 Principles and fundamental concepts of electricity

KT0102 Definitions, types, properties and applications of conductors, insulators and semi-conductors

KT0103 Concepts, theories and principles of Electrical Circuits

KT0104 Calculations on basic electrical circuits (resistance, voltage and current) and power

KT0105 Basic principles and calculation of magnetism

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Describe, calculate and interpret fundamental concepts of electricity (Electro motive force, Potential difference, Resistance)using the correct units of measurement and definitions. | List types of materials used for conductors, insulators and semi-conductors and describe their mechanical and electrical properties and applications. | Describe the factors that influences the resistance of a material | Define and explain, using the correct units of measurement ohms law of electricity | Define and explain using the correct units of measurement Kirchoff's law of electricity | Manipulate formula to calculate voltage, current and resistance in series/parallel circuits | Name, describe and explain the different types of magnets and their properties | Explain fundamental magnetic concepts by naming the five characteristics of magnetic lines of force and explaining the relationship between flux and flux density | Describe, with the aid of a drawing, the application of the right-hand grip rule and show how a magnetic field is established when an electrical current flows through a conductor, by using formula, calculate the force on a current carrying conductor. | Describe, with the aid of drawings, the effect on the magnetic field around a current carrying conductor when placed in a uniform magnetic field. Fleming's left-hand rule must be demonstrated. | Describe and explain the various forces or methods that can be used to alter magnetic fields, and describe the changes that take place (Weight 35%)

TOPIC KT0101: PRINCIPLES AND FUNDAMENTAL CONCEPTS OF ELECTRICITY

Principles and fundamental concepts of electricity

practice elementary first aid safety practice fundamentals of Electricity This is sound producing device

PRACTICE ELEMENTARY FIRST AID SAFETY PRACTICE

practice elementary first aid safety practice fundamentals of Electricity This is sound producing device

---

PRACTICE ELEMENTARY FIRST AID SAFETY PRACTICE

Concepts, theories and principles of Electrical Circuits

practice elementary first aid safety practice the SI unit of resistance is “Ohm” and is represented by Ω (Ω is Greek alphabet “Omega”) it can be defined using Ohm’s Law (to be discussed later): If a potential difference of 1 volt is applied to the ends of a conductor, and a current of 1 ampere flows through it, it is said that the value of the resistance is 1 ohm prefixes Some of the prefixes and their abbreviations which are commonly used are tabulated Fundamentals of Electricity Activity 1

PRACTICE ELEMENTARY FIRST AID SAFETY PRACTICE

Calculations on basic electrical circuits (resistance, voltage and current) and power

practice elementary first aid safety practice fundamentals of Electricity This is sound producing device this produces buzz sound when the voltage is applied The symbol represents the light bulb

---

THE BULB GLOWS WHEN REQUIRED VOLTAGE IS APPLIED

PRACTICE ELEMENTARY FIRST AID SAFETY PRACTICE

Basic principles and calculation of magnetism

practice elementary first aid safety practice fundamentals of Electricity This is sound producing device

---

PRACTICE ELEMENTARY FIRST AID SAFETY PRACTICE

TOPIC KT0102: DEFINITIONS, TYPES, PROPERTIES AND APPLICATIONS OF CONDUCTORS, INSULATORS AND SEMI-CONDUCTORS

KM-03-: Electrical principles of appliances (35%) Guidelines for Topics KM-03-: Fundamentals of electricity (35%) Topic elements to be covered include: Principles and fundamental concepts of electricity Definitions, types, properties and applications of conductors, insulators and Concepts, theories and principles of Electrical Circuits Calculations on basic electrical circuits (resistance, voltage and current) and Basic principles and calculation of magnetism Internal Asse

electrical fire. (02hrs.) Use of fire extinguishers. Practice elementary first aid safety practice. Hazard identification and Rescue a person and Personal safety and factory Response to emergencies Disposal procedure of e.g. power failure, system waste materials. (01Hrs.) failure and fire etc. (03 hrs.) Use of personal

protective Practice on cleanliness and procedure to maintain it. Identify trade tools and Concept of Standards and advantages of BIS/ISI. Trade tools specifications

12.Practice safe methods of lifting and han-trade tools specifications. Handling of tools & equipment. (03Hrs.) Introduction to National Electrical 13.Select proper tools for operation and pre- Code-2011. (02 hrs.) cautions in operation. (03Hrs.) 14.Care & maintenance of trade tools. 15.Operations of allied trade tools. (05 Hrs.) Allied trades: Introduction to fitting tools, safety precautions. Descrip16.Workshop practice on filing and operation of files, hammers, chisels hack hacksawing. (05Hrs

TOPIC KT0103: CONCEPTS, THEORIES AND PRINCIPLES OF ELECTRICAL CIRCUITS

KM-03-: Electrical principles of appliances (35%) Guidelines for Topics KM-03-: Fundamentals of electricity (35%) Topic elements to be covered include: Principles and fundamental concepts of electricity Definitions, types, properties and applications of conductors, insulators and Concepts, theories and principles of Electrical Circuits Calculations on basic electrical circuits (resistance, voltage and current) and Basic principles and calculation of magnetism Internal Asse

671101000-KM-07: Electrical supply systems and components, L4, Cr 31 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory of electrical supply systems and components required to practise as an Electrician

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

TOPIC KT0104: CALCULATIONS ON BASIC ELECTRICAL CIRCUITS (RESISTANCE, VOLTAGE AND CURRENT) AND POWER

KM-03-: Electrical principles of appliances (35%) Guidelines for Topics KM-03-: Fundamentals of electricity (35%) Topic elements to be covered include: Principles and fundamental concepts of electricity Definitions, types, properties and applications of conductors, insulators and Concepts, theories and principles of Electrical Circuits Calculations on basic electrical circuits (resistance, voltage and current) and Basic principles and calculation of magnetism Internal Asse

electrical fire. (02hrs.) Use of fire extinguishers. Practice elementary first aid safety practice. Hazard identification and Rescue a person and Personal safety and factory Response to emergencies Disposal procedure of e.g. power failure, system waste materials. (01Hrs.) failure and fire etc. (03 hrs.) Use of personal protective Practice on cleanliness and procedure to maintain it. Identify trade tools and Concept of Standards and advantages of BIS/ISI. Trade tools specifications

671101000-KM-07: Electrical supply systems and components, L4, Cr 31 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory of electrical supply systems and components required to practise as an Electrician

TOPIC KT0105: BASIC PRINCIPLES AND CALCULATION OF MAGNETISM

KM-03-: Electrical principles of appliances (35%) Guidelines for Topics KM-03-: Fundamentals of electricity (35%) Topic elements to be covered include: Principles and fundamental concepts of electricity Definitions, types, properties and applications of conductors, insulators and Concepts, theories and principles of Electrical Circuits Calculations on basic electrical circuits (resistance, voltage and current) and Basic principles and calculation of magnetism Internal Asse

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with National Artisan Moderation Body (NAMB)/Assessment Quality Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) requirements Critical Topics to be Asse

IN SUMMARY

In this unit we have focussed on:

KT0101 Principles and fundamental concepts of electricity

KT0102 Definitions, types, properties and applications of conductors, insulators and semi-conductors

KT0103 Concepts, theories and principles of Electrical Circuits

KT0104 Calculations on basic electrical circuits (resistance, voltage and current) and power

Complete the workbook activity linked to this chapter, which is activity 3.

CHAPTER 5

KM-03-KT02: ELECTRONICS

In this Chapter we will discuss the following Topics:

KT0201 International Electrotechnical Commission (IEC) symbols for electronic components

KT0202 Principles, safety precautions, identification and basic function of electronic components

KT0203 Basic electronic circuits

KT0204 Principles of rectification, single phase and 3-phase

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Describe the various capacitors used and typical applications | Determine the value of capacitors using charts and calculate the value of capacitance of capacitors in series and parallel | Determine the value of resistors utilizing colour coding charts | Describe the various inductors used and typical applications | Describe the construction and uses of various semiconductor devices | Draw basic electronic circuits using International Electrotechnical Commission (IEC) symbols (Weight 30%)

TOPIC KT0201: INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC) SYMBOLS FOR ELECTRONIC COMPONENTS

International Electrotechnical Commission (IEC) symbols for electronic components

describe, with the aid of a drawing, the application of the right-hand grip rule and show how a magnetic field is established when an electrical current flows through a conductor, by using formula, calculate the force on a current carrying conductor describe, with the aid of drawings, the effect on the magnetic field around a current carrying conductor when placed in a uniform magnetic field basic Electronics Technology Correct Too much Too little Note solder Note solder amount solder pulling away running and from joint mating well Dry joint Good joint Iron too cool Iron too hot Figure 20.22 Soldered joints

Principles, safety precautions, identification and basic function of electronic components

describe, with the aid of a drawing, the application of the right-hand grip rule and show how a magnetic field is established when an electrical current flows through a conductor, by using formula, calculate the force on a current carrying conductor describe, with the aid of drawings, the effect on the magnetic field around a current carrying conductor when placed in a uniform magnetic field basic Electronics Technology Correct Too much

Too little Note solder Note solder amount solder pulling away running and from joint mating well Dry joint Good joint Iron too cool Iron too hot Figure 20.22 Soldered joints

Basic electronic circuits

describe, with the aid of a drawing, the application of the right-hand grip rule and show how a magnetic field is established when an electrical current flows through a conductor, by using formula, calculate the force on a current carrying conductor describe, with the aid of drawings, the effect on the magnetic field around a current carrying conductor when placed in a uniform magnetic field bias the transistor and Use transistor as an electronic switch and series voltage regulator operate and set the Basic concept of power required frequency using IC voltage regulators Digital Electronics - Binary numbers, logic gates and Make a printed circuit board for power supply construct simple circuits containing UJT for triggering and FET as an amplifier

Principles of rectification, single phase and 3-phase

describe, with the aid of a drawing, the application of the right-hand grip rule and show how a magnetic field is established when an electrical current flows through a conductor, by using formula, calculate the force on a current carrying conductor describe, with the aid of drawings, the effect on the magnetic field around a current carrying conductor when placed in a uniform magnetic field yellow 1st number Green 5% Brown 100V Violet 2nd number White 10% Red 250V Black 20% Yellow 400V pF or 47 nF 20% 250 V Figure 20.4 Capacitor colour coding

TOPIC KT0202: PRINCIPLES,SAFETY PRECAUTIONS, IDENTIFICATION AND BASIC FUNCTION OF ELECTRONIC COMPONENTS

Describe, with the aid of a drawing, the application of the right-hand grip rule and show how a magnetic field is established when an electrical current flows through a conductor, by using formula, calculate the force on a current carrying conductor. Describe, with the aid of drawings, the effect on the magnetic field around a current carrying conductor when placed in a uniform magnetic field

671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

Explain by drawing wave form diagrams, the differences between line and phase values. Calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current. Calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods

TOPIC KT0203: BASIC ELECTRONIC CIRCUITS

Describe, with the aid of a drawing, the application of the right-hand grip rule and show how a magnetic field is established when an electrical current flows through a conductor, by using formula, calculate the force on a current carrying conductor. Describe, with the aid of drawings, the effect on the magnetic field around a current carrying conductor when placed in a uniform magnetic field

KM-03-: Electrical principles of appliances (35%) Guidelines for Topics KM-03-: Fundamentals of electricity (35%) Topic elements to be covered include: Principles and fundamental concepts of electricity Definitions, types, properties and applications of conductors, insulators and Concepts, theories and principles of Electrical Circuits Calculations on basic electrical circuits (resistance, voltage and current) and Basic principles and calculation of magnetism Internal Asse

Check transistors for their amplifier. (12 Hrs.) functioning by identifying its type and terminals. (10 Bias the transistor and Use transistor as an electronic switch and series voltage regulator. Operate and set the Basic concept of power required frequency using IC voltage regulators Digital Electronics - Binary numbers, logic gates and Make a printed circuit board for power supply. Construct simple circuits containing UJT for triggering and FET as an amplifier. (05 Hrs

TOPIC KT0204: PRINCIPLES OF RECTIFICATION, SINGLE PHASE AND 3-PHASE

Describe, with the aid of a drawing, the application of the right-hand grip rule and show how a magnetic field is established when an electrical current flows through a conductor, by using formula, calculate the force on a current carrying conductor. Describe, with the aid of drawings, the effect on the magnetic field around a current carrying conductor when placed in a uniform magnetic field

671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 671101000-WM-05, Maintenance processes for electrical installation. and control Details of Knowledge Modules 671101000-KM-01: Health, Safety, Quality and Legislation, L4, Cr 5 The main focus of the learning in this knowledge module is to build an understanding of the health, safety and other legislative requirements for the practise of an Electrician

671101000-KM-02: Tools, Equipment and Materials, L4, Cr 8 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory required for the use of hand tools, power tools and measuring and testing instruments to practise as an Electrician

IN SUMMARY

In this unit we have focussed on:

KT0201 International Electrotechnical Commission (IEC) symbols for electronic components

KT0202 principles, safety precautions, identification and basic function of electronic components

KT0203 Basic electronic circuits

KT0204 Principles of rectification, single phase and 3-phase

Complete the workbook activity linked to this chapter, which is activity 4.

CHAPTER 6

KM-03-KT03: EARTHING AND BONDING

In this Chapter we will discuss the following Topics:

KT0301 Principles of earthing and bonding

KT0302 Methods of earthing and bonding on low voltage overhead lines, equipment and cables.

KT0303 Regulatory and statutory requirements related to earthing and bonding

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Define earthing and bonding and explain the purpose thereof | Explain the regulatory requirements for earthing and bonding | Explain the regulatory requirements for the earthing of neutral conductors on both the supplier and consumer side of an installation | Explain the term 'systems earthing' | Describe the earthing systems relevant to SANS 10142-1. | Describe how a common earth electrode is used in reticulation circuits | Describe the provision of earthing for underground cables and overhead lines | Explain the various processes of measuring, testing and calculating earthing and bonding system values. (Weight 35%)

TOPIC KT0301: PRINCIPLES OF EARTHING AND BONDING

Principles of earthing and bonding

describe, with the aid of a drawing, the application of the right-hand grip rule and show how a magnetic field is established when an electrical current flows through a conductor, by using formula, calculate the force on a current carrying conductor describe, with the aid of drawings, the effect on the magnetic field around a current carrying conductor when placed in a uniform magnetic field the consumer’s mains equipment is normally fixed close to the point at which the supply cable enters the building to meet the requirements of the IEE Regulations it must provide: protection against electric shock (Chapter 41) protection against overcurrent (Chapter 43) isolation and switching (Chapter 53)

TOPIC KT0302: METHODS OF EARTHING AND BONDING ON LOW VOLTAGE OVERHEAD LINES, EQUIPMENT AND CABLES.

Regulations and statutory requirements for wiring of premises Electrical diagrams and symbols Electrical components and their applications Internal Assessment Criteria and Weight List, identify and explain the

meaning of all standard International Electrotechnical Commission (IEC) wiring symbols given on work drawings Identify electrical components and draw schematic diagrams of installations State and explain the safety purpose of earthing, fuse, circuit breakers and earth leakage Describe the

Methods of earthing and bonding on low voltage overhead lines, equipment and Regulatory and statutory requirements related to earthing and bonding Internal Assessment Criteria and Weight Define earthing and bonding and explain the purpose thereof Explain the regulatory requirements for earthing and bonding Explain the regulatory requirements for the earthing of neutral conductors on both the supplier and consumer side of an installation Explain the term 'systems earthing' Describe the earthing systems

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

TOPIC KT0303: REGULATORY AND STATUTORY REQUIREMENTS RELATED TO EARTHING AND BONDING

Regulations and statutory requirements for wiring of premises Electrical diagrams and symbols Electrical components and their applications Internal Assessment Criteria and Weight List, identify and explain the meaning of all standard International Electrotechnical Commission (IEC) wiring symbols given on work drawings Identify electrical components and draw schematic diagrams of installations State and explain the safety purpose of earthing, fuse, circuit breakers and earth leakage Describe the

Methods of earthing and bonding on low voltage overhead lines, equipment and Regulatory and statutory requirements related to earthing and bonding Internal Assessment Criteria and Weight Define earthing and bonding and explain the purpose thereof Explain the regulatory requirements for earthing and bonding Explain the regulatory requirements for the earthing of neutral conductors on both the supplier and consumer side of an installation Explain the term 'systems earthing' Describe the earthing systems

Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with National Artisan Moderation Body (NAMB)/Assessment Quality Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) requirements Critical Topics to be Assessed Externally for the Knowledge Module 671101000-KM-05: Wire ways and wiring, L4, Cr 11 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module

IN SUMMARY

In this unit we have focussed on:

KT0301 Principles of earthing and bonding

KT0302 Methods of earthing and bonding on low voltage overhead lines, equipment and cables.

KT0303 Regulatory and statutory requirements related to earthing and bonding

Complete the workbook activity linked to this chapter, which is activity 6.

CHAPTER 7

KM-04-KT01: INTRODUCTION TO THE WORLD OF WORK AND THE ELECTRICAL TRADE

In this Chapter we will discuss the following Topics:

KT0101 The Electricians world of work

KT0102 Career opportunities of qualified Electricians

KT0103 The Electrician's responsibilities and duties

KT0104 Legislation relating to apprentices in the Electrical trade

KT0105 Trade test requirements

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

The job environment and workplace roles of a Electrician are accurately described and explained

The profile of a Electrician is described with respect to industry descriptions, career path progression and requirements

Legal aspects that includes the employment contract and roles and responsibilities of the employer and employee are explained

The use of applicable legislation relating to the electrical trade is described

Trade test methodology, requirements and assessment procedure leading up to the trade assessment are explained (Weight 100%)

TOPIC KT0101: THE ELECTRICIANS WORLD OF WORK

The Electricians world of work

671101000-KM-04: Industry Context, L3, Cr 2 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the industry context to practise as an Electrician kM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician

TOPIC KT0102: CAREER OPPORTUNITIES OF QUALIFIED ELECTRICIANS

Career opportunities of qualified Electricians

671101000-KM-04: Industry Context, L3, Cr 2 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the industry context to practise as an Electrician kM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician to register and hire them as an electrical apprentice

TOPIC KT0103: THE ELECTRICIAN'S RESPONSIBILITIES AND DUTIES

The Electrician's responsibilities and duties

671101000-KM-04: Industry Context, L3, Cr 2 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the industry context to practise as an Electrician the legal responsibilities of both employers and employees you, as an employee (or worker) in the electrotechnical industry, in order to assist your employer to obey the law as an Employee you must: obey all lawful and reasonable requests; behave in a sensible and responsible way at work; work with care and reasonable skill your Employer must: take care for your safety ; not ask you to do anything unlawful or unreasonable; pay agreed wages; not change your contract of employment without your agreement to promote adequate health and safety measures the employer must consult with the employees' safety representatives in companies which employ more than 20 people this is normally undertaken by forming a safety committee which is made up of a safety officer and employee representatives, usually nominated by a trade union your Employer must: take care for your safety ; not ask you to do anything unlawful or unreasonable; pay agreed wages; not change your contract of employment without your agreement to promote adequate health and safety measures the employer must consult with the employees' safety representatives in companies which employ more than 20 people this is normally undertaken by forming a safety committee which is made up of a safety officer and employee representatives, usually nominated by a trade union

TOPIC KT0104: LEGISLATION RELATING TO APPRENTICES IN THE ELECTRICAL TRADE

Legislation relating to apprentices in the Electrical trade

671101000-KM-04: Industry Context, L3, Cr 2 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the industry context to practise as an Electrician kM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician

TOPIC KT0105: TRADE TEST REQUIREMENTS

Trade test requirements

671101000-KM-04: Industry Context, L3, Cr 2 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the industry context to practise as an Electrician kM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician teacher's supervision, energize and test their circuits students will likely work in small groups, depending on the availability of material and the number of walls available

TEACHER'S SUPERVISION, ENERGIZE AND TEST THEIR CIRCUITS

IN SUMMARY

In this unit we have focussed on:

KT0101 The Electricians world of work

KT0102 Career opportunities of qualified Electricians

KT0103 The Electrician's responsibilities and duties

KT0104 Legislation relating to apprentices in the Electrical trade

KT0105 Trade test requirements

Complete the workbook activities linked to this chapter, which are activities 1 – 5.

CHAPTER 8

KM-05-KT01: WIRE WAYS

In this Chapter we will discuss the following Topics:

KT0101 Definition, types, purpose and applications of wireways

KT0102 General provisions for wireways as prescribed in the SANS 10142-1

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Define the term wireway as given in the SANS 10142-1 Code of Practice | Describe different types of wireways and their applications | Describe and explain the provisions in the SANS 10142-1 Code of Practice for the correct and safe installation and use of the various types of wireways | Describe factors that influences the selection of wireways (Weight 35%)

TOPIC KT0101: DEFINITION, TYPES, PURPOSE AND APPLICATIONS OF WIREWAYS

Definition, types, purpose and applications of wireways

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1

General provisions for wireways as prescribed in the SANS 10142-1

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1

TOPIC KT0102: GENERAL PROVISIONS FOR WIREWAYS AS PRESCRIBED IN THE SANS 10142-1

Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with National Artisan Moderation Body (NAMB)/Assessment Quality Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) requirements Critical Topics to be Assessed Externally for the Knowledge Module 671101000-KM-05: Wire ways and wiring, L4, Cr 11 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module

PA0201 Use Tools, Materials and Equipment PA0202 Use Personal Protective Equipment AK0201 Methods and techniques for using personal protective equipment AK0202 Methods and techniques of using tools AK0203 Methods and techniques of using power tools AK0204 Methods and techniques of ensuring workshop / worksite safety Internal Assessment Criteria Tools, material and equipment were safely and correctly used while adhering to worksite and workshop safety procedures

Cable racks, ladders and trays were correctly installed as prescribed by the manufacturer's Trunking and ducting were correctly installed as prescribed by the manufacturer's The correct methods for terminating and joining wireways were used in accordance with the installation of wireways reflected the specific selection of fastening and support requirements of the installation. The installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1

IN SUMMARY

In this unit we have focussed on:

KT0101 Definition, types, purpose and applications of wireways

KT0102 General provisions for wireways as prescribed in the SANS 10142-1

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 9

KM-05-KT02: WIRING OF INSTALLATIONS

In this Chapter we will discuss the following Topics:

KT0201 Regulations and statutory requirements for wiring of premises

KT0202 Electrical diagrams and symbols

KT0203 Electrical components and their applications

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

List, identify and explain the meaning of all standard International Electrotechnical | Commission (IEC) wiring symbols given on work drawings | Identify electrical components and draw schematic diagrams of installations | State and explain the safety purpose of earthing, fuse, circuit breakers and earth leakage protection unit | Describe the principles of operation of various control systems | Describe the principles of operation of single and three phase circuit breakers and core balance earth leakage relays (wound primaries and straight primaries with tripping relay) | Describe the purpose of load distribution, lightning arrestors and energy control units. (Weight 35%)

TOPIC KT0201: REGULATIONS AND STATUTORY REQUIREMENTS FOR WIRING OF PREMISES

Regulations and statutory requirements for wiring of premises Electrical diagrams and symbols Electrical components and their applications Internal Assessment Criteria and Weight List, identify and explain the meaning of all standard International Electrotechnical Commission (IEC) wiring symbols given on work drawings Identify electrical components and draw schematic diagrams of installations State and explain the safety purpose of earthing, fuse, circuit breakers and earth leakage Describe the

Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with National Artisan Moderation Body (NAMB)/Assessment Quality Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) requirements Critical Topics to be Assessed Externally for the Knowledge Module 671101000-KM-05: Wire ways and wiring, L4, Cr 11 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module

KM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician

TOPIC KT0202: ELECTRICAL DIAGRAMS AND SYMBOLS

Regulations and statutory requirements for wiring of premises Electrical diagrams and symbols Electrical components and their applications Internal Assessment Criteria and Weight List, identify and explain the meaning of all standard International Electrotechnical Commission (IEC) wiring symbols given on work drawings Identify electrical components and draw schematic diagrams of installations State and explain the safety purpose of earthing, fuse, circuit breakers and earth leakage Describe the

KM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician

Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with National Artisan Moderation Body (NAMB)/Assessment Quality Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) requirements Critical Topics to be Assessed Externally for the Knowledge Module 671101000-KM-05: Wire ways and wiring, L4, Cr 11 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module

TOPIC KT0203: ELECTRICAL COMPONENTS AND THEIR APPLICATIONS

Regulations and statutory requirements for wiring of premises Electrical diagrams and symbols Electrical components and their applications Internal Assessment Criteria and Weight List, identify and explain the meaning of all standard International Electrotechnical Commission (IEC) wiring symbols given on work drawings Identify electrical components and draw schematic diagrams of installations State and explain the safety purpose of earthing, fuse, circuit breakers and earth leakage Describe the

KM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician

Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with National Artisan Moderation Body (NAMB)/Assessment Quality Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) requirements Critical Topics to be Assessed Externally for the Knowledge Module 671101000-KM-05: Wire

ways and wiring, L4, Cr 11 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module

IN SUMMARY

In this unit we have focussed on:

KT0201 Regulations and statutory requirements for wiring of premises

KT0202 Electrical diagrams and symbols

KT0203 Electrical components and their applications

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 10

KM-05-KT03: EARTHING AND BONDING

In this Chapter we will discuss the following Topics:

KT0301 Principles of earthing and bonding

KT0302 Methods of earthing and bonding on low voltage overhead lines, equipment and cables.

KT0303 Regulatory and statutory requirements related to earthing and bonding

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Define earthing and bonding and explain the purpose thereof | Explain the regulatory requirements for earthing and bonding | Explain the regulatory requirements for the earthing of neutral conductors on both the supplier and consumer side of an installation | Explain the term 'systems earthing' | Describe the earthing systems relevant to SANS 10142-1. | Describe how a common earth electrode is used in reticulation circuits | Describe the provision of earthing for underground cables and overhead lines | Explain the various processes of measuring, testing and calculating earthing and bonding system values. (Weight 30%)

TOPIC KT0301: PRINCIPLES OF EARTHING AND BONDING

Principles of earthing and bonding

describe, with the aid of a drawing, the application of the right-hand grip rule and show how a magnetic field is established when an electrical current flows through a conductor, by using formula, calculate the force on a current carrying conductor describe, with the aid of drawings, the effect on the magnetic field around a current carrying conductor when placed in a uniform magnetic field

TOPIC KT0302: METHODS OF EARTHING AND BONDING ON LOW VOLTAGE OVERHEAD LINES, EQUIPMENT AND CABLES.

Regulations and statutory requirements for wiring of premises Electrical diagrams and symbols Electrical components and their applications Internal Assessment Criteria and Weight List, identify and explain the meaning of all standard International Electrotechnical Commission (IEC) wiring symbols given on work drawings Identify electrical components and draw schematic diagrams of installations State and explain the safety purpose of earthing, fuse, circuit breakers and earth leakage Describe the

Methods of earthing and bonding on low voltage overhead lines, equipment and Regulatory and statutory requirements related to earthing and bonding Internal Assessment Criteria and Weight Define earthing and bonding and explain the purpose thereof Explain the regulatory requirements for earthing and bonding Explain the regulatory requirements for the earthing of neutral conductors on both the supplier and consumer side of an installation Explain the term 'systems earthing' Describe the earthing systems

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

TOPIC KT0303: REGULATORY AND STATUTORY REQUIREMENTS RELATED TO EARTHING AND BONDING

Regulations and statutory requirements for wiring of premises Electrical diagrams and symbols Electrical components and their applications Internal Assessment Criteria and Weight List, identify and explain the meaning of all standard International Electrotechnical Commission (IEC) wiring symbols given on work drawings Identify electrical components and draw schematic diagrams of installations State and explain the safety purpose of earthing, fuse, circuit breakers and earth leakage Describe the

Methods of earthing and bonding on low voltage overhead lines, equipment and Regulatory and statutory requirements related to earthing and bonding Internal Assessment Criteria and Weight Define earthing and bonding and explain the purpose thereof Explain the regulatory requirements for earthing and bonding Explain the regulatory requirements for the earthing of neutral conductors on both the supplier and consumer side of an installation Explain the term 'systems earthing' Describe the earthing systems

Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with National Artisan Moderation Body (NAMB)/Assessment Quality Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) requirements Critical Topics to be Assessed Externally for the Knowledge Module 671101000-KM-05: Wire ways and wiring, L4, Cr 11 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module

IN SUMMARY

In this unit we have focussed on:

KT0301 Principles of earthing and bonding

KT0302 Methods of earthing and bonding on low voltage overhead lines, equipment and cables.

KT0303 Regulatory and statutory requirements related to earthing and bonding

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 11

KM-06-KT01: ROTATING ELECTRICAL MACHINERY - AC MOTORS

In this Chapter we will discuss the following Topics:

KT0101 Construction of alternating current motors

KT0102 Principle of operation of alternating current motors

KT0103 Configuration of motor connections

KT0104 Types of single phase and three phase alternating current motors

KT0105 Application of alternating current motors

KT0106 Testing principles of single phase and three phase alternating current motors

KT0107 Protection of motors

KT0108 Calculation of motor properties

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

List the different types of alternating current motors and describe their construction | Explain the principle of operation of alternating current motors | Characteristics of single phase and three phase motors | Describe and explain how tests are conducted on alternating current motors | Describe with the aid of diagrams, the configuration of motor connections | Describe and explain protection devices used for alternating current motors | Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits. | Compare the advantages and disadvantages of single- and three-phase motors | Calculate alternating current motor properties (Weight 35%)

TOPIC KT0101: CONSTRUCTION OF ALTERNATING CURRENT MOTORS

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits

KM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician

TOPIC KT0102: PRINCIPLE OF OPERATION OF ALTERNATING CURRENT MOTORS

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

Electrical drawings Introduction to Programmable Logic Controllers (PLC's) Introduction to Soft starters Introduction to Variable speed drives Internal Assessment Criteria and Weight Describe disconnectors, relays, timers and contactors in terms of construction and operating principles with reference to the contacts, operating coils (where applicable) and Describe, with the aid of labelled drawings, the principle of operation of the over current and earth leakage protection relays.

Application of alternating current motors Testing principles of single phase and three phase alternating current motors Protection of motors Calculation of motor properties Internal Assessment Criteria and Weight List the different types of alternating current motors and describe their construction Explain the principle of operation of alternating current motors Characteristics of single phase and three phase motors Describe and explain how tests are conducted on alternating current moto

TOPIC KT0103: CONFIGURATION OF MOTOR CONNECTIONS

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

KM-06-: Rotating Electrical Machinery - DC motors (35%) Topic elements to be covered include: Construction of direct current motors Principles of operation for direct current motors Configuration of direct current motor

connections Types and application of direct current motors Testing principles of direct current motors Protection of direct current motors Calculation of direct current motor properties Internal Assessment Criteria and Weight List the different ty

Electrical drawings Introduction to Programmable Logic Controllers (PLC's) Introduction to Soft starters Introduction to Variable speed drives Internal Assessment Criteria and Weight Describe disconnectors, relays, timers and contactors in terms of construction and operating principles with reference to the contacts, operating coils (where applicable) and Describe, with the aid of labelled drawings, the principle of operation of the over current and earth leakage protection relays.

TOPIC KT0104: TYPES OF SINGLE PHASE AND THREE PHASE ALTERNATING CURRENT MOTORS

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

671101000-KM-07: Electrical supply systems and components, L4, Cr 31 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory of electrical supply systems and components required to practise as an Electrician

Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits

TOPIC KT0105: APPLICATION OF ALTERNATING CURRENT MOTORS

Application of alternating current motors Testing principles of single phase and three phase alternating current motors Protection of motors Calculation of motor properties Internal Assessment Criteria and Weight List the different types of alternating current motors and describe their construction Explain the principle of operation of alternating current motors Characteristics of single phase and three phase motors Describe and explain how tests are conducted on alternating current moto

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive

to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

KM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician

TOPIC KT0106: TESTING PRINCIPLES OF SINGLE PHASE AND THREE PHASE ALTERNATING CURRENT MOTORS

Application of alternating current motors Testing principles of single phase and three phase alternating current motors Protection of motors Calculation of motor properties Internal Assessment Criteria and Weight List the different types of alternating current motors and describe their construction Explain the principle of operation of alternating current motors Characteristics of single phase and three phase motors Describe and explain how tests are conducted on alternating current moto

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

671101000-KM-07: Electrical supply systems and components, L4, Cr 31 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory of electrical supply systems and components required to practise as an Electrician

TOPIC KT0107: PROTECTION OF MOTORS

Application of alternating current motors Testing principles of single phase and three phase alternating current motors Protection of motors Calculation of motor properties Internal Assessment Criteria and Weight List the different types of alternating current motors and describe their construction Explain the principle of operation of alternating current motors Characteristics of single phase and three phase motors Describe and explain how tests are conducted on alternating current moto

KM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

TOPIC KT0108: CALCULATION OF MOTOR PROPERTIES

Application of alternating current motors Testing principles of single phase and three phase alternating current motors Protection of motors Calculation of motor properties Internal Assessment Criteria and Weight List the different types of alternating current motors and describe their construction Explain the principle of operation of alternating current motors Characteristics of single phase and three phase motors Describe and explain how tests are conducted on alternating current moto

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

KM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician

IN SUMMARY

In this unit we have focussed on:

KT0101 Construction of alternating current motors

KT0102 Principle of operation of alternating current motors

KT0103 Configuration of motor connections

KT0104 Types of single phase and three phase alternating current motors

KT0105 Application of alternating current motors

KT0106 Testing principles of single phase and three phase alternating current motors

KT0107 Protection of motors

KT0108 Calculation of motor properties

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 12

KM-06-KT02: ROTATING ELECTRICAL MACHINERY - DC MOTORS

In this Chapter we will discuss the following Topics:

KT0201 Construction of direct current motors

KT0202 Principles of operation for direct current motors

KT0203 Configuration of direct current motor connections

KT0204 Types and application of direct current motors

KT0205 Testing principles of direct current motors

KT0206 Protection of direct current motors

KT0207 Calculation of direct current motor properties

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

List the different types of direct current motors and describe their construction | Explain the principle of operation of direct current motors | nt types of direct current motors | Describe and explain how various tests are conducted on direct current motors | Describe with the aid of diagrams, the configuration of motor connections, power circuits and control circuits. | Describe and explain protection devices used for direct current motors | Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation, short circuits. | Compare the advantages and disadvantages of different types of direct current motors | Calculate direct current motor properties (Weight 35%)

TOPIC KT0201: CONSTRUCTION OF DIRECT CURRENT MOTORS

Construction of direct current motors

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and

document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

Types and application of direct current motors

explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits

Protection of direct current motors

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

TOPIC KT0202: PRINCIPLES OF OPERATION FOR DIRECT CURRENT MOTORS

KM-06-: Rotating Electrical Machinery - DC motors (35%) Topic elements to be covered include: Construction of direct current motors Principles of operation for direct current motors Configuration of direct current motor connections Types and application of direct current motors Testing principles of direct current motors Protection of direct current motors Calculation of direct current motor properties Internal Assessment Criteria and Weight List the different ty

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

KM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician

TOPIC KT0203: CONFIGURATION OF DIRECT CURRENT MOTOR CONNECTIONS

KM-06-: Rotating Electrical Machinery - DC motors (35%) Topic elements to be covered include: Construction of direct current motors Principles of operation for direct current motors Configuration of direct current motor connections Types and application of direct current motors Testing principles of direct current motors Protection of direct current motors Calculation of direct current motor properties Internal Assessment Criteria and Weight List the different ty

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

KM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician

TOPIC KT0204: TYPES AND APPLICATION OF DIRECT CURRENT MOTORS

KM-06-: Rotating Electrical Machinery - DC motors (35%) Topic elements to be covered include: Construction of direct current motors Principles of operation for direct current motors Configuration of direct current motor connections Types and application of direct current motors Testing principles of direct current motors Protection of direct current motors Calculation of direct current motor properties Internal Assessment Criteria and Weight List the different ty

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

KM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician

TOPIC KT0205: TESTING PRINCIPLES OF DIRECT CURRENT MOTORS

KM-06-: Rotating Electrical Machinery - DC motors (35%) Topic elements to be covered include: Construction of direct current motors Principles of operation for direct current motors Configuration of direct current motor connections Types and application of direct current motors Testing principles of direct current motors Protection of direct current motors Calculation of direct current motor properties Internal Assessment Criteria and Weight List the different ty

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

KM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician

TOPIC KT0206: PROTECTION OF DIRECT CURRENT MOTORS

KM-06-: Rotating Electrical Machinery - DC motors (35%) Topic elements to be covered include: Construction of direct current motors Principles of operation for direct current motors Configuration of direct current motor connections Types and application of direct current motors Testing principles of direct current motors Protection of direct current motors Calculation of direct current motor properties Internal Assessment Criteria and Weight List the different ty

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

KM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician

TOPIC KT0207: CALCULATION OF DIRECT CURRENT MOTOR PROPERTIES

KM-06-: Rotating Electrical Machinery - DC motors (35%) Topic elements to be covered include: Construction of direct current motors Principles of operation for direct current motors Configuration of direct current motor connections Types and application of direct current motors Testing principles of direct current motors Protection of direct current motors Calculation of direct current motor properties Internal Assessment Criteria and Weight List the different ty

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

KM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician

IN SUMMARY

In this unit we have focussed on:

KT0201 Construction of direct current motors

KT0202 Principles of operation for direct current motors

KT0203 Configuration of direct current motor connections

KT0204 Types and application of direct current motors

KT0205 Testing principles of direct current motors

KT0206 Protection of direct current motors

KT0207 Calculation of direct current motor properties

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 13

KM-06-KT03: ROTATING ELECTRICAL MACHINERY - ALTERNATORS AND GENERATORS

In this Chapter we will discuss the following Topics:

KT0301 Construction of alternators and generators

KT0302 Principle of operation of alternators and generators

KT0303 Configuration of alternator and generator connections

KT0304 Types and application of alternators and generators

KT0305 Testing principles of alternators and generators

KT0306 Protection of alternators and generators

KT0307 Calculate properties of alternators and generators

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

List the different types of alternators and generators and describe their construction | Explain the principle of operation of alternators and generators | Explain the fundamental differences between alternators and generators | nt types of alternators and generators | Describe and explain how tests are conducted on alternators and generators | Describe with the aid of diagrams, the configuration of alternator and generator connections, power circuits and control circuits. | Describe and explain protection devices used for alternators and generators | Explain how time delay and current rating of overload protection devices influence their use in protecting alternators and generators from damage in the case of locked rotors, overload during operation and short circuits. (Weight 30%)

TOPIC KT0301: CONSTRUCTION OF ALTERNATORS AND GENERATORS

KM-06-: Rotating Electrical Machinery - DC motors (35%) Topic elements to be covered include: Construction of direct current motors Principles of operation for direct current motors Configuration of direct current motor connections Types and application of direct current motors Testing principles of direct current motors Protection of direct current motors Calculation of direct current motor properties Internal Assessment Criteria and Weight List the different ty

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

KM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician

TOPIC KT0302: PRINCIPLE OF OPERATION OF ALTERNATORS AND GENERATORS

KM-06-: Rotating Electrical Machinery - DC motors (35%) Topic elements to be covered include: Construction of direct current motors Principles of operation for direct current motors Configuration of direct current motor connections Types and application of direct current motors Testing principles of direct current motors Protection of direct current motors Calculation of direct current motor properties Internal Assessment Criteria and Weight List the different ty

Types and application of alternators and generators Testing principles of alternators and generators Protection of alternators and generators Calculate properties of alternators and generators Internal Assessment Criteria and Weight List the different types of alternators and generators and describe their construction Explain the principle of operation of alternators and generators Explain the fundamental differences between alternators and generators.

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited.

TOPIC KT0303: CONFIGURATION OF ALTERNATOR AND GENERATOR CONNECTIONS

KM-06-: Rotating Electrical Machinery - DC motors (35%) Topic elements to be covered include: Construction of direct current motors Principles of operation for direct current motors Configuration of direct current motor connections Types and application of direct current motors Testing principles of direct current motors Protection of direct current motors Calculation of direct current motor properties Internal Assessment Criteria and Weight List the different ty

Types and application of alternators and generators Testing principles of alternators and generators Protection of alternators and generators Calculate properties of alternators and generators Internal Assessment Criteria and Weight List the different types of alternators and generators and describe their construction Explain the principle of operation of alternators and generators Explain the fundamental differences between alternators and generators.

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

TOPIC KT0304: TYPES AND APPLICATION OF ALTERNATORS AND GENERATORS

Types and application of alternators and generators Testing principles of alternators and generators Protection of alternators and generators Calculate properties of alternators and generators Internal Assessment Criteria and Weight List the different types of alternators and generators and describe their construction Explain the principle of operation of alternators and generators Explain the fundamental differences between alternators and generators

KM-06-: Rotating Electrical Machinery - DC motors (35%) Topic elements to be covered include: Construction of direct current motors Principles of operation for direct current motors Configuration of direct current motor connections Types and application of direct current motors Testing principles of direct current motors Protection of direct current motors Calculation of direct current motor properties Internal Assessment Criteria and Weight List the different ty

Describe the provision of earthing for underground cables and overhead lines Explain the various processes of measuring, testing and calculating earthing and bonding Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with Nat

TOPIC KT0305: TESTING PRINCIPLES OF ALTERNATORS AND GENERATORS

Types and application of alternators and generators Testing principles of alternators and generators Protection of alternators and generators Calculate properties of alternators and generators Internal Assessment Criteria and

Weight List the different types of alternators and generators and describe their construction Explain the principle of operation of alternators and generators Explain the fundamental differences between alternators and generators

KM-06-: Rotating Electrical Machinery - DC motors (35%) Topic elements to be covered include: Construction of direct current motors Principles of operation for direct current motors Configuration of direct current motor connections Types and application of direct current motors Testing principles of direct current motors Protection of direct current motors Calculation of direct current motor properties Internal Assessment Criteria and Weight List the different ty

Internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures. Portable power tools were correctly used, as per workshop practices and safety The correct personal protective equipment was used and complied with the workshop PM-05-PS04: Install equipment and control systems Scope of Practical Skill Given equipment and control systems Includes: Batteries, Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rot

TOPIC KT0306: PROTECTION OF ALTERNATORS AND GENERATORS

Types and application of alternators and generators Testing principles of alternators and generators Protection of alternators and generators Calculate properties of alternators and generators Internal Assessment Criteria and Weight List the different types of alternators and generators and describe their construction Explain the principle of operation of alternators and generators Explain the fundamental differences between alternators and generators

KM-06-: Rotating Electrical Machinery - DC motors (35%) Topic elements to be covered include: Construction of direct current motors Principles of operation for direct current motors Configuration of direct current motor connections Types and application of direct current motors Testing principles of direct current motors Protection of direct current motors Calculation of direct current motor properties Internal Assessment Criteria and Weight List the different ty

Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rotating electrical machinery - AC, Rotating Electrical Machinery - DC, Rotating electrical machinery - Alternators and Generators, Protection devices, luminaries, current transformers, potential transformers and shunts according to the design specifications

TOPIC KT0307: CALCULATE PROPERTIES OF ALTERNATORS AND GENERATORS

Types and application of alternators and generators Testing principles of alternators and generators Protection of alternators and generators Calculate properties of alternators and generators Internal Assessment Criteria and Weight List the different types of alternators and generators and describe their construction Explain the principle of operation of alternators and generators Explain the fundamental differences between alternators and generators

KM-06-: Rotating Electrical Machinery - DC motors (35%) Topic elements to be covered include: Construction of direct current motors Principles of operation for direct current motors Configuration of direct current motor connections Types and application of direct current motors Testing principles of direct current motors Protection of direct current motors Calculation of direct current motor properties Internal Assessment Criteria and Weight List the different ty

KM-03-: Electrical principles of appliances (35%) 671101000-KM-04: Industry Context, L3, Cr 2 The main focus of the learning in this knowledge module is to build an understanding of the industry context of a Electrician

IN SUMMARY

In this unit we have focussed on:

KT0301 Construction of alternators and generators

KT0302 Principle of operation of alternators and generators

KT0303 Configuration of alternator and generator connections

KT0304 Types and application of alternators and generators

KT0305 Testing principles of alternators and generators

KT0306 Protection of alternators and generators

KT0307 Calculate properties of alternators and generators

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 14

KM-07-KT01: CONCEPTS, THEORIES AND PRINCIPLES OF SUPPLY SYSTEMS

In this Chapter we will discuss the following Topics:

KT0101 Theories and concepts of alternating current

KT0102 Fundamental principles of alternating current

KT0103 Alternating current generation (distribution systems theory)

KT0104 Characteristics and calculations from alternating current waveforms

KT0105 Theories, concepts and principles of direct current sources

KT0106 Sources of direct current

KT0107 Calculations of direct and alternating current circuits

KT0108 Construction and operating principles of direct current generators

KT0109 Characteristics, sources and generation of renewable energy

KT0110 Relevant legislative requirements

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Explain the principles of generation by using Fleming's right hand rule and Faraday's laws

Explain the generation and differences between single and three phase alternating current by using wave forms and vector diagrams.

Explain the load balancing principles in a three phase supply system.

Explain by drawing wave form diagrams, the differences between line and phase values.

Calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account

Explain the generation principles of direct current.

Calculate power and energy in direct and alternating current circuits

Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods.

Describe with the aid of drawings the components of direct current generators and their functions.

Explain with the aid of drawings the operating principles of direct current generators. (Weight 20%)

TOPIC KT0101: THEORIES AND CONCEPTS OF ALTERNATING CURRENT

671101000-KM-07: Electrical supply systems and components, L4, Cr 31 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory of electrical supply systems and components required to practise as an Electrician

KM-03-: Electrical principles of appliances (35%) Guidelines for Topics KM-03-: Fundamentals of electricity (35%) Topic elements to be covered include: Principles and fundamental concepts of electricity Definitions, types, properties and applications of conductors, insulators and Concepts, theories and principles of Electrical Circuits Calculations on basic electrical circuits (resistance, voltage and current) and Basic principles and calculation of magnetism Internal Asse

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

TOPIC KT0102: FUNDAMENTAL PRINCIPLES OF ALTERNATING CURRENT

Fundamental principles of alternating current

describe the operation and functions of different types of low voltage protective devices including overload relays, fuses, circuit breakers and earth leakage protection devices explain by drawing wave form diagrams, the differences between line and phase values calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods

TOPIC KT0103: ALTERNATING CURRENT GENERATION (DISTRIBUTION SYSTEMS THEORY)

a load connected to phase and neutral of a star-connected three phase four-wire supply system from the local sub-station would have a voltage of: c the phase voltage of a star connected load is 100 V the phase voltage of a delta connected load is 100 V the phase current of a star connected load is 100 A three-phase 400 V supply is used for supplying small industrial and commercial loads such as garages, schools and blocks of fl ats

TOPIC KT0104: CHARACTERISTICS AND CALCULATIONS FROM ALTERNATING CURRENT WAVEFORMS

Characteristics and calculations from alternating current waveforms

explain by drawing wave form diagrams, the differences between line and phase values calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods

TOPIC KT0105: THEORIES, CONCEPTS AND PRINCIPLES OF DIRECT CURRENT SOURCES

Theories, concepts and principles of direct current sources

explain by drawing wave form diagrams, the differences between line and phase values calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods

TOPIC KT0106: SOURCES OF DIRECT CURRENT

Sources of direct current

explain by drawing wave form diagrams, the differences between line and phase values calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods

TOPIC KT0107: CALCULATIONS OF DIRECT AND ALTERNATING CURRENT CIRCUITS

explain by drawing wave form diagrams, the differences between line and phase values calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods

TOPIC KT0108: CONSTRUCTION AND OPERATING PRICIPLES OF DIRECT CURRENT GENERATORS

Construction and operating principles of direct current generators

explain by drawing wave form diagrams, the differences between line and phase values calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods

TOPIC KT0109: CHARICTERISTICS, SOURCES AND GENERATION OF RENEWABLE ENERGY

Characteristics, sources and generation of renewable energy

explain by drawing wave form diagrams, the differences between line and phase values calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation

TOPIC KT0110: RELEVANT LEGLISATIVE REQUIMENTS

Relevant legislative requirements

explain by drawing wave form diagrams, the differences between line and phase values calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into

account Explain the generation principles of direct current calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods

IN SUMMARY

In this unit we have focussed on:

KT0101 Theories and concepts of alternating current

KT0102 Fundamental principles of alternating current

KT0103 Alternating current generation (distribution systems theory)

KT0104 Characteristics and calculations from alternating current waveforms

KT0105 Theories, concepts and principles of direct current sources

KT0106 Sources of direct current

KT0107 Calculations of direct and alternating current circuits

KT0108 Construction and operating principles of direct current generators

KT0109 Characteristics, sources and generation of renewable energy

KT0110 Relevant legislative requirements

Complete the workbook activities linked to this chapter, which are activities 1 – 10.

CHAPTER 15

KM-07-KT02: BATTERIES

In this Chapter we will discuss the following Topics:

KT0201 Fundamentals of battery operation

KT0202 Types and classification of batteries

KT0203 Safety precautions of batteries

KT0204 Care and maintenance of batteries

KT0205 Battery components and construction

KT0206 Disposal of batteries

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Explain the terms potential difference, electromotive force, relative density and capacity | Describe, with the aid of a diagram, the fundamentals of battery construction | Describe and explain the various types of batteries and cells | Describe the advantages and disadvantages of primary and secondary cells for particular applications | Describe and explain the correct procedures to care, maintain, store and dispose of batteries and the hazards and safety precautions associated with batteries and battery rooms | Name the instruments used for testing batteries, describe how they are used and explain the purpose of each test | Explain, with the aid of drawings, the discharging and charging action of lead-acid cells (Weight 10%)

TOPIC KT0201: FUNDEMENTALS OF BATTERY OPERATION

Fundamentals of battery operation

explain by drawing wave form diagrams, the differences between line and phase values calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods chemical effect of electric Hrs) current and Laws of batteries and solar 66 install, test and main57.Use of various types of cells hrs.) Chemical effect of electric current Skill 50 Hrs.; tenancy of batteries and Laws of electrolysis 58.Practice on grouping of cells for special nd solar cell install, test and main57.Use of various types of cells

Types and classification of batteries

explain by drawing wave form diagrams, the differences between line and phase values calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods current level Cells are connected in series to a

Safety precautions of batteries

explain by drawing wave form diagrams, the differences between line and phase values calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods chemical effect of electric Hrs) current and Laws of batteries and solar 66 install, test and main57.Use of various types of cells hrs.) Chemical effect of electric current Skill 50 Hrs.; tenancy of batteries and Laws of electrolysis 58.Practice on grouping of cells for space and solar cell install, test and main57.Use of various types of cells

Care and maintenance of batteries

explain by drawing wave form diagrams, the differences between line and phase values calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods & 1.4.44 Capacitors - types - functions, grouping and uses (QR Code Pg no.80) * 80 Power, energy and power factor in AC single & 1.5.51 Power factor - Improvement of power factor 108 3-Phase AC fundamentals 110 Primary cells and secondary cells (QR Code Pg no.120)* 4 120 Battery charging method - Battery charger (QR Code Pg chemical effect of electric Hrs) current and Laws of batteries and solar 66

Battery components and construction

explain by drawing wave form diagrams, the differences between line and phase values calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods chemical effect of electric Hrs) current and Laws of batteries and solar 66

Disposal of batteries

explain by drawing wave form diagrams, the differences between line and phase values calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods install, test and main57.Use of various types of cells hrs.) Chemical effect of electric current Skill 50 Hrs.; tenancy of batteries and Laws of electrolysis 58.Practice on grouping of cells for space and solar cell install, test and main57.Use of various types of cells

TOPIC KT0202: TYPES AND CLASSIFICATION OF BATTERIES

Explain by drawing wave form diagrams, the differences between line and phase values. Calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current. Calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods

Name the instruments used for testing batteries, describe how they are used and explain the purpose of each test Explain, with the aid of drawings, the discharging and charging action of lead-acid cells Topic elements to be covered include: Theories of single phase wound transformers Types of single phase transformers including single wound, double wound and auto-transformers and their applications

A battery is a device that stores chemical energy and converts it to electrical energy, it is a common source of electrical power, however, it differs in the type of current supplied by it. A battery supplies a Direct Current (DC) as opposed to the Alternating Current supplied to our homes. Batteries are a portable source of electricity, think of your cell phone, torches, electric cars, portable power tools and much more

TOPIC KT0203: SAFETY PRECAUTIONS OF BATTERIES

Explain by drawing wave form diagrams, the differences between line and phase values. Calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current. Calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods

phase is short-circuited in three phase four wire system and compare with healthy system.(10 hrs) 22-23 Project work / Industrial visit Prepare and assemble a test board with switches, plug socket, lamp holder etc. Temperature controlled system for switching ‘ON’ and ‘OFF’ of any circuit using bimetallic strip. Series/

parallel combinational circuits 27-28 Install, test and 65. Use of various types of cells. Chemical effect of electric Hrs) current and Laws of batteries and solar 66

Install, test and main57.Use of various types of cells. (08 Hrs.) Chemical effect of electric current Skill 50 Hrs.; tenancy of batteries and Laws of electrolysis. 58.Practice on grouping of cells for space and solar cell. field voltage and current under differ Explanation of Anodes and Cath Knowledge 10 (NOS: PSS/N6001) Ent conditions and care. (12 Hrs.) 59.Prepare and practice on battery charger Types of cells, advantages / and details of charging circuit. (12 vantages and their applications

TOPIC KT0204: CARE AND MAINTENANCE OF BATTERIES

Explain by drawing wave form diagrams, the differences between line and phase values. Calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current. Calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods

Power : Electrician (NSQF Revised - 2022) - Related Theory for Exercise 1.6.59 129 Power Related Theory for Exercise 1.6.60 Electrician - Cells and Batteries Care and maintenance of batteries Objectives: At the end of this lesson you shall be able to state the guidelines for care and maintenance of batteries and installation Scan the QR Code to view state the precaution to be followed while charging and discharging of battery

Care and maintenance of batteries : The lead acid batteries must be operated under the right conditions if The rate of charging at the end of the period called finish rate they are to function properly. Regular maintenance is most important. It must not exceed the value necessary in order to maintain proper conditions and thus recommended by the manufacturer. prolong the life of the battery. During recharging, the lead acid battery produces flammable The battery should not be discharged beyond the minimum gases

TOPIC KT0205: BATTERY COMPONENTS AND CONSTRUCTION

Explain by drawing wave form diagrams, the differences between line and phase values. Calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current. Calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods

televisions, GPS units, circuit board components, or small electrical connectors, desk-top and laptop computers, to transfer information quickly. It is a very light, soft, ductile and malleable conductor. It does not corrode and is easy to work on or terminate. Aluminium is used for outdoor activity, such as overhead transmission and Tin

is a metallic conductor, which is malleable, very soft and ductile and it offers a high resistance to acid attack. Tin is used as a solder when mixed with lead

A battery is a device that stores chemical energy and converts it to electrical energy, it is a common source of electrical power, however, it differs in the type of current supplied by it. A battery supplies a Direct Current (DC) as opposed to the Alternating Current supplied to our homes. Batteries are a portable source of electricity, think of your cell phone, torches, electric cars, portable power tools and much more

TOPIC KT0206: DISPOSAL OF BATTERIES

Explain by drawing wave form diagrams, the differences between line and phase values. Calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account Explain the generation principles of direct current. Calculate power and energy in direct and alternating current circuits Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods

Install, test and main57.Use of various types of cells. (08 Hrs.) Chemical effect of electric current Skill 50 Hrs.; tenancy of batteries and Laws of electrolysis. 58.Practice on grouping of cells for space and solar cell. field voltage and current under differ Explanation of Anodes and Cath Knowledge 10 (NOS: PSS/N6001) Ent conditions and care. (12 Hrs.) 59.Prepare and practice on battery charger Types of cells, advantages / dissuading and details of charging circuit. (12 vantages and their applications

technical requirements. The alternating current supply batteries (Figs 1a and 1b) and DC generators (dynamos). (AC) and the direct current supply (DC). (Fig 1C) __ DC is represented by this symbol. Direct voltage is of constant magnitude (amplitude). It remains at the same amplitude from the moment of AC is represented by this symbol. switching on to the moment of switching off. The polarity of the voltage source does not change. (Fig 2) Power : Electrician (NSQF - Revised 2022) - Related Theory for Exercise 1.2

IN SUMMARY

In this unit we have focussed on:

KT0201 Fundamentals of battery operation

KT0202 Types and classification of batteries

KT0203 Safety precautions of batteries

KT0204 Care and maintenance of batteries

KT0205 Battery components and construction

KT0206 Disposal of batteries

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 16

KM-07-KT03: TRANSFORMERS

In this Chapter we will discuss the following Topics:

KT0301 Theories of single phase wound transformers

KT0302 Types of single phase transformers including single wound, double wound and auto-transformers and their applications.

KT0303 Fundamentals of transformer construction

KT0305 Principles of single phase transformer operation

KT0306 Principles of single phase auto-transformer operation

KT0307 Transformer losses

KT0308 Formulas and calculations on input and output of transformers

KT0310 Types of three phase transformers and their applications

KT0311 Construction of three phase transformers including open core, closed core and shell or divided core

KT0312 Principles of 3 phase transformer operation and configuration

KT0313 Principles of 3 phase auto transformer operation and configuration

KT0314 Transformer protection and phasing

KT0704 Transformer cooling systems

KT0709 Maintenance requirements of transformers

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Name and describe the types of single phase transformers | Describe, with the aid of drawings, the construction of single phase transformers | Describe and explain the cooling systems used on transformers | Describe, with the aid of drawings, the principle of operation of single phase transformers in terms of mutual inductance and the henry as the unit of inductance | Explain, with the aid of drawings, the principle of operation of single phase autotransformers | Describe types of transformer losses and explain their causes and effects | Calculate the terminal voltage, turns and current ratios for an single phase transformer. | Describe and explain

the maintenance requirements for single/three phase transformers | Describe, with the aid of diagrams, the construction of three phase transformers | Describe the various types of construction of three phase transformers - open core, closed core, and shell or divided core | Describe the basic transformer maintenance procedures | Calculate the relationship of the voltage, current, turns ratio's and power including true power, apparent power and power factor. | Describe and explain how inductance causes a phase angle between voltage and current and its relationship with the concept 'power factor' | Describe, with the aid of diagrams, three phase transformer connections | Calculate the line and phase voltages and currents relating to three-phase transformer connections | Explain the concept of, and need for, transformer tapings | Explain transformer protection and phasing (Weight 20%)

TOPIC KT0301: THEORIES OF SINGLE PHASE WOUND TRANSFORMERS

Theories of single phase wound transformers

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose identify the basic components of various measuring and test instruments and describe the basic principle of operation Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation

Types of single phase transformers including single wound, double wound and auto-transformers and their applications

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose identify the basic components of various measuring and test instruments and describe the basic principle of operation explain, with the aid of drawings, the principle of operation of single phase autotransformers Describe types of transformer losses and explain their causes and effects Calculate the terminal voltage, turns and current ratios for an single phase transformer Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation

Fundamentals of transformer construction

explain, with the aid of drawings, the principle of operation of single phase autotransformers Describe types of transformer losses and explain their causes and effects Calculate the terminal voltage, turns and current ratios

for an single phase transformer hrs.) Test single phase energy meter for its errors hrs.) Test single phase energy meter for its errors

Principles of single phase transformer operation

hrs.) and voltage drop effect of Knowledge 91 hrs.) Test single phase energy meter for its errors connect and test a single-phase auto- transformer determine the losses (iron loss and copper loss) and the regulation of a single-phase transformer at different loads carry out winding for small transformer of 1KVA rating hrs.) Test single phase energy meter for its errors

CONNECT AND TEST A SINGLE-PHASE AUTO- TRANSFORMER

Principles of single phase auto-transformer operation

series and parallel operation of Determine voltage regulation of singleN2407) phase transformer at different loads and power factors auto Transformer and instrument Perform series and parallel operation transformers (CT & PT) hrs.) Test single phase energy meter for its errors hrs.) Test single phase energy meter for its errors

Transformer losses

explain, with the aid of drawings, the principle of operation of single phase autotransformers Describe types of transformer losses and explain their causes and effects Calculate the terminal voltage, turns and current ratios for an single phase transformer connect and test a single-phase auto- transformer determine the losses (iron loss and copper loss) and the regulation of a single-phase transformer at different loads measure the current and voltage using CT and PT carry out winding for small transformer of 1KVA rating

CONNECT AND TEST A SINGLE-PHASE AUTO- TRANSFORMER

Formulas and calculations on input and output of transformers

objectives: At the end of this lesson you shall be able to explain the principle of the operation of a transformer derive the EMF equation of a two-winding transformer derive the transformation ratio of a transformer let us consider an ideal transformer (Fig 1) whose secondary is open and whose primary is connected to a sinusoidal voltage V the transformers work on the principle of mutual induction of Faraday's law of electro - magnetic induction

Types of three phase transformers and their applications

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose identify the basic components of various measuring and test instruments and describe the basic principle of operation Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection

of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation

Construction of three phase transformers including open core, closed core and shell or divided core

explain, with the aid of drawings, the principle of operation of single phase autotransformers Describe types of transformer losses and explain their causes and effects Calculate the terminal voltage, turns and current ratios for an single phase transformer Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose identify the basic components of various measuring and test instruments and describe the basic principle of operation Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation

Principles of 3 phase transformer operation and configuration

connect and test a single-phase auto- transformer determine the losses (iron loss and copper loss) and the regulation of a single-phase transformer at different loads carry out winding for small transformer of 1KVA rating briefly describe what we mean by a schedule of work

CONNECT AND TEST A SINGLE-PHASE AUTO- TRANSFORMER

Principles of 3 phase auto transformer operation and configuration

series and parallel operation of Determine voltage regulation of singleN2407) phase transformer at different loads and power factors auto Transformer and instrument Perform series and parallel operation transformers (CT & PT) voltage Regulation and Auto Transformer and regulation of single-phase instrument transformers (CT transformer at different loads and power factors perform series and parallel operation of two single phase transformers verify the terminals and accessories of three phase transformer HT and LT Perform 3 phase Method of connecting three operation (i) delta-delta, single phase transformers for three phase operation

Transformer protection and phasing

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose identify the basic components of various measuring and test instruments and describe the basic principle of operation Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit identify the basic components of various measuring and test instruments and describe the basic principle of operation

explain, with the aid of drawings, the principle of operation of single phase autotransformers Describe types of transformer losses and explain their causes and effects Calculate the terminal voltage, turns and current ratios for an single phase transformer series and parallel operation of Determine voltage regulation of singleN2407) phase transformer at different loads and power factors auto Transformer and instrument Perform series and parallel operation transformers (CT & PT)

Maintenance requirements of transformers

explain, with the aid of drawings, the principle of operation of single phase autotransformers Describe types of transformer losses and explain their causes and effects Calculate the terminal voltage, turns and current ratios for an single phase transformer objectives: At the end of this lesson you shall be able to explain the principle of the operation of a transformer derive the EMF equation of a two-winding transformer derive the transformation ratio of a transformer let us consider an ideal transformer (Fig 1) whose secondary is open and whose primary is connected to a sinusoidal voltage V the transformers work on the principle of mutual induction of Faraday's law of electro - magnetic induction

TOPIC KT0302: TYPES OF SINGLE PHASE TRANSFORMERS INCLUDING SINGLE WOUND, DOUBLE WOUND AND AUTO-TRANSFORMERS AND THEIR APPLICATIONS.

Name the instruments used for testing batteries, describe how they are used and explain the purpose of each test Explain, with the aid of drawings, the discharging and charging action of lead-acid cells Topic elements to be covered include: Theories of single phase wound transformers Types of single phase transformers including single wound, double wound and auto-transformers and their applications

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit. Internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose. Identify the basic components of various measuring and test instruments and describe the basic principle of operation

671101000-KM-07: Electrical supply systems and components, L4, Cr 31 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory of electrical supply systems and components required to practise as an Electrician

TOPIC KT0303: FUNDAMENTALS OF TRANSFORMER CONSTRUCTION

Name the instruments used for testing batteries, describe how they are used and explain the purpose of each test Explain, with the aid of drawings, the discharging and charging action of lead-acid cells Topic elements to be covered include: Theories of single phase wound transformers Types of single phase transformers including single wound, double wound and auto-transformers and their applications

Explain, with the aid of drawings, the principle of operation of single phase autotransformers Describe types of transformer losses and explain their causes and effects Calculate the terminal voltage, turns and current ratios for an single phase transformer

to street lighting and stadium lighting, trenched cable systems, direct current Context: Energy environments Transformers; distribution enclosures; distribution systems, earthing systems, metering units, , single phase and three phase AC motors and general lighting used in domestic, industrial and emergency areas , battery charging systems, metering systems and associated medium voltage and HV switchgear under dead conditions, street lighting and stadium lighting, overhead line systems; Access to the following cont.

TOPIC KT0305: PRINCIPLES OF SINGLE PHASE TRANSFORMER OPERATION

Name the instruments used for testing batteries, describe how they are used and explain the purpose of each test Explain, with the aid of drawings, the discharging and charging action of lead-acid cells Topic elements to be covered include: Theories of single phase wound transformers Types of single phase transformers including single wound, double wound and auto-transformers and their applications

This module presents power calculations for single-phase and three-phase AC circuits and includes the power triangle concept. This module describes the operating characteristics of AC generators and includes terminology, methods of voltage production, and methods of paralleling AC generation sources. This module describes the basic operation and application of voltage regulators. This module explains the theory of operation of AC motors and discusses the various types of AC motors and their application

Professional instruments. instruments. (10 Hrs.) and voltage drop effect of Knowledge 91. Determine errors in ammeter in circuits. 05Hrs. resistance measurement by Extension of range and voltage drop method. (8 calibration of measuring Hrs.) instruments. (05 hrs.) Test single phase energy meter for its errors. (7 Hrs.) Professional Plan and carry out 93. Dismantle and assemble Working principles and Skill 75 Hrs

TOPIC KT0306: PRINCIPLES OF SINGLE PHASE AUTO-TRANSFORMER OPERATION

Name the instruments used for testing batteries, describe how they are used and explain the purpose of each test Explain, with the aid of drawings, the discharging and charging action of lead-acid cells Topic elements to be covered include: Theories of single phase wound transformers Types of single phase transformers including single wound, double wound and auto-transformers and their applications

Connect and test a single-phase auto- transformer. Determine the losses (iron loss and copper loss) and the regulation of a single-phase transformer at different loads. Measure the current and voltage using CT and PT. Carry out winding for small transformer of 1KVA rating. Test the transformer oil with oil testing kit. Connect 3 single phase transformers for 3 phase operation of delta delta /delta-star /star-star /star-delta

This module presents power calculations for single-phase and three-phase AC circuits and includes the power triangle concept. This module describes the operating characteristics of AC generators and includes terminology, methods of voltage production, and methods of paralleling AC generation sources. This module describes the basic operation and application of voltage regulators. This module explains the theory of operation of AC motors and discusses the various types of AC motors and their application

TOPIC KT0307: TRANSFORMER LOSSES

Name the instruments used for testing batteries, describe how they are used and explain the purpose of each test Explain, with the aid of drawings, the discharging and charging action of lead-acid cells Topic elements to be covered include: Theories of single phase wound transformers Types of single phase transformers including single wound, double wound and auto-transformers and their applications

Explain, with the aid of drawings, the principle of operation of single phase autotransformers Describe types of transformer losses and explain their causes and effects Calculate the terminal voltage, turns and current ratios for an single phase transformer

Transformer losses - OC and SC test - efficiency - Voltage Regulation Objectives: At the end of this lesson you shall be able to state the type of losses occurred in transformer explain Iron (No - load) losses and copper (load) losses in transformer. Losses These losses are minimised by using steel of high silicon content (from 1.0 to 4.0 percent) for the core and by using There are two type of losses occurred in the transformer very thin laminations

TOPIC KT0308: FORMULAS AND CALCULATIONS ON INPUT AND OUTPUT OF TRANSFORMERS

Name the instruments used for testing batteries, describe how they are used and explain the purpose of each test Explain, with the aid of drawings, the discharging and charging action of lead-acid cells Topic elements to be

covered include: Theories of single phase wound transformers Types of single phase transformers including single wound, double wound and auto-transformers and their applications

671101000-KM-07: Electrical supply systems and components, L4, Cr 31 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory of electrical supply systems and components required to practise as an Electrician

the primary winding of the transformer and the ammeter connected to the Transformers are very efficient (more than 80%) because they do secondary winding. The ammeter is calibrated to take account of the turns not have any moving parts. ratio of the transformer, so that the ammeter displays the actual current being taken by the load when the ammeter is actually only taking a small proportion of the load current. 2:1 turns ratio Primary coil Secondary coil V □ 230 V supply V □ 115 V output A step down transformer

TOPIC KT0310: TYPES OF THREE PHASE TRANSFORMERS AND THEIR APPLICATIONS

Name the instruments used for testing batteries, describe how they are used and explain the purpose of each test Explain, with the aid of drawings, the discharging and charging action of lead-acid cells Topic elements to be covered include: Theories of single phase wound transformers Types of single phase transformers including single wound, double wound and auto-transformers and their applications

671101000-KM-07: Electrical supply systems and components, L4, Cr 31 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory of electrical supply systems and components required to practise as an Electrician

Instrument transformers Types and functions of panel mounted electrical measuring and testing Safe use of panel mounted measuring and test instruments Protection of measuring instruments when connected in a circuit. Internal Assessment Criteria and Weight Identify different types of electrical measuring instruments including fixed and portable and state the purpose. Identify the basic components of various measuring and test instruments and describe the basic principle of operation

TOPIC KT0311: CONSTRUCTION OF THREE PHASE TRANSFORMERS INCLUDING OPEN CORE, CLOSED CORE AND SHELL OR DIVIDED CORE

Name the instruments used for testing batteries, describe how they are used and explain the purpose of each test Explain, with the aid of drawings, the discharging and charging action of lead-acid cells Topic elements to be covered include: Theories of single phase wound transformers Types of single phase transformers including single wound, double wound and auto-transformers and their applications

Explain, with the aid of drawings, the principle of operation of single phase autotransformers Describe types of transformer losses and explain their causes and effects Calculate the terminal voltage, turns and current ratios for an single phase transformer

the leakage reactance to the minimum. In order to simplify the insulation problem, generally a low voltage winding Service conditions including whether voltage (secondary) is put next to the core

TOPIC KT0312: PRINCIPLES OF 3 PHASE TRANSFORMER OPERATION AND CONFIGURATION

Name the instruments used for testing batteries, describe how they are used and explain the purpose of each test Explain, with the aid of drawings, the discharging and charging action of lead-acid cells Topic elements to be covered include: Theories of single phase wound transformers Types of single phase transformers including single wound, double wound and auto-transformers and their applications

This module presents power calculations for single-phase and three-phase AC circuits and includes the power triangle concept. This module describes the operating characteristics of AC generators and includes terminology, methods of voltage production, and methods of paralleling AC generation sources. This module describes the basic operation and application of voltage regulators. This module explains the theory of operation of AC motors and discusses the various types of AC motors and their application

Connect and test a single-phase auto- transformer. Determine the losses (iron loss and copper loss) and the regulation of a single-phase transformer at different loads. Measure the current and voltage using CT and PT. Carry out winding for small transformer of 1KVA rating. Test the transformer oil with oil testing kit. Connect 3 single phase transformers for 3 phase operation of delta delta /delta-star /star-star /star-delta

TOPIC KT0313: PRINCIPLES OF 3 PHASE AUTO TRANSFORMER OPERATION AND CONFIGURATION

Name the instruments used for testing batteries, describe how they are used and explain the purpose of each test Explain, with the aid of drawings, the discharging and charging action of lead-acid cells Topic elements to be covered include: Theories of single phase wound transformers Types of single phase transformers including single wound, double wound and auto-transformers and their applications

Connect and test a single-phase auto- transformer. Determine the losses (iron loss and copper loss) and the regulation of a single-phase transformer at different loads. Measure the current and voltage using CT and PT. Carry out winding for small transformer of 1KVA rating. Test the transformer oil with oil testing kit. Connect 3 single phase transformers for 3 phase operation of delta delta /delta-star /star-star /star-delta

This module presents power calculations for single-phase and three-phase AC circuits and includes the power triangle concept. This module describes the operating characteristics of AC generators and includes terminology, methods of voltage production, and methods of paralleling AC generation sources. This module describes the basic operation and application of voltage regulators. This module explains the theory of operation of AC motors and discusses the various types of AC motors and their application

TOPIC KT0314: TRANSFORMER PROTECTION AND PHASING

Name the instruments used for testing batteries, describe how they are used and explain the purpose of each test Explain, with the aid of drawings, the discharging and charging action of lead-acid cells Topic elements to be covered include: Theories of single phase wound transformers Types of single phase transformers including single wound, double wound and auto-transformers and their applications

Calculate the line and phase voltages and currents relating to three-phase transformer Explain the concept of, and need for, transformer tapings Explain transformer protection and phasing KM-07-: Types of cables and applications (20%) Topic elements to be covered include: Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables

Context: Railway environments Transformers; distribution enclosures; distribution systems, earthing systems, metering units, , single phase and three phase AC motors and general lighting used in domestic, industrial and emergency areas , battery charging systems, metering systems and associated medium voltage and HV switchgear under dead conditions ,to street lighting and stadium lighting, trenched cable systems, direct current Context: Energy environments Transformers; distribution enclosures; distribution systems

TOPIC KT0704: TRANSFORMER COOLING SYSTEMS

Name the instruments used for testing batteries, describe how they are used and explain the purpose of each test Explain, with the aid of drawings, the discharging and charging action of lead-acid cells Topic elements to be covered include: Theories of single phase wound transformers Types of single phase transformers including single wound, double wound and auto-transformers and their applications

bell transformers. Because it has no moving parts, a transformer can have a very high efficiency. Large power transformers, used on electrical distribution systems, can have an efficiency of better than 90%. Large power transformers need cooling to take the heat generated away Typical oil filled power transformer. from the core. This is often achieved by totally immersing the core and windings in insulating oil. A sketch of an oil immersed transformer can be seen in Fig. 5.18

Context: Railway environments Transformers; distribution enclosures; distribution systems, earthing systems, metering units, , single phase and three phase AC motors and general lighting used in domestic, industrial and

emergency areas , battery charging systems, metering systems and associated medium voltage and HV switchgear under dead conditions ,to street lighting and stadium lighting, trenched cable systems, direct current Context: Energy environments Transformers; distribution enclosures; distribution systems

TOPIC KT0709: MAINTENANCE REQUIREMENTS OF TRANSFORMERS

Name the instruments used for testing batteries, describe how they are used and explain the purpose of each test Explain, with the aid of drawings, the discharging and charging action of lead-acid cells Topic elements to be covered include: Theories of single phase wound transformers Types of single phase transformers including single wound, double wound and auto-transformers and their applications

Explain, with the aid of drawings, the principle of operation of single phase autotransformers Describe types of transformer losses and explain their causes and effects Calculate the terminal voltage, turns and current ratios for an single phase transformer

Glossary of terms Spring toggle bolts A spring toggle bolt provides one method of fixing to hollow partition walls which are usually faced with plasterboard and a plaster skimming. Static electricity Static electricity is a voltage charge which builds up to many thousands of volts between two surfaces when they rub together

IN SUMMARY

In this unit we have focussed on:

KT0301 Theories of single phase wound transformers

KT0302 Types of single phase transformers including single wound, double wound and auto-transformers and their applications.

KT0303 Fundamentals of transformer construction

KT0305 Principles of single phase transformer operation

KT0306 Principles of single phase auto-transformer operation

KT0307 Transformer losses

KT0308 Formulas and calculations on input and output of transformers

KT0310 Types of three phase transformers and their applications

KT0311 Construction of three phase transformers including open core, closed core and shell or divided core

KT0312 Principles of 3 phase transformer operation and configuration

KT0313 Principles of 3 phase auto transformer operation and configuration

KT0314 Transformer protection and phasing

KT0704 Transformer cooling systems

KT0709 Maintenance requirements of transformers

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 17

KM-07-KT04: TYPES OF CABLES AND APPLICATIONS

In this Chapter we will discuss the following Topics:

KT0401 Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables.

KT0402 Cable materials and their functions and characteristics

KT0403 Identification of cable characteristics and properties

KT0404 Applications of various types of cable

KT0405 Installation methods and safe use of cables

KT0406 Safe transport and storage of cables

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Describe, with the aid of sketches, the different types of cables and their construction, and explain the function of the various materials used | Describe the characteristics and applications of the various types of cables and compare their advantages and disadvantages. | Describe the factors effecting the efficiency of cables | Describe and explain the provisions in the SANS 10142-1 Code of Practice for the correct and safe installation and use of the various types of cables. | Describe the precautions required when transporting and storing cables (Weight 20%)

TOPIC KT0401: TYPES OF CABLE CONSTRUCTION INCLUDING ARMOURED AND UN-ARMOURED, INSULATED AND UN-INSULATED, SINGLE AND MULTI CORE CABLES.

Calculate the line and phase voltages and currents relating to three-phase transformer Explain the concept of, and need for, transformer tapings Explain transformer protection and phasing KM-07-: Types of cables and applications (20%) Topic elements to be covered include: Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables

671101000-KM-07: Electrical supply systems and components, L4, Cr 31 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory of electrical supply systems and components required to practise as an Electrician

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

TOPIC KT0402: CABLE MATERIALS AND THEIR FUNCTIONS AND CHARACTERISTICS

Calculate the line and phase voltages and currents relating to three-phase transformer Explain the concept of, and need for, transformer tapings Explain transformer protection and phasing KM-07-: Types of cables and applications (20%) Topic elements to be covered include: Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables

671101000-KM-07: Electrical supply systems and components, L4, Cr 31 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory of electrical supply systems and components required to practise as an Electrician

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

TOPIC KT0403: IDENTIFICATION OF CABLE CHARACTERISTICS AND PROPERTIES

Calculate the line and phase voltages and currents relating to three-phase transformer Explain the concept of, and need for, transformer tapings Explain transformer protection and phasing KM-07-: Types of cables and applications (20%) Topic elements to be covered include: Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables

671101000-KM-07: Electrical supply systems and components, L4, Cr 31 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory of electrical supply systems and components required to practise as an Electrician

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

TOPIC KT0404: APPLICATIONS OF VARIOUS TYPES OF CABLE

Calculate the line and phase voltages and currents relating to three-phase transformer Explain the concept of, and need for, transformer tapings Explain transformer protection and phasing KM-07-: Types of cables and applications (20%) Topic elements to be covered include: Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

671101000-KM-07: Electrical supply systems and components, L4, Cr 31 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory of electrical supply systems and components required to practise as an Electrician

TOPIC KT0405: INSTALLATION METHODS AND SAFE USE OF CABLES

Calculate the line and phase voltages and currents relating to three-phase transformer Explain the concept of, and need for, transformer tapings Explain transformer protection and phasing KM-07-: Types of cables and applications (20%) Topic elements to be covered include: Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

671101000-KM-07: Electrical supply systems and components, L4, Cr 31 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory of electrical supply systems and components required to practise as an Electrician

TOPIC KT0406: SAFE TRANSPORT AND STORAGE OF CABLES

Calculate the line and phase voltages and currents relating to three-phase transformer Explain the concept of, and need for, transformer tapings Explain transformer protection and phasing KM-07-: Types of cables and applications (20%) Topic elements to be covered include: Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

671101000-KM-07: Electrical supply systems and components, L4, Cr 31 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory of electrical supply systems and components required to practise as an Electrician

IN SUMMARY

In this unit we have focussed on:

KT0401 Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables.

KT0402 Cable materials and their functions and characteristics

KT0403 Identification of cable characteristics and properties

KT0404 Applications of various types of cable

KT0405 Installation methods and safe use of cables

KT0406 Safe transport and storage of cables

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 18

KM-07-KT05: SWITCHGEAR AND CONTROL GEAR

In this Chapter we will discuss the following Topics:

KT0501 Principles of operation of switchgear and control gear

KT0502 Components of switchgear and control gear systems and the application thereof.

KT0503 Electrical drawings

KT0504 Introduction to Programmable Logic Controllers (PLC's)

KT0505 Introduction to Soft starters

KT0506 Introduction to Variable speed drives

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Describe disconnectors, relays, timers and contactors in terms of construction and operating principles with reference to the contacts, operating coils (where applicable) and operating mechanisms

Describe, with the aid of labelled drawings, the principle of operation of the over current and earth leakage protection relays

Describe the following terminology in terms of low voltage circuit breakers: moulded cases, positive indication, trip position, factory sealed, thermal magnetic tripping, quick make, quick break, trip free mechanism, interpole barriers.

Describe the basic principles of operation for programmable logic controllers

Describe the basic principles of operation for soft starters

Describe the basic principles of operation for variable speed drives (Weight 20%)

TOPIC KT0501: PRINCIPLES OF OPERATION OF SWITCHGEAR AND CONTROL GEAR

Calculate the line and phase voltages and currents relating to three-phase transformer Explain the concept of, and need for, transformer tapings Explain transformer protection and phasing KM-07-: Types of cables and

applications (20%) Topic elements to be covered include: Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

671101000-KM-07: Electrical supply systems and components, L4, Cr 31 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory of electrical supply systems and components required to practise as an Electrician

TOPIC KT0502: COMPONENTS OF SWITCHGEAR AND CONTROL GEAR SYSTEMS AND THE APPLICATION THEREOF.

Calculate the line and phase voltages and currents relating to three-phase transformer Explain the concept of, and need for, transformer tapings Explain transformer protection and phasing KM-07-: Types of cables and applications (20%) Topic elements to be covered include: Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables

671101000-KM-07: Electrical supply systems and components, L4, Cr 31 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory of electrical supply systems and components required to practise as an Electrician

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

TOPIC KT0503: ELECTRICAL DRAWINGS

Electrical drawings Introduction to Programmable Logic Controllers (PLC’s) Introduction to Soft starters Introduction to Variable speed drives Internal Assessment Criteria and Weight Describe disconnectors, relays, timers and contactors in terms of construction and operating principles with reference to the contacts, operating coils (where applicable) and Describe, with the aid of labelled drawings, the principle of operation of the over current and earth leakage protection relays.

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

TOPIC KT0504: INTRODUCTION TO PROGRAMMABLE LOGIC CONTROLLERS (PLC'S)

Electrical drawings Introduction to Programmable Logic Controllers (PLC's) Introduction to Soft starters Introduction to Variable speed drives Internal Assessment Criteria and Weight Describe disconnectors, relays, timers and contactors in terms of construction and operating principles with reference to the contacts, operating coils (where applicable) and Describe, with the aid of labelled drawings, the principle of operation of the over current and earth leakage protection relays.

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

Calculate the line and phase voltages and currents relating to three-phase transformer Explain the concept of, and need for, transformer tapings Explain transformer protection and phasing KM-07-: Types of cables and applications (20%) Topic elements to be covered include: Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables

TOPIC KT0505: INTRODUCTION TO SOFT STARTERS

Electrical drawings Introduction to Programmable Logic Controllers (PLC's) Introduction to Soft starters Introduction to Variable speed drives Internal Assessment Criteria and Weight Describe disconnectors, relays, timers and contactors in terms of construction and operating principles with reference to the contacts, operating coils (where applicable) and Describe, with the aid of labelled drawings, the principle of operation of the over current and earth leakage protection relays.

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

Calculate the line and phase voltages and currents relating to three-phase transformer Explain the concept of, and need for, transformer tapings Explain transformer protection and phasing KM-07-: Types of cables and applications (20%) Topic elements to be covered include: Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables

TOPIC KT0506: INTRODUCTION TO VARIABLE SPEED DRIVES

Electrical drawings Introduction to Programmable Logic Controllers (PLC's) Introduction to Soft starters Introduction to Variable speed drives Internal Assessment Criteria and Weight Describe disconnectors, relays, timers and contactors in terms of construction and operating principles with reference to the contacts, operating coils (where applicable) and Describe, with the aid of labelled drawings, the principle of operation of the over current and earth leakage protection relays.

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

Calculate the line and phase voltages and currents relating to three-phase transformer Explain the concept of, and need for, transformer tapings Explain transformer protection and phasing KM-07-: Types of cables and applications (20%) Topic elements to be covered include: Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables

IN SUMMARY

In this unit we have focussed on:

KT0501 Principles of operation of switchgear and control gear

KT0502 Components of switchgear and control gear systems and the application thereof.

KT0503 Electrical drawings

KT0504 Introduction to Programmable Logic Controllers (PLC's)

KT0505 Introduction to Soft starters

Complete the workbook activities linked to this chapter, which are activities 1 – 6.

CHAPTER 19

KM-07-KT06: LIGHTING SYSTEMS

In this Chapter we will discuss the following Topics:

KT0601 Principles of Illumination

KT0602 Types of luminaries and lighting systems

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

List the different types of alternating current motors and describe their construction | Explain the principle of operation of alternating current motors | Characteristics of single phase and three phase motors | Describe and explain how tests are conducted on alternating current motors | Describe with the aid of diagrams, the configuration of motor connections | Describe and explain protection devices used for alternating current motors | Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits. | Compare the advantages and disadvantages of single- and three-phase motors | Calculate alternating current motor properties (Weight 10%)

TOPIC KT0601: PRINCIPLES OF ILLUMINATION

Types of luminaries and lighting systems

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified

TOPIC KT0602: TYPES OF LUMINARIES AND LIGHTING SYSTEMS

Electrical drawings Introduction to Programmable Logic Controllers (PLC’s) Introduction to Soft starters Introduction to Variable speed drives Internal Assessment Criteria and Weight Describe disconnectors, relays, timers and contactors in terms of construction and operating principles with reference to the contacts, operating coils (where applicable) and Describe, with the aid of labelled drawings, the principle of operation of the over current and earth leakage protection relays.

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

671101000-KM-07: Electrical supply systems and components, L4, Cr 31 Purpose of the Knowledge Modules The main focus of the learning in this knowledge module is to build an understanding of the theory of electrical supply systems and components required to practise as an Electrician

IN SUMMARY

In this unit we have focussed on:

KT0601 Principles of Illumination

KT0602 Types of luminaries and lighting systems

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 20

KM-08-KT01: LOW VOLTAGE PROTECTION

In this Chapter we will discuss the following Topics:

KT0101 Purpose and application of low voltage protection

KT0102 Types of low voltage protection

KT0103 Low voltage protection parameters and statutory requirements

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Name and describe the types of low voltage protective devices | Describe the operation and functions of different types of low voltage protective devices including overload relays, fuses, circuit breakers and earth leakage protection devices. | Explain, with the aid of circuit diagrams, how single-and three phase electrical installation. are protected. | Describe the effect of adverse conditions on the operational characteristics of protective devices (Weight 100%)

TOPIC KT0101: PURPOSE AND APPLICATION OF LOW VOLTAGE PROTECTION

Purpose and application of low voltage protection

explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits describe the operation and functions of different types of low voltage protective devices including overload relays, fuses, circuit breakers and earth leakage protection devices there are many different types and sizes of cables in the industry, here we'll only cover a few that we may recognize I have inserted a Cable to Current table that gives guidelines to the amount of Amps a cable can withstand as well as the voltage drop of the cable on Page 25

Types of low voltage protection

explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits describe the operation and functions of different types of low voltage protective devices including overload relays, fuses, circuit breakers and earth leakage protection devices practice on routine, care/ Types of battery charging, maintenance and testing of Safety precautions, test batteries determine the number of Basic principles of Electro solar cells in series / parallel plating and cathodic for given power protection requirement hrs.)

Grouping of cells for specified voltage and Principle and operation of solar cell practice on routine, care/ Types of battery charging, maintenance and testing of Safety precautions, test batteries

Low voltage protection parameters and statutory requirements

explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits describe the operation and functions of different types of low voltage protective devices including overload relays, fuses, circuit breakers and earth leakage protection devices rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified

TOPIC KT0102: TYPES OF LOW VOLTAGE PROTECTION

Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

Describe the operation and functions of different types of low voltage protective devices including overload relays, fuses, circuit breakers and earth leakage protection devices. Explain, with the aid of circuit diagrams, how single-and three phase electrical installation. Describe the effect of adverse conditions on the operational characteristics of protective Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a

TOPIC KT0103: LOW VOLTAGE PROTECTION PARAMETERS AND STATUTORY REQUIREMENTS

Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits

Optimal functioning equipment, control systems and installations in compliance with statutory and mandatory requirements Fault find and repair electrical control systems and electrical installation. Maintenance processes for equipment, control systems and electrical installation. List of Knowledge Modules for which Specifications are included 671101000-KM-01: Health, Safety, Quality and Legislation (NQF Level: 4) Credits: 5 671101000-KM-02: Tools, Equipment and Materials (NQF Level: 4) Credits: 8 671101000-KM-03: E

Describe the operation and functions of different types of low voltage protective devices including overload relays, fuses, circuit breakers and earth leakage protection devices. Explain, with the aid of circuit diagrams, how single-and three phase electrical installation. Describe the effect of adverse conditions on the operational characteristics of protective Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a

IN SUMMARY

In this unit we have focussed on:

KT0101 Purpose and application of low voltage protection

KT0102 Types of low voltage protection

KT0103 Low voltage protection parameters and statutory requirements

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 21

KM-09-KT01: FAULT FINDING

In this Chapter we will discuss the following Topics:

KT0101 Fault finding principles for electrical circuits

KT0102 Fault finding techniques for electrical circuits

KT0103 Safety during fault finding

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Explain fault finding principles and techniques for alternating and direct current main supply circuits | Explain fault finding principles and techniques for alternating and direct current control circuits (Weight 100%)

TOPIC KT0101: FAULTFINDING PRINCIPLES FOR ELECTRICAL CIRCUITS

Describe the operation and functions of different types of low voltage protective devices including overload relays, fuses, circuit breakers and earth leakage protection devices. Explain, with the aid of circuit diagrams, how single-and three phase electrical installation. Describe the effect of adverse conditions on the operational characteristics of protective Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

TOPIC KT0102: FAULTFINDING TECHNIQUES FOR ELECTRICAL CIRCUITS

Describe the operation and functions of different types of low voltage protective devices including overload relays, fuses, circuit breakers and earth leakage protection devices. Explain, with the aid of circuit diagrams, how single-and three phase electrical installation. Describe the effect of adverse conditions on the operational characteristics of protective Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

§ Measurements and observations are interpreted and captured accurately according to manufacturers specifications § All the relevant units/variables are recorded accurately § Hand-over documents are completed as per hand-over procedures § Demonstrate knowledge and understanding of commissioning control systems The ability to maintain and repair electrical equipment, control systems and installations Associated Assessment Criteria § Correct work instructions are followed and correct documentation including checklist

TOPIC KT0103: SAFETY DURING FAULTFINDING

Describe the operation and functions of different types of low voltage protective devices including overload relays, fuses, circuit breakers and earth leakage protection devices. Explain, with the aid of circuit diagrams, how single-and three phase electrical installation. Describe the effect of adverse conditions on the operational characteristics of protective Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a

§ Demonstrate knowledge and understanding of maintaining and repairing electrical equipment, control systems and installations Assessment Standards for Phases No phase tests are applicable Assessment Standards for Part Qualifications No Part Qualifications are applicable No assignments must be produced for external evaluation No internal assessments to be re-assessed during external assessment Qualified and registered assessor Minimum of five (5) years' experience as qualified electrician Registered with NAMB as an

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses

AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

IN SUMMARY

In this unit we have focussed on:

KT0101 Fault finding principles for electrical circuits

KT0102 Fault finding techniques for electrical circuits

KT0103 Safety during fault finding

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 22

PM-01-PS01: SELECT, USE AND CARE OF ENGINEERING HAND TOOLS

In this Chapter we will discuss the following Topics:

AK0101 Workshop procedures including housekeeping practices according to statutory requirements.

AK0102 Techniques of maintaining hand tools

AK0103 Techniques to correctly store hand tools

PA0101 Select engineering hand tools for serviceability.

PA0102 Use engineering hand tools.

PA0103 Care for engineering hand tools.

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Unsafe and faulty tools were identified and marked for repair/maintenance or replacement according to set procedures | The correct application of tools were used to measure and mark out steel, copper, aluminium, wood, plastics and synthetic materials. | The correct application of tools were used for cutting steel, copper, aluminium, wood, plastics and synthetic materials as per manufacturers specifications. | The correct application of tools was used to loosen or fasten a range of different types and sizes of nuts or bolts and different types and sizes of screws as per manufacturer's specifications. | The correct application of tools was used to strip or cut a range of different types and sizes of electrical wire as per manufacturer’s specifications. | Sharp edges on chisel/punch type tool heads with "mushroom effect" were removed to ensure that they can be used safely; then tips with burs or chips are sharpened for effective use. | Flat Screw driver tips with burs or chips were filed and prepared to a manufacturer's specification ensuring that they are serviceable | The adjusting/moving mechanisms of shifting spanners, gas pump pliers, vice grip, dividers, crimping tools, Stilson wrench were lubricated to ensure free movement and serviceability. | Engineering files cutting edge were cleaned using a wire brush to ensure serviceability. | Scribers, dividers and centre punches were sharpened to ensure serviceability.

TOPIC AK0101: WORKSHOP PROCEDURES INCLUDING HOUSEKEEPING PRACTICES ACCORDING TO STATUTORY REQUIREMENTS.

PA0101 - Select engineering hand tools for serviceability

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools the correct application of tools were used for cutting steel, copper, aluminium, wood, plastics and synthetic materials as per manufacturers specifications power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

AK0101 - Workshop procedures including housekeeping practices according to statutory requirements

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

PA0102 - Use engineering hand tools

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools the correct application of tools were used for cutting steel, copper, aluminium, wood, plastics and synthetic materials as per manufacturers specifications power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

AK0102 - Techniques of maintaining hand tools

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools the correct application of tools were used for cutting steel, copper, aluminium, wood, plastics and synthetic materials as per manufacturers specifications

PA0103 - Care for engineering hand tools

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools the correct application of tools were used for cutting steel, copper, aluminium, wood, plastics and synthetic materials as per manufacturers specifications power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

AK0103 - Techniques to correctly store hand tools

the correct application of tools were used for cutting steel, copper, aluminium, wood, plastics and synthetic materials as per manufacturers specifications 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC AK0102: TECHNIQUES OF MAINTAINING HAND TOOLS

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

TOPIC AK0103: TECHNIQUES TO CORRECTLY STORE HAND TOOLS

AK0103 Techniques to correctly store hand tools Internal Assessment Criteria Unsafe and faulty tools were identified and marked for repair/maintenance or replacement according to set procedures The correct application of tools were used to measure and mark out steel, copper, aluminium, wood, plastics and synthetic materials. The correct application of tools were used for cutting steel, copper, aluminium, wood, plastics and synthetic materials as per manufacturers specifications

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

TOPIC PA0101: SELECT ENGINEERING HAND TOOLS FOR SERVICEABILITY.

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC PA0102: USE ENGINEERING HAND TOOLS.

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

TOPIC PA0103: CARE FOR ENGINEERING HAND TOOLS.

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and

PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

IN SUMMARY

In this unit we have focussed on:

AK0101 Workshop procedures including housekeeping practices according to statutory requirements.

AK0102 Techniques of maintaining hand tools

AK0103 Techniques to correctly store hand tools

PA0101 Select engineering hand tools for serviceability.

PA0102 Use engineering hand tools.

PA0103 Care for engineering hand tools.

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 23

PM-01-PS02: SELECT, USE AND CARE FOR POWER TOOLS

In this Chapter we will discuss the following Topics:

AK0201 Workshop procedures including housekeeping practices according to statutory requirements.

AK0202 Safety practices relating to the use of power tools including the use of personal protective equipment, electrical and fire protection

AK0203 Types and uses of fixed and portable power tools

PA0201 Select a power tool pertaining to specific job requirements.

PA0202 Use fixed power tools.

PA0203 Use portable power tools.

PA0204 Care for and store power tools and their accessories.

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Power tools were selected according to the task requirements. | Pre-operational check was carried out in terms of safety requirements and function. | Hazards associated with the use of power tools were recognised and necessary precautions taken according to workshop procedures. | Faulty and unsafe tools were identified and reported in accordance with standard operating procedures | Personal safety equipment was selected and used according to tool requirement. | Fixed and portable power tools were selected and set up according to job requirements. | All guards and securing mechanisms were effectively utilised in terms of job requirements. | Fixed and portable power tools were used safely and in accordance with manufacturer's specifications. | Power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications. | Loose items were secured and minor defects repaired in accordance to workshop procedures. | Portable power tools and accessories were stored in accordance to workshop procedures.

TOPIC AK0201: WORKSHOP PROCEDURES INCLUDING HOUSEKEEPING PRACTICES ACCORDING TO STATUTORY REQUIREMENTS.

PA0201 - Select a power tool pertaining to specific job requirements

scribers, dividers and centre punches were sharpened to ensure serviceability 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning

in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures scribers, dividers and centre punches were sharpened to ensure serviceability power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

AK0201 - Workshop procedures including housekeeping practices according to statutory requirements

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

PA0202 - Use fixed power tools

scribers, dividers and centre punches were sharpened to ensure serviceability 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures scribers, dividers and centre punches were sharpened to ensure serviceability power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

AK0202 - Safety practices relating to the use of power tools including the use of personal protective equipment, electrical and fire protection

scribers, dividers and centre punches were sharpened to ensure serviceability 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning

in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures scribers, dividers and centre punches were sharpened to ensure serviceability power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

PA0203 - Use portable power tools

scribers, dividers and centre punches were sharpened to ensure serviceability 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures scribers, dividers and centre punches were sharpened to ensure serviceability power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

AK0203 - Types and uses of fixed and portable power tools

scribers, dividers and centre punches were sharpened to ensure serviceability 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures scribers, dividers and centre punches were sharpened to ensure serviceability power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

PA0204 - Care for and store power tools and their accessories

scribers, dividers and centre punches were sharpened to ensure serviceability 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures scribers, dividers and centre punches were sharpened to ensure serviceability power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

TOPIC AK0202: SAFETY PRACTICES RELATING TO THE USE OF POWER TOOLS INCLUDING THE USE OF PERSONAL PROTECTIVE EQUIPMENT, ELECTRICAL AND FIRE PROTECTION

Scribers, dividers and centre punches were sharpened to ensure serviceability. PM-01-PS02: Select, use and care for power tools Scope of Practical Skill Given work instructions, work area, documents, templates, drilling machines, grinders, and saws standard operating procedures and, statutory requirements, you should be PA0201 Select a power tool pertaining to specific job requirements. PA0202 Use fixed power tools. PA0203 Use portable power tools

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

TOPIC AK0203: TYPES AND USES OF FIXED AND PORTABLE POWER TOOLS

Scribers, dividers and centre punches were sharpened to ensure serviceability. PM-01-PS02: Select, use and care for power tools Scope of Practical Skill Given work instructions, work area, documents, templates, drilling machines, grinders, and saws standard operating procedures and, statutory requirements, you should be PA0201 Select a power tool pertaining to specific job requirements. PA0202 Use fixed power tools. PA0203 Use portable power tools

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC PA0201: SELECT A POWER TOOL PERTAINING TO SPECIFIC JOB REQUIREMENTS.

Scribers, dividers and centre punches were sharpened to ensure serviceability. PM-01-PS02: Select, use and care for power tools Scope of Practical Skill Given work instructions, work area, documents, templates, drilling machines, grinders, and saws standard operating procedures and, statutory requirements, you should be PA0201 Select a power tool pertaining to specific job requirements. PA0202 Use fixed power tools. PA0203 Use portable power tools

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

TOPIC PA0202: USE FIXED POWER TOOLS.

Scribers, dividers and centre punches were sharpened to ensure serviceability. PM-01-PS02: Select, use and care for power tools Scope of Practical Skill Given work instructions, work area, documents, templates, drilling machines, grinders, and saws standard operating procedures and, statutory requirements, you should be PA0201 Select a power tool pertaining to specific job requirements. PA0202 Use fixed power tools. PA0203 Use portable power tools

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC PA0203: USE PORTABLE POWER TOOLS.

Scribers, dividers and centre punches were sharpened to ensure serviceability. PM-01-PS02: Select, use and care for power tools Scope of Practical Skill Given work instructions, work area, documents, templates, drilling machines, grinders, and saws standard operating procedures and, statutory requirements, you should be PA0201 Select a power tool pertaining to specific job requirements. PA0202 Use fixed power tools. PA0203 Use portable power tools

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC PA0204: CARE FOR AND STORE POWER TOOLS AND THEIR ACCESSORIES.

Scribers, dividers and centre punches were sharpened to ensure serviceability. PM-01-PS02: Select, use and care for power tools Scope of Practical Skill Given work instructions, work area, documents, templates, drilling machines, grinders, and saws standard operating procedures and, statutory requirements, you should be PA0201 Select a power tool pertaining to specific job requirements. PA0202 Use fixed power tools. PA0203 Use portable power tools

The learning will enable learners to demonstrating an understanding of: KM-07-: Concepts, theories and principles of supply Systems (20%) KM-07-: Types of cables and applications (20%) KM-07-: Switchgear and control gear (20%) KM-07-: Lighting systems (10%) 671101000-KM-08: Low Voltage protection, L4, Cr5 The main focus of the learning in this knowledge module is to build an understanding of the theory of low voltage protection required to practise as an Electrician

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

IN SUMMARY

In this unit we have focussed on:

AK0201 Workshop procedures including housekeeping practices according to statutory requirements.

AK0202 Safety practices relating to the use of power tools including the use of personal protective equipment, electrical and fire protection

AK0203 Types and uses of fixed and portable power tools

PA0201 Select a power tool pertaining to specific job requirements.

PA0202 Use fixed power tools.

PA0203 Use portable power tools.

PA0204 Care for and store power tools and their accessories.

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 24

PM-01-PS03: SELECT, USE AND CARE FOR ELECTRICAL MEASURING INSTRUMENTS

In this Chapter we will discuss the following Topics:

AK0301 Handling procedures related to portable measuring instrument.

AK0302 Basic operating principles of portable electrical measuring instruments.

AK0303 Types, applications and functions of portable electrical measuring instruments.

PA0301 Identify and select portable electrical measuring instruments.

PA0302 Use and interpret portable electrical measuring instrument readings.

PA0303 Care for portable electrical measuring instruments.

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Job instructions were interpreted correctly and sequence of operation is determined. | Portable electrical measuring instruments were correctly selected according to workshop procedures and to meet safety and job requirements | Unsafe and faulty measuring instruments were identified visually and marked for repair or replacement. | Portable measuring instruments were read correctly and voltage, amperage, and resistance readings were recorded as per workshop procedures. | Portable electrical equipment was checked for correct operation and functionality. | Measuring instruments were correctly set up for application. | Results were recorded on appropriate documentation. | Results were accurately interpreted against the specifications of the job requirements. | Electrical measuring instruments were used in accordance with their specifications. | Electrical measuring instruments were placed and stored in accordance with manufactures specifications and workshop standards.

TOPIC AK0301: HANDLING PROCEDURES RELATED TO PORTABLE MEASURING INSTRUMENT.

PA0301 - Identify and select portable electrical measuring instruments

power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures 671101000-PM-01,

671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

AK0301 - Handling procedures related to portable measuring instrument

power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

PA0302 - Use and interpret portable electrical measuring instrument readings

power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools readings were recorded as per workshop procedures portable electrical equipment was checked for correct operation and functionality measuring instruments were correctly set up for application results were accurately interpreted against the specifications of the job requirements power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

READINGS WERE RECORDED AS PER WORKSHOP PROCEDURES

portable electrical equipment was checked for correct operation and functionality

AK0302 - Basic operating principles of portable electrical measuring instruments

power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop

Portable power tools and accessories were stored in accordance to workshop procedures 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

PA0303 - Care for portable electrical measuring instruments

power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

AK0303 - Types, applications and functions of portable electrical measuring instruments

power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools readings were recorded as per workshop procedures portable electrical equipment was checked for correct operation and functionality measuring instruments were correctly set up for application results were accurately interpreted against the specifications of the job requirements power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

READINGS WERE RECORDED AS PER WORKSHOP PROCEDURES

portable electrical equipment was checked for correct operation and functionality

TOPIC AK0302: BASIC OPERATING PRINCIPLES OF PORTABLE ELECTRICAL MEASURING INSTRUMENTS.

Power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications. Loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

TOPIC AK0303: TYPES, APPLICATIONS AND FUNCTIONS OF PORTABLE ELECTRICAL MEASURING INSTRUMENTS.

Power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications. Loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC PA0301: IDENTIFY AND SELECT PORTABLE ELECTRICAL MEASURING INSTRUMENTS.

Power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications. Loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC PA0302: USE AND INTERPRET PORTABLE ELECTRICAL MEASURING INSTRUMENT READINGS.

Power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications. Loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC PA0303: CARE FOR PORTABLE ELECTRICAL MEASURING INSTRUMENTS.

Power tools were inspected, cleaned and lubricated after use according to workshop practices and/or manufactures specifications. Loose items were secured and minor defects repaired in accordance to workshop Portable power tools and accessories were stored in accordance to workshop procedures

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

IN SUMMARY

In this unit we have focussed on:

AK0301 Handling procedures related to portable measuring instrument.

AK0302 Basic operating principles of portable electrical measuring instruments.

AK0303 Types, applications and functions of portable electrical measuring instruments.

PA0301 Identify and select portable electrical measuring instruments.

PA0302 Use and interpret portable electrical measuring instrument readings.

PA0303 Care for portable electrical measuring instruments.

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 25

PM-01-PS04: PERFORM SOLDERING ACTIVITIES

In this Chapter we will discuss the following Topics:

AK0401 Soldering techniques and applications

AK0402 Safety practices relating to the use of soldering equipment including the use of personal protective equipment.

AK0403 De-soldering methods

PA0401 Solider various wire joints

PA0402 Solder components on a PC or Vero board

PA0403 De-solder components

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Work area was inspected for safe working conditions and corrective action is taken where required. | Applicable soldering equipment was selected as required by task. | Soldering material was correctly selected as required by the task. | All connections were cleaned from any dirt or oxidation. | Tinning of wire and connections was done according to manufacturer's specifications. | Connections were soldered according to set specifications and techniques. | Components were laid out on the circuit board according to the circuit diagram. | Components were soldered according to soldering techniques.

TOPIC AK0401: SOLDERING TECHNIQUES AND APPLICATIONS

PA0401 - Solider various wire joints

readings were recorded as per workshop procedures portable electrical equipment was checked for correct operation and functionality measuring instruments were correctly set up for application 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables

READINGS WERE RECORDED AS PER WORKSHOP PROCEDURES

portable electrical equipment was checked for correct operation and functionality

AK0401 - Soldering techniques and applications

readings were recorded as per workshop procedures portable electrical equipment was checked for correct operation and functionality measuring instruments were correctly set up for application 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables

---

READINGS WERE RECORDED AS PER WORKSHOP PROCEDURES

portable electrical equipment was checked for correct operation and functionality

PA0402 - Solder components on a PC or Vero board

readings were recorded as per workshop procedures portable electrical equipment was checked for correct operation and functionality measuring instruments were correctly set up for application 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

---

READINGS WERE RECORDED AS PER WORKSHOP PROCEDURES

portable electrical equipment was checked for correct operation and functionality

AK0402 - Safety practices relating to the use of soldering equipment including the use of personal protective equipment

readings were recorded as per workshop procedures portable electrical equipment was checked for correct operation and functionality 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables

---

READINGS WERE RECORDED AS PER WORKSHOP PROCEDURES

portable electrical equipment was checked for correct operation and functionality

PA0403 - De-solder components

readings were recorded as per workshop procedures portable electrical equipment was checked for correct operation and functionality measuring instruments were correctly set up for application 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

READINGS WERE RECORDED AS PER WORKSHOP PROCEDURES

portable electrical equipment was checked for correct operation and functionality

AK0403 - De-soldering methods

readings were recorded as per workshop procedures portable electrical equipment was checked for correct operation and functionality measuring instruments were correctly set up for application 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

READINGS WERE RECORDED AS PER WORKSHOP PROCEDURES

portable electrical equipment was checked for correct operation and functionality

TOPIC AK0402: SAFETY PRACTICES RELATING TO THE USE OF SOLDERING EQUIPMENT INCLUDING THE USE OF PERSONAL PROTECTIVE EQUIPMENT.

readings were recorded as per workshop procedures. Portable electrical equipment was checked for correct operation and functionality. Measuring instruments were correctly set up for application. Results were recorded on appropriate documentation. Results were accurately interpreted against the specifications of the job requirements. Electrical measuring instruments were used in accordance with their specifications

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

Internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures. Portable power tools were correctly used, as per workshop practices and safety The correct personal protective equipment was used and complied with the workshop PM-05-PS04: Install equipment and control systems

Scope of Practical Skill Given equipment and control systems Includes: Batteries, Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rot

TOPIC AK0403: DE-SOLDERING METHODS

readings were recorded as per workshop procedures. Portable electrical equipment was checked for correct operation and functionality. Measuring instruments were correctly set up for application. Results were recorded on appropriate documentation. Results were accurately interpreted against the specifications of the job requirements. Electrical measuring instruments were used in accordance with their specifications

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC PA0401: SOLIDER VARIOUS WIRE JOINTS

readings were recorded as per workshop procedures. Portable electrical equipment was checked for correct operation and functionality. Measuring instruments were correctly set up for application. Results were recorded on appropriate documentation. Results were accurately interpreted against the specifications of the job requirements. Electrical measuring instruments were used in accordance with their specifications

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

TOPIC PA0402: SOLDER COMPONENTS ON A PC OR VERO BOARD

readings were recorded as per workshop procedures. Portable electrical equipment was checked for correct operation and functionality. Measuring instruments were correctly set up for application. Results were recorded on appropriate documentation. Results were accurately interpreted against the specifications of the job requirements. Electrical measuring instruments were used in accordance with their specifications

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC PA0403: DE-SOLDER COMPONENTS

readings were recorded as per workshop procedures. Portable electrical equipment was checked for correct operation and functionality. Measuring instruments were correctly set up for application. Results were recorded on appropriate documentation. Results were accurately interpreted against the specifications of the job requirements. Electrical measuring instruments were used in accordance with their specifications

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

IN SUMMARY

In this unit we have focussed on:

AK0401 Soldering techniques and applications

AK0402 Safety practices relating to the use of soldering equipment including the use of personal protective equipment.

AK0403 De-soldering methods

PA0401 Solider various wire joints

PA0402 Solder components on a PC or Vero board

PA0403 De-solder components

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 26

PM-01-PS05: CARRY OUT BASIC ELECTRIC ARC WELDING IN AN ELECTRICAL ENVIRONMENT

In this Chapter we will discuss the following Topics:

AK0501 Names, applications and functions of electric arc welding machines and equipment.

AK0502 Handling procedures related to electric arc welding equipment.

AK0503 Application of various types of electrodes and current settings during the electric arc welding process.

AK0504 Safe work procedures and workshop practices.

AK0505 Workshop procedures including housekeeping practices according to statutory requirements.

PA0501 Prepare the electric arc welding equipment for the task.

PA0502 Prepare the metal surfaces for electric arc welding.

PA0503 Apply basic electric arc welding techniques.

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

The applicable safety equipment was obtained and inspected as per safe work procedures and statutory requirements. | The electric arc welding equipment was set up and cable connections checked for tightness to ensure effective operation as per safe work procedures. | The cleaning materials for preparing the metal surfaces to be welded were correctly applied as per task requirements. | The appropriate personal protective equipment was worn as per safe work procedures. | The edges of the metals welded were correctly bevelled where needed as per specific task requirements. | The contact surfaces of the earth clamp, metals earthed and the electrode holder were cleaned thoroughly to ensure proper current flow during the welding process. | The required current setting on the welding machine was correctly selected for electric arc welding according to application requirements. | The joining of metals by means of electric arc welding was carried out and correct electrode positioning and movement are demonstrated according to application requirements. | Hot metal surfaces were allowed to cool down and all slag removed from the welding joints as per safe work procedures. | Electric arc welding machine, equipment and materials were correctly stored as per applicable safety standards. | The work area was cleaned and waste materials disposed of, according to local housekeeping standards. | Safety equipment was cleaned and stored according to statutory requirements.

TOPIC AK0501: NAMES, APPLICATIONS AND FUNCTIONS OF ELECTRIC ARC WELDING MACHINES AND EQUIPMENT.

PA0501 - Prepare the electric arc welding equipment for the task

components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

AK0501 - Names, applications and functions of electric arc welding machines and equipment

components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

PA0502 - Prepare the metal surfaces for electric arc welding

components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

AK0502 - Handling procedures related to electric arc welding equipment

components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools

PA0503 - Apply basic electric arc welding techniques

components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

AK0503 - Application of various types of electrodes and current settings during the electric arc welding process

components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

AK0504 - Safe work procedures and workshop practices

components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables the work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements

AK0505 - Workshop procedures including housekeeping practices according to statutory requirements

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables

TOPIC AK0502: HANDLING PROCEDURES RELATED TO ELECTRIC ARC WELDING EQUIPMENT.

Components were soldered according to soldering techniques. PM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC AK0503: APPLICATION OF VARIOUS TYPES OF ELECTRODES AND CURRENT SETTINGS DURING THE ELECTRIC ARC WELDING PROCESS.

Components were soldered according to soldering techniques. PM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC AK0504: SAFE WORK PROCEDURES AND WORKSHOP PRACTICES.

Components were soldered according to soldering techniques. PM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables

applicable safety standards. The work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC AK0505: WORKSHOP PROCEDURES INCLUDING HOUSEKEEPING PRACTICES ACCORDING TO STATUTORY REQUIREMENTS.

Components were soldered according to soldering techniques. PM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

applicable safety standards. The work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements

TOPIC PA0501: PREPARE THE ELECTRIC ARC WELDING EQUIPMENT FOR THE TASK.

Components were soldered according to soldering techniques. PM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC PA0502: PREPARE THE METAL SURFACES FOR ELECTRIC ARC WELDING.

Components were soldered according to soldering techniques. PM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC PA0503: APPLY BASIC ELECTRIC ARC WELDING TECHNIQUES.

Components were soldered according to soldering techniques. PM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

IN SUMMARY

In this unit we have focussed on:

AK0501 Names, applications and functions of electric arc welding machines and equipment.

AK0502 Handling procedures related to electric arc welding equipment.

AK0503 Application of various types of electrodes and current settings during the electric arc welding process.

AK0504 Safe work procedures and workshop practices.

AK0505 Workshop procedures including housekeeping practices according to statutory requirements.

PA0501 Prepare the electric arc welding equipment for the task.

PA0502 Prepare the metal surfaces for electric arc welding.

PA0503 Apply basic electric arc welding techniques.

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 27

PM-01-PS06: CARRY OUT BASIC GAS CUTTING IN AN ELECTRICAL ENVIRONMENT

In this Chapter we will discuss the following Topics:

AK0601 Names, applications and functions of gas cutting equipment.

AK0602 Handling procedures related to gas cutting equipment.

AK0604 Safe work procedures and workshop practices.

AK0605 Workshop procedures including housekeeping practices according to statutory requirements.

PA0601 Prepare for gas cutting.

PA0602 Apply basic gas cutting process.

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

The applicable safety equipment was obtained and inspected as per safe work procedures and statutory requirements. | The gas cutting equipment was set up and checked for gas leaks and correct cylinder pressures according to safe work procedures and industry requirements. | The relevant cutting material is limited to mild steel only (maximum thickness of 6mm), required for the task was ensured to be ready and available, and nozzles were selected correctly to match the application. | The gas cylinders were handled safely as per safe work procedures and industry requirements. | Safety and proper workplace practices were maintained at all times. | The oxygen and acetylene gas pressures on the regulator were correctly set for gas cutting according to application requirements. | The gas torch was lit up in the correct sequence and the gas mixture set as per safe work procedures and industry requirements. | The metal was cut with the gas cutting torch and the correct nozzle position- and movement was demonstrated according to application requirements. | Hot metal surfaces were allowed to cool down as per safe work procedures. | Gas cylinder valves were closed and gas pressure in the hoses released as per safe work procedures. | Gas cutting equipment and materials were correctly stored as per applicable safety standards. | The work area was cleaned and waste materials disposed of according to local housekeeping standards. | Provider Accreditation Requirements for the Module

TOPIC AK0601: NAMES, APPLICATIONS AND FUNCTIONS OF GAS CUTTING EQUIPMENT.

PA0601 - Prepare for gas cutting

the work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements

AK0601 - Names, applications and functions of gas cutting equipment

the work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements

PA0602 - Apply basic gas cutting process

the work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables the work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements

AK0602 - Handling procedures related to gas cutting equipment

the work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables the work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements

AK0604 - Safe work procedures and workshop practices

the work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements

AK0605 - Workshop procedures including housekeeping practices according to statutory requirements

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables the work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements

TOPIC AK0602: HANDLING PROCEDURES RELATED TO GAS CUTTING EQUIPMENT.

applicable safety standards. The work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC AK0604: SAFE WORK PROCEDURES AND WORKSHOP PRACTICES.

applicable safety standards. The work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements

Components were soldered according to soldering techniques. PM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC AK0605: WORKSHOP PROCEDURES INCLUDING HOUSEKEEPING PRACTICES ACCORDING TO STATUTORY REQUIREMENTS.

applicable safety standards. The work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

Components were soldered according to soldering techniques. PM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables

TOPIC PA0601: PREPARE FOR GAS CUTTING.

applicable safety standards. The work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC PA0602: APPLY BASIC GAS CUTTING PROCESS.

applicable safety standards. The work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

IN SUMMARY

In this unit we have focussed on:

AK0601 Names, applications and functions of gas cutting equipment.

AK0602 Handling procedures related to gas cutting equipment.

AK0604 Safe work procedures and workshop practices.

AK0605 Workshop procedures including housekeeping practices according to statutory requirements.

PA0601 Prepare for gas cutting.

PA0602 Apply basic gas cutting process.

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 28

PM-02-PS01: UNDERTAKE A PRELIMINARY RISK ASSESSMENT

In this Chapter we will discuss the following Topics:

AK0101 Safety procedures

AK0102 Statutory requirements for hazardous conditions/material/substance.

AK0103 Risk assessment procedure for specific context

AK0104 Inspection procedure

AK0105 Recording/reporting procedure

AK0106 Approval procedures

PA0101 Conduct a preliminary risk assessment.

PA0102 Identify the hazards.

PA0103 Relate the risk to the hazard.

PA0104 Conduct mitigation of the risk.

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Risk assessments, including work related hazards and ergonomics, are carried out and reports completed as per statutory requirements. | Evidence of mitigation is observed in accordance with standard operating procedures | Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures

TOPIC AK0101: SAFETY PROCEDURES

PA0101 - Conduct a preliminary risk assessment

pA0103 Relate the risk to the hazard 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

AK0101 - Safety procedures

pA0103 Relate the risk to the hazard rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment

PA0102 - Identify the hazards

pA0103 Relate the risk to the hazard ✚ Given to the employer who will use them to work out how much to pay ✚ Used to record daily activities on site ✚ Given to the employer who will use them to work out how much to pay ✚ Used to record daily activities on site

AK0102 - Statutory requirements for hazardous conditions/material/substance

pA0103 Relate the risk to the hazard results were accurately interpreted against the specifications of the job requirements

PA0103 - Relate the risk to the hazard

pA0103 Relate the risk to the hazard 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment ✚ Given to the employer who will use them to work out how much to pay ✚ Used to record daily activities on site 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment ✚ Given to the employer who will use them to work out how much to pay ✚ Used to record daily activities on site

AK0103 - Risk assessment procedure for specific context

PA0103 RELATE THE RISK TO THE HAZARD

PA0104 - Conduct mitigation of the risk

PA0103 RELATE THE RISK TO THE HAZARD

AK0104 - Inspection procedure

PA0103 RELATE THE RISK TO THE HAZARD

AK0105 - Recording/reporting procedure

pA0103 Relate the risk to the hazard the correct electrical testing and fault finding procedures were applied defects were rectified as per installation requirements repairs were recorded and reported as per reporting procedures the correct electrical testing and fault finding procedures were applied

DEFECTS WERE RECTIFIED AS PER INSTALLATION REQUIREMENTS

repairs were recorded and reported as per reporting procedures

AK0106 - Approval procedures

PA0103 RELATE THE RISK TO THE HAZARD

TOPIC AK0102: STATUTORY REQUIREMENTS FOR HAZARDOUS CONDITIONS/MATERIAL/SUBSTANCE.

PM-02-PS01: Undertake a preliminary risk assessment Scope of Practical Skill Given work instructions, checklists, work area, case study, drawings, documents, templates, forms, safety and quality principles, standard operating procedures and, statutory requirements, you should be able to: PA0101 Conduct a preliminary risk assessment. PA0102 Identify the hazards. PA0103 Relate the risk to the hazard. PA0104 Conduct mitigation of the risk

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC AK0103: RISK ASSESSMENT PROCEDURE FOR SPECIFIC CONTEXT

PM-02-PS01: Undertake a preliminary risk assessment Scope of Practical Skill Given work instructions, checklists, work area, case study, drawings, documents, templates, forms, safety and quality principles, standard operating procedures and, statutory requirements, you should be able to: PA0101 Conduct a preliminary risk assessment. PA0102 Identify the hazards. PA0103 Relate the risk to the hazard. PA0104 Conduct mitigation of the risk

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and

PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation. Laton.

TOPIC AK0104: INSPECTION PROCEDURE

PM-02-PS01: Undertake a preliminary risk assessment Scope of Practical Skill Given work instructions, checklists, work area, case study, drawings, documents, templates, forms, safety and quality principles, standard operating procedures and, statutory requirements, you should be able to: PA0101 Conduct a preliminary risk assessment. PA0102 Identify the hazards. PA0103 Relate the risk to the hazard. PA0104 Conduct mitigation of the risk

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation

TOPIC AK0105: RECORDING/REPORTING PROCEDURE

PM-02-PS01: Undertake a preliminary risk assessment Scope of Practical Skill Given work instructions, checklists, work area, case study, drawings, documents, templates, forms, safety and quality principles, standard operating procedures and, statutory requirements, you should be able to: PA0101 Conduct a preliminary risk assessment. PA0102 Identify the hazards. PA0103 Relate the risk to the hazard. PA0104 Conduct mitigation of the risk.

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you

will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

TOPIC AK0106: APPROVAL PROCEDURES

PM-02-PS01: Undertake a preliminary risk assessment Scope of Practical Skill Given work instructions, checklists, work area, case study, drawings, documents, templates, forms, safety and quality principles, standard operating procedures and, statutory requirements, you should be able to: PA0101 Conduct a preliminary risk assessment. PA0102 Identify the hazards. PA0103 Relate the risk to the hazard. PA0104 Conduct mitigation of the risk.

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

Test and inspect electrical equipment, control systems and installations. Commission control systems and installations. Maintain and repair electrical equipment, control systems and installations.

TOPIC PA0101: CONDUCT A PRELIMINARY RISK ASSESSMENT.

PM-02-PS01: Undertake a preliminary risk assessment Scope of Practical Skill Given work instructions, checklists, work area, case study, drawings, documents, templates, forms, safety and quality principles, standard operating procedures and, statutory requirements, you should be able to: PA0101 Conduct a preliminary risk assessment. PA0102 Identify the hazards. PA0103 Relate the risk to the hazard. PA0104 Conduct mitigation of the risk.

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC PA0102: IDENTIFY THE HAZARDS.

PM-02-PS01: Undertake a preliminary risk assessment Scope of Practical Skill Given work instructions, checklists, work area, case study, drawings, documents, templates, forms, safety and quality principles, standard operating procedures and, statutory requirements, you should be able to: PA0101 Conduct a preliminary risk assessment. PA0102 Identify the hazards. PA0103 Relate the risk to the hazard. PA0104 Conduct mitigation of the risk.

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC PA0103: RELATE THE RISK TO THE HAZARD.

PM-02-PS01: Undertake a preliminary risk assessment Scope of Practical Skill Given work instructions, checklists, work area, case study, drawings, documents, templates, forms, safety and quality principles, standard operating procedures and, statutory requirements, you should be able to: PA0101 Conduct a preliminary risk assessment. PA0102 Identify the hazards. PA0103 Relate the risk to the hazard. PA0104 Conduct mitigation of the risk.

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and

PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

TOPIC PA0104: CONDUCT MITIGATION OF THE RISK.

PM-02-PS01: Undertake a preliminary risk assessment Scope of Practical Skill Given work instructions, checklists, work area, case study, drawings, documents, templates, forms, safety and quality principles, standard operating procedures and, statutory requirements, you should be able to: PA0101 Conduct a preliminary risk assessment. PA0102 Identify the hazards. PA0103 Relate the risk to the hazard. PA0104 Conduct mitigation of the risk.

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

671101000-PM-03, Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr30 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

IN SUMMARY

In this unit we have focussed on:

AK0101 Safety procedures

AK0102 Statutory requirements for hazardous conditions/material/substance.

AK0103 Risk assessment procedure for specific context

AK0104 Inspection procedure

AK0105 Recording/reporting procedure

AK0106 Approval procedures

PA0101 Conduct a preliminary risk assessment.

PA0102 Identify the hazards.

PA0103 Relate the risk to the hazard.

PA0104 Conduct mitigation of the risk.

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 29

PM-02-PS02: READ AND INTERPRET ELECTRICAL DRAWINGS

In this Chapter we will discuss the following Topics:

AK0201 Methods of interpreting scales

AK0202 Techniques of interpreting diagrams

AK0203 Procedures for evaluating the work site.

PA0201 Identify signs and symbols.

PA0202 Read and interpret electrical drawings.

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Signs and symbols were correctly identified according to the IEC(International Electrical compliance standards) | Electrical diagrams were correctly interpreted, using schematics, plans, wiring and layout representations according to worksite requirements.

TOPIC AK0201: METHODS OF INTERPRETING SCALES

PA0201 - Identify signs and symbols.

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

AK0201 - Methods of interpreting scales

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was

completed that included health and safety considerations in accordance with risk management procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

PA0202 - Read and interpret electrical drawings.

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures.

AK0202 - Techniques of interpreting diagrams

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

AK0203 - Procedures for evaluating the work site.

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

TOPIC AK0202: TECHNIQUES OF INTERPRETING DIAGRAMS

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify

risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures.

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications.

TOPIC AK0203: PROCEDURES FOR EVALUATING THE WORK SITE.

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures.

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

TOPIC PA0201: IDENTIFY SIGNS AND SYMBOLS

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified

through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures.

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

TOPIC PA0202: READ AND INTERPRET ELECTRICAL DRAWINGS

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures.

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

IN SUMMARY

In this unit we have focussed on:

AK0201 Methods of interpreting scales

AK0202 Techniques of interpreting diagrams

AK0203 Procedures for evaluating the work site.

PA0201 Identify signs and symbols.

PA0202 Read and interpret electrical drawings.

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 30

PM-02-PS03: PLAN THE WORKSITE SET-UP

In this Chapter we will discuss the following Topics:

AK0201 Knowledge of tools, material and equipment required to perform various functions within the worksite.

AK0202 Knowledge of human resources required to perform various functions within the worksite.

AK0203 Analytic techniques for interpreting work instructions

AK0204 Knowledge of workflow processes

PA0201 Compile a list of tools, material and equipment required.

PA0202 Compile a list of human resources required.

PA0203 Compile a workflow diagram

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

The relevant tools, material and equipment were correctly listed and quantities recorded in accordance with the task specification | The correct human resources were listed that meet the job specification | The work flow / work order was correctly analysed and applied in accordance with the task specification | Provider Accreditation Requirements for the Module

TOPIC AK0201: KNOWLEDGE OF TOOLS, MATERIAL AND EQUIPMENT REQUIRED TO PERFORM VARIOUS FUNCTIONS WITHIN THE WORKSITE

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry

out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

thickness of 3mm), Gas cutting is limited to mild steel only (maximum thickness of Human Resource Requirements: Trainers should have a minimum requirement to include a level 4 qualification and a level 5 facilitator qualification that includes competencies related to installing, testing, connecting,, maintaining and altering/repairing electrical equipment, wiring and control systems or equivalent or at least 5 years relevant experience working within an electrical-related industry with a recognized Tr

TOPIC AK0202: KNOWLEDGE OF HUMAN RESOURCES REQUIRED TO PERFORM VARIOUS FUNCTIONS WITHIN THE WORKSITE

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

thickness of 3mm), Gas cutting is limited to mild steel only (maximum thickness of Human Resource Requirements: Trainers should have a minimum requirement to include a level 4 qualification and a level 5 facilitator qualification that includes competencies related to installing, testing, connecting, maintaining and altering/repairing electrical equipment, wiring and control systems or equivalent or at least 5 years relevant experience working within an electrical-related industry with a recognized Tr

TOPIC AK0203: ANALYTIC TECHNIQUES FOR INTERPRETING WORK INSTRUCTIONS

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs

and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

PA0201 Use Tools, Materials and Equipment PA0202 Use Personal Protective Equipment AK0201 Methods and techniques for using personal protective equipment AK0202 Methods and techniques of using tools AK0203 Methods and techniques of using power tools AK0204 Methods and techniques of ensuring workshop / worksite safety Internal Assessment Criteria Tools, material and equipment were safely and correctly used while adhering to worksite and workshop safety procedures

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

TOPIC AK0204: KNOWLEDGE OF WORKFLOW PROCESSES

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

thickness of 3mm), Gas cutting is limited to mild steel only (maximum thickness of Human Resource Requirements: Trainers should have a minimum requirement to include a level 4 qualification and a level 5 facilitator qualification that includes competencies related to installing, testing, connecting.ssioning, maintaining and altering/repairing electrical equipment, wiring and control systems or equivalent or at least 5 years relevant experience working within an electrical-related industry with a recognized Tr

TOPIC PA0201: COMPILE A LIST OF TOOLS, MATERIAL AND EQUIPMENT REQUIRED

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

thickness of 3mm), Gas cutting is limited to mild steel only (maximum thickness of Human Resource Requirements: Trainers should have a minimum requirement to include a level 4 qualification and a level 5 facilitator qualification that includes competencies related to installing, testing, connecting.ssioning, maintaining and altering/repairing electrical equipment, wiring and control systems or equivalent or at least 5 years relevant experience working within an electrical-related industry with a recognized Tr

TOPIC PA0202: COMPILE A LIST OF HUMAN RESOURCES REQUIRED

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

thickness of 3mm), Gas cutting is limited to mild steel only (maximum thickness of Human Resource Requirements: Trainers should have a minimum requirement to include a level 4 qualification and a level 5 facilitator qualification that includes competencies related to installing, testing, connecting.ssioning, maintaining and altering/repairing electrical equipment, wiring and control systems or equivalent or at least 5 years relevant experience working within an electrical-related industry with a recognized Tr

TOPIC PA0203: COMPILE A WORKFLOW DIAGRAM

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

thickness of 3mm), Gas cutting is limited to mild steel only (maximum thickness of Human Resource Requirements: Trainers should have a minimum requirement to include a level 4 qualification and a level 5 facilitator qualification that includes competencies related to installing, testing, connecting.ssioning, maintaining and altering/repairing electrical equipment, wiring and control systems or equivalent or at least 5 years relevant experience working within an electrical-related industry with a recognized Tr

IN SUMMARY

In this unit we have focussed on:

AK0201 Knowledge of tools, material and equipment required to perform various functions within the worksite.

AK0202 Knowledge of human resources required to perform various functions within the worksite.

AK0203 Analytic techniques for interpreting work instructions

AK0204 Knowledge of workflow processes

PA0201 Compile a list of tools, material and equipment required.

PA0202 Compile a list of human resources required.

PA0203 Compile a workflow diagram.

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 31

PM-03-PS01: PROCURE RESOURCES

In this Chapter we will discuss the following Topics:

AK0101 Procurement procedures

AK0102 Resource determination methods

AK0103 Techniques and procedures for requisitioning

PA0101 Obtain human resources required.

PA0102 Requisition (obtain) tools and equipment required.

PA0103 Requisition (obtain) material and consumables required.

PA0104 Determine functionality of tools and testing equipment.

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

The Human resources were determined and listed to meet the specific needs of the activity and are fit for purpose | Tools and testing instruments required were determined and procured to meet the specific needs of the activity and are fit for purpose | Equipment were procured to meet the specific needs of the activity and are fit for purpose | Material requirements were procured to meet the specific needs of the activity and are fit for purpose | Consumable requirements were procured to meet the specific needs of the activity and are fit for purpose

TOPIC AK0101: PROCUREMENT PROCEDURES

671101000-PM-03, Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr30 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools.

TOPIC AK0102: RESOURCE DETERMINATION METHODS

671101000-PM-03, Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr30 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC AK0103: TECHNIQUES AND PROCEDURES FOR REQUISITIONING

671101000-PM-03, Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr30 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC PA0101: OBTAIN HUMAN RESOURCES REQUIRED.

671101000-PM-03, Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr30 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

TOPIC PA0102: REQUISITION (OBTAIN) TOOLS AND EQUIPMENT REQUIRED.

671101000-PM-03, Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr30 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC PA0103: REQUISITION (OBTAIN) MATERIAL AND CONSUMABLES REQUIRED.

671101000-PM-03, Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr30 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools.

TOPIC PA0104: DETERMINE FUNCTIONALITY OF TOOLS AND TESTING EQUIPMENT

671101000-PM-03, Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr30 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

PM-01-PS03: Select, use and care for electrical measuring instruments PM-01-PS04: Perform soldering activities PM-01-PS05: Carry out basic electric arc welding in an electrical environment PM-01-PS06: Carry out basic gas cutting in an electrical environment 671101000-PM-02, Plan worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated wo

IN SUMMARY

In this unit we have focussed on:

AK0101 Procurement procedures

AK0102 Resource determination methods

AK0103 Techniques and procedures for requisitioning

PA0101 Obtain human resources required.

PA0102 Requisition (obtain) tools and equipment required.

PA0103 Requisition (obtain) material and consumables required.

PA0104 Determine functionality of tools and testing equipment.

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 32

PM-03-PS02: PREPARE AND MOVE RESOURCES

In this Chapter we will discuss the following Topics:

AK0201 Basic lifting techniques

AK0202 Use of material safety data sheet

PA0201 Load and off load resources

PA0202 Procure type and size of moving equipment.

PA0203 Stack and secure tools, material and equipment

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

AC 69

AC 70

AC 71

AC 72

AC 73

TOPIC AK0201: BASIC LIFTING TECHNIQUES

AK0201 - Basic lifting techniques

always use a mechanical aid to transport a load Clear obstacles out of the lifting area safe systems of working Use the manual lifting techniques described above and avoid sudden or jerky movements u se gloves when manual handling to avoid injury from rough or Take special care when moving loads wrapped in grease or Never move a load over other people or walk under a suspended load

TOPIC AK0202: USE OF MATERIAL SAFETY DATA SHEET

Equipment were procured to meet the specific needs of the activity and are fit for purpose Material requirements were procured to meet the specific needs of the activity and are fit Consumable requirements were procured to meet the specific needs of the activity and are fit for purpose PM-03-PS02: Prepare and move

resources Scope of Practical Skill Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, you should be able to: PA0201

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

PA0201 Use Tools, Materials and Equipment PA0202 Use Personal Protective Equipment AK0201 Methods and techniques for using personal protective equipment AK0202 Methods and techniques of using tools AK0203 Methods and techniques of using power tools AK0204 Methods and techniques of ensuring workshop / worksite safety Internal Assessment Criteria Tools, material and equipment were safely and correctly used while adhering to worksite and workshop safety procedures

TOPIC PA0201: LOAD AND OFF LOAD RESOURCES

Equipment were procured to meet the specific needs of the activity and are fit for purpose Material requirements were procured to meet the specific needs of the activity and are fit Consumable requirements were procured to meet the specific needs of the activity and are fit for purpose PM-03-PS02: Prepare and move resources Scope of Practical Skill Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, you should be able to: PA0201

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

Scribers, dividers and centre punches were sharpened to ensure serviceability. PM-01-PS02: Select, use and care for power tools Scope of Practical Skill Given work instructions, work area, documents, templates, drilling machines, grinders, and saws standard operating procedures and, statutory requirements, you should be PA0201 Select a power tool pertaining to specific job requirements. PA0202 Use fixed power tools. PA0203 Use portable power tools.

TOPIC PA0202: PROCURE TYPE AND SIZE OF MOVING EQUIPMENT

Equipment were procured to meet the specific needs of the activity and are fit for purpose Material requirements were procured to meet the specific needs of the activity and are fit Consumable requirements were

procured to meet the specific needs of the activity and are fit for purpose PM-03-PS02: Prepare and move resources Scope of Practical Skill Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, you should be able to: PA0201

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

Scribers, dividers and centre punches were sharpened to ensure serviceability. PM-01-PS02: Select, use and care for power tools Scope of Practical Skill Given work instructions, work area, documents, templates, drilling machines, grinders, and saws standard operating procedures and, statutory requirements, you should be PA0201 Select a power tool pertaining to specific job requirements. PA0202 Use fixed power tools. PA0203 Use portable power tools

TOPIC PA0203: STACK AND SECURE TOOLS, MATERIAL AND EQUIPMENT

Equipment were procured to meet the specific needs of the activity and are fit for purpose Material requirements were procured to meet the specific needs of the activity and are fit Consumable requirements were procured to meet the specific needs of the activity and are fit for purpose PM-03-PS02: Prepare and move resources Scope of Practical Skill Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, you should be able to: PA0201

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

Scribers, dividers and centre punches were sharpened to ensure serviceability. PM-01-PS02: Select, use and care for power tools Scope of Practical Skill Given work instructions, work area, documents, templates, drilling machines, grinders, and saws standard operating procedures and, statutory requirements, you should be PA0201 Select a power tool pertaining to specific job requirements. PA0202 Use fixed power tools. PA0203 Use portable power tools

IN SUMMARY

In this unit we have focussed on:

AK0201 Basic lifting techniques

AK0202 Use of material safety data sheet

PA0201 Load and off load resources

PA0202 Procure type and size of moving equipment

PA0203 Stack and secure tools, material and equipment

Complete the workbook activities linked to this chapter, which are activities 1 – 5.

CHAPTER 33

PM-03-PS03: PREPARE WORK SITE AND EQUIPMENT

In this Chapter we will discuss the following Topics:

AK0301 Lock out procedures

AK0302 Relevant subsections of SANS 10142-1

AK0303 Installation requirements

AK0304 Site specific rules and regulations

PA0301 Undertake safety checks

PA0302 Source temporary electrical supplies for use

PA0303 Position equipment for installation

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

A safety report was completed that contain all relevant areas of the installation | Equipment and components were correctly positioned according to the installation requirements | The relevant subsections of SANS 10142-1 were correctly interpreted and applied | Provider Accreditation Requirements for the Module

TOPIC AK0301: LOCK OUT PROCEDURES

AK0301 - Lock out procedures

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

PA0302 - Source temporary electrical supplies for use

the correct electrical testing and fault finding procedures were applied defects were rectified as per installation requirements the correct electrical testing and fault finding procedures were applied

DEFECTS WERE RECTIFIED AS PER INSTALLATION REQUIREMENTS

AK0302 - Relevant subsections of SANS 10142-1

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

PA0303 - Position equipment for installation

test and inspect electrical equipment, control systems and installations maintain and repair electrical equipment, control systems and installations test and inspect electrical equipment, control systems and installations maintain and repair electrical equipment, control systems and installations

AK0303 - Installation requirements

the correct electrical testing and fault finding procedures were applied defects were rectified as per installation requirements the correct electrical testing and fault finding procedures were applied

DEFECTS WERE RECTIFIED AS PER INSTALLATION REQUIREMENTS

AK0304 - Site specific rules and regulations

test and inspect electrical equipment, control systems and installations maintain and repair electrical equipment, control systems and installations test and inspect electrical equipment, control systems and installations maintain and repair electrical equipment, control systems and installations

TOPIC AK0302: RELEVANT SUBSECTIONS OF SANS 10142-1

Equipment were procured to meet the specific needs of the activity and are fit for purpose Material requirements were procured to meet the specific needs of the activity and are fit Consumable requirements were procured to meet the specific needs of the activity and are fit for purpose PM-03-PS02: Prepare and move resources Scope of Practical Skill Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, you should be able to: PA0201

PA0201 Use Tools, Materials and Equipment PA0202 Use Personal Protective Equipment AK0201 Methods and techniques for using personal protective equipment AK0202 Methods and techniques of using tools AK0203 Methods and techniques of using power tools AK0204 Methods and techniques of ensuring workshop / worksite safety Internal Assessment Criteria Tools, material and equipment were safely and correctly used while adhering to worksite and workshop safety procedures

The relevant subsections of SANS 10142-1 were correctly interpreted and applied Provider Accreditation Requirements for the Module Demonstrate access to: Categories of tools, equipment and processes in order for learners to be able to prepare worksite set up for installing, wiring and connecting electrical equipment and control Human Resource Requirements: Trainers should have a minimum requirement to include a level 4 qualification and a level 5 facilitator qualification that includes competencies related to insta

TOPIC AK0303: INSTALLATION REQUIREMENTS

Equipment were procured to meet the specific needs of the activity and are fit for purpose Material requirements were procured to meet the specific needs of the activity and are fit Consumable requirements were procured to meet the specific needs of the activity and are fit for purpose PM-03-PS02: Prepare and move resources Scope of Practical Skill Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, you should be able to: PA0201

manufacturer's specifications. The correct electrical testing and fault finding procedures were applied. Defects were rectified as per installation requirements. Repairs were recorded and reported as per reporting procedures

regulations and codes. § A test report that confirms the correctness of the installation is completed in accordance with relevant regulations and codes and relevant workshop § Personal protective equipment are selected and correctly used in accordance with the relevant subsection of the OHS Act and the Mines Health and Safety § The selected equipment are isolated and locked out

TOPIC AK0304: SITE SPECIFIC RULES AND REGULATIONS

Equipment were procured to meet the specific needs of the activity and are fit for purpose Material requirements were procured to meet the specific needs of the activity and are fit Consumable requirements were procured to meet the specific needs of the activity and are fit for purpose PM-03-PS02: Prepare and move resources Scope of Practical Skill Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, you should be able to: PA0201

PA0201 Use Tools, Materials and Equipment PA0202 Use Personal Protective Equipment AK0201 Methods and techniques for using personal protective equipment AK0202 Methods and techniques of using tools AK0203 Methods and techniques of using power tools AK0204 Methods and techniques of ensuring workshop / worksite safety Internal Assessment Criteria Tools, material and equipment were safely and correctly used while adhering to worksite and workshop safety procedures

Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rotating electrical machinery - AC, Rotating Electrical Machinery - DC, Rotating electrical machinery -

Alternators and Generators, Protection devices, luminaries, current transformers, potential transformers and shunts according to the design specifications

TOPIC PA0301: UNDERTAKE SAFETY CHECKS

Equipment were procured to meet the specific needs of the activity and are fit for purpose Material requirements were procured to meet the specific needs of the activity and are fit Consumable requirements were procured to meet the specific needs of the activity and are fit for purpose PM-03-PS02: Prepare and move resources Scope of Practical Skill Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, you should be able to: PA0201

PA0201 Use Tools, Materials and Equipment PA0202 Use Personal Protective Equipment AK0201 Methods and techniques for using personal protective equipment AK0202 Methods and techniques of using tools AK0203 Methods and techniques of using power tools AK0204 Methods and techniques of ensuring workshop / worksite safety Internal Assessment Criteria Tools, material and equipment were safely and correctly used while adhering to worksite and workshop safety procedures

Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rotating electrical machinery - AC, Rotating Electrical Machinery - DC, Rotating electrical machinery - Alternators and Generators, Protection devices, luminaries, current transformers, potential transformers and shunts according to the design specifications

TOPIC PA0302: SOURCE TEMPORARY ELECTRICAL SUPPLIES FOR USE

Equipment were procured to meet the specific needs of the activity and are fit for purpose Material requirements were procured to meet the specific needs of the activity and are fit Consumable requirements were procured to meet the specific needs of the activity and are fit for purpose PM-03-PS02: Prepare and move resources Scope of Practical Skill Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, you should be able to: PA0201

Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rotating electrical machinery - AC, Rotating Electrical Machinery - DC, Rotating electrical machinery - Alternators and Generators, Protection devices, luminaries, current transformers, potential transformers and shunts according to the design specifications

manufacturer’s specifications. The correct electrical testing and fault finding procedures were applied. Defects were rectified as per installation requirements. Repairs were recorded and reported as per reporting procedures

TOPIC PA0303: POSITION EQUIPMENT FOR INSTALLATION

Equipment were procured to meet the specific needs of the activity and are fit for purpose Material requirements were procured to meet the specific needs of the activity and are fit Consumable requirements were procured to meet the specific needs of the activity and are fit for purpose PM-03-PS02: Prepare and move resources Scope of Practical Skill Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, you should be able to: PA0201

PA0201 Use Tools, Materials and Equipment PA0202 Use Personal Protective Equipment AK0201 Methods and techniques for using personal protective equipment AK0202 Methods and techniques of using tools AK0203 Methods and techniques of using power tools AK0204 Methods and techniques of ensuring workshop / worksite safety Internal Assessment Criteria Tools, material and equipment were safely and correctly used while adhering to worksite and workshop safety procedures

671101000-PM-03, Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr30 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

IN SUMMARY

In this unit we have focussed on:

AK0301 Lock out procedures

AK0302 Relevant subsections of SANS 10142-1

AK0303 Installation requirements

AK0304 Site specific rules and regulations

PA0301 Undertake safety checks

PA0302 Source temporary electrical supplies for use

PA0303 Position equipment for installation

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 34

PM-04-PS01: CONFIRM THE SELECTION AND INSTALLATION METHOD OF THE SELECTED WIRE WAYS

In this Chapter we will discuss the following Topics:

AK0101 Relevant subsections of SANS 10142-1

AK0102 Installation methods and techniques

AK0103 Hazard and risk procedures

PA0101 Select the correct wireways

PA0102 Select the method of installation

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

The correct wire ways were selected according to SANS 10142-1 that meet the operational conditions of the installation. | The correct installation methods were selected according to SANS 10142-1, the relevant job instruction and environmental requirements.

TOPIC AK0101: RELEVANT SUBSECTIONS OF SANS 10142-1

PA0101 - Select the correct wireways

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

AK0101 - Relevant subsections of SANS 10142-1

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

PA0102 - Select the method of installation

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures 671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to installing wire-ways 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures 671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to installing wire-ways

AK0102 - Installation methods and techniques

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

AK0103 - Hazard and risk procedures

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills

within a simulated work environment risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

TOPIC AK0102: ISTALLATION METHODS AND TECHNIQUES

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to installing wire-ways

TOPIC AK0103: HAZARD AND RISK PROCEDURES

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

PM-02-PS01: Undertake a preliminary risk assessment Scope of Practical Skill Given work instructions, checklists, work area, case study, drawings, documents, templates, forms, safety and quality principles, standard operating procedures and, statutory requirements, you should be able to: PA0101 Conduct a preliminary risk assessment. PA0102 Identify the hazards. PA0103 Relate the risk to the hazard. PA0104 Conduct mitigation of the risk

TOPIC PA0101: SELECT THE CORRECT WIREWAYS

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to installing wire-ways

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC PA0102: SELECT THE METHOD OF INSTALLATION

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to installing wire-ways

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

IN SUMMARY

In this unit we have focussed on:

AK0101 Relevant subsections of SANS 10142-1

AK0102 Installation methods and techniques

AK0103 Hazard and risk procedures

PA0101 Select the correct wireways

PA0102 Select the method of installation

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 35

PM-04-PS02: USE OF INSTALLATION TOOLS AND EQUIPMENT

In this Chapter we will discuss the following Topics:

AK0201 Methods and techniques for using personal protective equipment

AK0202 Methods and techniques of using tools

AK0203 Methods and techniques of using power tools

AK0204 Methods and techniques of ensuring workshop / worksite safety

PA0201 Use Tools, Materials and Equipment

PA0202 Use Personal Protective Equipment

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Tools, material and equipment were safely and correctly used while adhering to worksite and workshop safety procedures. | Personal protective equipment was correctly selected and used according to statutory requirements. | Power tools were selected and used correctly and safely while adhering fully to the workshop safety regulations.

TOPIC AK0201: METHODS AND TECHNIQUES FOR USING PERSONAL PROTECTIVE EQUIPMENT

PA0201 Use Tools, Materials and Equipment PA0202 Use Personal Protective Equipment AK0201 Methods and techniques for using personal protective equipment AK0202 Methods and techniques of using tools AK0203 Methods and techniques of using power tools AK0204 Methods and techniques of ensuring workshop / worksite safety Internal Assessment Criteria Tools, material and equipment were safely and correctly used while adhering to worksite and workshop safety procedures

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

TOPIC AK0202: METHODS AND TECHNIQUES OF USING TOOLS

PA0201 Use Tools, Materials and Equipment PA0202 Use Personal Protective Equipment AK0201 Methods and techniques for using personal protective equipment AK0202 Methods and techniques of using tools AK0203 Methods and techniques of using power tools AK0204 Methods and techniques of ensuring workshop / worksite safety Internal Assessment Criteria Tools, material and equipment were safely and correctly used while adhering to worksite and workshop safety procedures

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

TOPIC AK0203: METHODS AND TECHNIQUES OF USING POWER TOOLS

PA0201 Use Tools, Materials and Equipment PA0202 Use Personal Protective Equipment AK0201 Methods and techniques for using personal protective equipment AK0202 Methods and techniques of using tools AK0203 Methods and techniques of using power tools AK0204 Methods and techniques of ensuring workshop / worksite safety Internal Assessment Criteria Tools, material and equipment were safely and correctly used while adhering to worksite and workshop safety procedures

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

TOPIC AK0204: METHODS AND TECHNIQUES OF ENSURING WORKSHOP / WORKSITE SAFETY

PA0201 Use Tools, Materials and Equipment PA0202 Use Personal Protective Equipment AK0201 Methods and techniques for using personal protective equipment AK0202 Methods and techniques of using tools AK0203 Methods and techniques of using power tools AK0204 Methods and techniques of ensuring workshop / worksite safety Internal Assessment Criteria Tools, material and equipment were safely and correctly used while adhering to worksite and workshop safety procedures

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

TOPIC PA0201: USE TOOLS, MATERIALS AND EQUIPMENT

PA0201 Use Tools, Materials and Equipment PA0202 Use Personal Protective Equipment AK0201 Methods and techniques for using personal protective equipment AK0202 Methods and techniques of using tools AK0203 Methods and techniques of using power tools AK0204 Methods and techniques of ensuring workshop / worksite safety Internal Assessment Criteria Tools, material and equipment were safely and correctly used while adhering to worksite and workshop safety procedures

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

Equipment were procured to meet the specific needs of the activity and are fit for purpose Material requirements were procured to meet the specific needs of the activity and are fit Consumable requirements were procured to meet the specific needs of the activity and are fit for purpose PM-03-PS02: Prepare and move resources Scope of Practical Skill Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, you should be able to: PA0201

TOPIC PA0202: USE PERSONAL PROTECTIVE EQUIPMENT

PA0201 Use Tools, Materials and Equipment PA0202 Use Personal Protective Equipment AK0201 Methods and techniques for using personal protective equipment AK0202 Methods and techniques of using tools AK0203 Methods and techniques of using power tools AK0204 Methods and techniques of ensuring workshop / worksite safety Internal Assessment Criteria Tools, material and equipment were safely and correctly used while adhering to worksite and workshop safety procedures

Enlist any three hazards found at workplace Why safety laws are necessary? Answer the following question in detail Write a note on importance of occupational health and safety Explain the personal protective equipment for protecting hands, face, ear, eyes and foot Explain health and safety precautions, regulations/guidelines. Explain the types of Fire Extinguisher and their uses. Describe health and safety regulations. List PPE available and required to work in lab/ Institute

Scribers, dividers and centre punches were sharpened to ensure serviceability. PM-01-PS02: Select, use and care for power tools Scope of Practical Skill Given work instructions, work area, documents, templates, drilling machines, grinders, and saws standard operating procedures and, statutory requirements, you should be PA0201 Select a power tool pertaining to specific job requirements. PA0202 Use fixed power tools. PA0203 Use portable power tools

IN SUMMARY

In this unit we have focussed on:

AK0201 Methods and techniques for using personal protective equipment

AK0202 Methods and techniques of using tools

AK0203 Methods and techniques of using power tools

AK0204 Methods and techniques of ensuring workshop / worksite safety

PA0201 Use Tools, Materials and Equipment

PA0202 Use Personal Protective Equipment

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 36

PM-04-PS03: DESIGN AND INSTALL WIRE WAYS

In this Chapter we will discuss the following Topics:

AK0301 Methods and techniques for using personal protective equipment

AK0302 Methods and techniques of using tools

AK0303 Methods and techniques of using power tools

AK0304 Methods and techniques ensuring workshop / worksite Safety

AK0305 Relevant subsections of SANS 10142-1

AK0306 Safety and performance standards of wireways

PA0301 Design the layout of wireways

PA0302 Install wire ways

PA0303 Terminate and join the wireways

PA0304 Install wire way accessories

PA0305 Bond wireways

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Conduit was correctly installed and bent to the correct radius as prescribed in SANS

10142-1

Cable racks, ladders and trays were correctly installed as prescribed by the manufacturer's specifications

Trunking and ducting were correctly installed as prescribed by the manufacturer's specifications

The correct methods for terminating and joining wireways were used in accordance with the installation requirements.

The installation of wireways reflected the specific selection of fastening and support requirements of the installation.

The installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1.

The correct methods for the installation of wireways were selected that fully complies with

SANS 10142-1.

The wire ways were installed neatly (vertically, horizontally and offsets parallel and similar) and complies with industry acceptable standard.

Provider Accreditation Requirements for the Module

TOPIC AK0301: METHODS AND TECHNIQUES FOR USING PERSONAL PROTECTIVE EQUIPMENT

AK0301 - Methods and techniques for using personal protective equipment

the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC AK0302: METHODS AND TECHNIQUES OF USING TOOLS

AK0302 - Methods and techniques of using tools

the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC AK0303: METHODS AND TECHNIQUES OF USING POWER TOOLS

AK0303 - Methods and techniques of using power tools

the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the

Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC AK0304: METHODS AND TECHNIQUES ENSURING WORKSHOP / WORKSITE SAFETY

AK0304 - Methods and techniques ensuring workshop / worksite Safety

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures

TOPIC AK0305: RELEVANT SUBSECTIONS OF SANS 10142-1

AK0305 - Relevant subsections of SANS 10142-1

the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC AK0306: SAFETY AND PERFORMANCE STANDARDS OF WIREWAYS

AK0306 - Safety and performance standards of wireways

the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 requirements and comply with industry acceptable standards electrical conductors were correctly connected and neatly secured in accordance with statutory requirements and industry acceptable standards the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1

TOPIC PA0301: DESIGN THE LAYOUT OF WIREWAYS

PA0301 - Design the layout of wireways

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1

TOPIC PA0302: INSTALL WIRE WAYS

PA0302 - Install wire ways

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment 671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to installing wire-ways 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment 671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to installing wire-ways

TOPIC PA0303: TERMINATE AND JOIN THE WIREWAYS

PA0201 Use Tools, Materials and Equipment PA0202 Use Personal Protective Equipment AK0201 Methods and techniques for using personal protective equipment AK0202 Methods and techniques of using tools AK0203 Methods and techniques of using power tools AK0204 Methods and techniques of ensuring workshop / worksite safety Internal Assessment Criteria Tools, material and equipment were safely and correctly used while adhering to worksite and workshop safety procedures

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to installing wire-ways

TOPIC PA0304: INSTALL WIRE WAY ACCESSORIES

the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to installing wire-ways the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to installing wire-ways

TOPIC PA0305: BOND WIREWAYS

PA0305 - Bond wireways

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1

IN SUMMARY

In this unit we have focussed on:

AK0301 Methods and techniques for using personal protective equipment

AK0302 Methods and techniques of using tools

AK0303 Methods and techniques of using power tools

AK0304 Methods and techniques ensuring workshop / worksite Safety

AK0305 Relevant subsections of SANS 10142-1

AK0306 Safety and performance standards of wireways

PA0301 Design the layout of wireways

PA0302 Install wire ways

PA0303 Terminate and join the wireways

PA0304 Install wire way accessories

Complete the workbook activities linked to this chapter, which are activities 1 – 11.

CHAPTER 37

PM-05-PS01: IDENTIFY HAZARDS WITHIN INSTALLATION SCOPE OF PRACTICAL SKILL

In this Chapter we will discuss the following Topics:

AK0101 Risk assessment techniques/procedures

AK0102 Health and safety reporting procedures

AK0103 Environmental procedures for the disposal of hazardous substances

AK0104 Identification of hazardous substances

AK0105 Electrical safety procedures for working in close proximity to live equipment

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. | Risks were mitigated and or rectified through the implementation of the correct risk management procedures. | A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures.

TOPIC AK0101: RISK ASSESSMENT TECHNIQUES/PROCEDURES

AK0101 - Risk assessment techniques/procedures

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

AK0102 - Health and safety reporting procedures

PA0103 RELATE THE RISK TO THE HAZARD

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

AK0104 - Identification of hazardous substances

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures pA0103 Relate the risk to the hazard risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

AK0105 - Electrical safety procedures for working in close proximity to live equipment

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

TOPIC AK0102: HEALTH AND SAFETY REPORTING PROCEDURES

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

PM-02-PS01: Undertake a preliminary risk assessment Scope of Practical Skill Given work instructions, checklists, work area, case study, drawings, documents, templates, forms, safety and quality principles, standard operating procedures and, statutory requirements, you should be able to: PA0101 Conduct a preliminary risk assessment. PA0102 Identify the hazards. PA0103 Relate the risk to the hazard. PA0104 Conduct mitigation of the risk

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

TOPIC AK0103: ENVIRONMENTAL PROCEDURES FOR THE DISPOSAL OF HAZARDOUS SUBSTANCES

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

PM-02-PS01: Undertake a preliminary risk assessment Scope of Practical Skill Given work instructions, checklists, work area, case study, drawings, documents, templates, forms, safety and quality principles, standard operating procedures and, statutory requirements, you should be able to: PA0101 Conduct a preliminary risk assessment. PA0102 Identify the hazards. PA0103 Relate the risk to the hazard. PA0104 Conduct mitigation of the risk

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC AK0104: IDENTIFICATION OF HAZARDOUS SUBSTANCES

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

PM-02-PS01: Undertake a preliminary risk assessment Scope of Practical Skill Given work instructions, checklists, work area, case study, drawings, documents, templates, forms, safety and quality principles, standard operating procedures and, statutory requirements, you should be able to: PA0101 Conduct a preliminary risk assessment. PA0102 Identify the hazards. PA0103 Relate the risk to the hazard. PA0104 Conduct mitigation of the risk

TOPIC AK0105: ELECTRICAL SAFETY PROCEDURES FOR WORKING IN CLOSE PROXIMITY TO LIVE EQUIPMENT

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

IN SUMMARY

In this unit we have focussed on:

AK0101 Risk assessment techniques/procedures

AK0102 Health and safety reporting procedures

AK0103 Environmental procedures for the disposal of hazardous substances

AK0104 Identification of hazardous substances

AK0105 Electrical safety procedures for working in close proximity to live equipment

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 38

PM-05-PS02: CONFIRM THE SELECTION AND INSTALLATION METHOD OF THE ELECTRICAL EQUIPMENT AND CONTROL SYSTEMS

In this Chapter we will discuss the following Topics:

AK0201 Relevant subsections of SANS 10142-1

AK0202 Knowledge of electrical components and switchgear

AK0203 Installation techniques of electrical components and switchgear

AK0204 Read and interpret electrical drawings and installation specifications

AK0205 Knowledge of electronic components

AK0206 Read and interpret electronic circuit diagrams

AK0207 Installation techniques of electronic components

PA0201 Select equipment and control system according to design specifications.

PA0202 Interpret drawings

PA0203 Use manufacturer specifications

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Drawings and/or manufacturer specifications were correctly interpreted and the correct installation method was selected that meet the installation specifications, for the given equipment and control systems and complies fully with SANS 10142-1. | The correct equipment and control systems were selected including Batteries, Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rotating electrical machinery - AC, Rotating Electrical | Machinery - DC, Rotating electrical machinery - Alternators and Generators, Protection devices, luminaries, current transformers, potential transformers and shunts according to the design specifications. | The correct installation methods were selected for Batteries, Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rotating electrical machinery - AC, Rotating Electrical Machinery - DC, Rotating electrical machinery - Alternators and Generators, Protection devices, luminaries, current transformers, potential transformers and shunts according to the design specifications. | The correct electronic components were selected including

resistors, diodes, Zener diodes, capacitors, transistors according to the design specifications. | The correct installation method was selected for resistors, diodes, Zener diodes, capacitors, transistors according to the design specifications.

TOPIC AK0201: RELEVANT SUBSECTIONS OF SANS 10142-1

PA0201 - Select equipment and control system according to design specifications

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

AK0201 - Relevant subsections of SANS 10142-1

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

PA0202 - Interpret drawings

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures connect electrical power to audio and visual communication equipment, signalling devices, and heating and cooling systems electrician Overview of the Electrical Trade NOC # 7242 Industrial Electricians Industrial electricians install, maintain, test, troubleshoot, modify and repair industrial electrical equipment and associated electrical and electronic controls risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures electrician Overview of the Electrical Trade NOC # 7242 Industrial Electricians

Industrial electricians install, maintain, test, troubleshoot, modify and repair industrial electrical equipment and associated electrical and electronic controls

AK0202 - Knowledge of electrical components and switchgear

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

PA0203 - Use manufacturer specifications

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

AK0203 - Installation techniques of electrical components and switchgear

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

AK0204 - Read and interpret electrical drawings and installation specifications

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

AK0205 - Knowledge of electronic components

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures

AK0206 - Read and interpret electronic circuit diagrams

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures connect electrical power to audio and visual communication equipment, signalling devices, and heating and cooling systems electrician Overview of the Electrical Trade NOC # 7242 Industrial Electricians Industrial electricians install, maintain, test, troubleshoot, modify and repair industrial electrical equipment and associated electrical and electronic controls risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures electrician Overview of the Electrical Trade NOC # 7242 Industrial Electricians Industrial electricians install, maintain, test, troubleshoot, modify and repair industrial electrical equipment and associated electrical and electronic controls

AK0207 - Installation techniques of electronic components

risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures

TOPIC AK0202: KNOWLEDGE OF ELECTRICAL COMPONENTS AND SWITCHGEAR

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation’s functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

Equipment were procured to meet the specific needs of the activity and are fit for purpose Material requirements were procured to meet the specific needs of the activity and are fit Consumable requirements were procured to meet the specific needs of the activity and are fit for purpose PM-03-PS02: Prepare and move resources Scope of Practical Skill Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, you should be able to: PA0201

TOPIC AK0203: INSTALLATION TECHNIQUES OF ELECTRICAL COMPONENTS AND SWITCHGEAR

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

TOPIC AK0204: READ AND INTERPRET ELECTRICAL DRAWINGS AND INSTALLATION SPECIFICATIONS

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

TOPIC AK0205: KNOWLEDGE OF ELECTRONIC COMPONENTS

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rotating electrical machinery - AC, Rotating Electrical Machinery - DC, Rotating electrical machinery - Alternators and Generators, Protection devices, luminaries, current transformers, potential transformers and shunts according to the design specifications

TOPIC AK0206: READ AND INTERPRET ELECTRONIC CIRCUIT DIAGRAMS

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

Overview of the Electrical Trade Electrician NOC # 7241 Construction Electrician Construction electricians lay out, assemble, install, test, troubleshoot, and repair electrical wiring, fixtures, control devices, and related equipment in buildings and other structures

TOPIC AK0207: INSTALLATION TECHNIQUES OF ELECTRONIC COMPONENTS

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified

through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rotating electrical machinery - AC, Rotating Electrical Machinery - DC, Rotating electrical machinery - Alternators and Generators, Protection devices, luminaries, current transformers, potential transformers and shunts according to the design specifications

Internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures. Portable power tools were correctly used, as per workshop practices and safety The correct personal protective equipment was used and complied with the workshop PM-05-PS04: Install equipment and control systems Scope of Practical Skill Given equipment and control systems Includes: Batteries, Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rot

TOPIC PA0201: SELECT EQUIPMENT AND CONTROL SYSTEM ACCORDING TO DESIGN SPECIFICATIONS.

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation’s functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

TOPIC PA0202: INTERPRET DRAWINGS

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

Overview of the Electrical Trade Electrician NOC # 7241 Construction Electrician Construction electricians lay out, assemble, install, test, troubleshoot, and repair electrical wiring, fixtures, control devices, and related equipment in buildings and other structures

replace faulty components. Connect electrical power to audio and visual communication equipment, signalling devices, and heating and cooling systems. Conduct preventive maintenance programs and keep maintenance records. Electrician Overview of the Electrical Trade NOC # 7242 Industrial Electricians Industrial electricians install, maintain, test, troubleshoot, modify and repair industrial electrical equipment and associated electrical and electronic controls

TOPIC PA0203: USE MANUFATURER SPECIFICATIONS

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

IN SUMMARY

In this unit we have focussed on:

AK0201 Relevant subsections of SANS 10142-1

AK0202 Knowledge of electrical components and switchgear

AK0203 Installation techniques of electrical components and switchgear

AK0204 Read and interpret electrical drawings and installation specifications

AK0205 Knowledge of electronic components

AK0206 Read and interpret electronic circuit diagrams

AK0207 Installation techniques of electronic components

PA0201 Select equipment and control system according to design specifications.

PA0202 Interpret drawings

PA0203 Use manufacturer specifications

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 39

PM-05-PS03: USE INSTALLATION TOOLS

In this Chapter we will discuss the following Topics:

AK0301 Procedure for using personal protective equipment

AK0302 Techniques for identifying, inspecting, using, maintaining and caring for hand tools

AK0303 Techniques for identifying, selecting, using, maintaining and caring for power tools

AK0304 Workshop safety procedures and practices

PA0301 Use installation tools.

PA0302 Use task specific personal protective equipment

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Hand tools were correctly used, as per workshop practices and safety procedures. | Portable power tools were correctly used, as per workshop practices and safety procedures. | The correct personal protective equipment was used and complied with the workshop safety procedures.

TOPIC AK0301: PROCEDURE FOR USING PERSONAL PROTECTIVE EQUIPMENT

PA0301 - Use installation tools

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

AK0302 - Techniques for identifying, inspecting, using, maintaining and caring for hand tools

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools

AK0303 - Techniques for identifying, selecting, using, maintaining and caring for power tools

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the you will be practicing skills related to preparing and using of hand, power and measuring tools

AK0304 - Workshop safety procedures and practices

internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures

TOPIC AK0302: TECHNIQUES FOR INDENTIFYING, INSPECTING, USING, MAINTAINING AND CARING FOR HAND TOOLS

Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rotating electrical machinery - AC, Rotating Electrical Machinery - DC, Rotating electrical machinery - Alternators and Generators, Protection devices, luminaries, current transformers, potential transformers and shunts according to the design specifications

manufacturer’s specifications. The correct electrical testing and fault finding procedures were applied. Defects were rectified as per installation requirements. Repairs were recorded and reported as per reporting procedures

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC AK0303: TECHNIQUES FOR INDENTIFYING, SELECTING, USING, MAINTAINING AND CARING FOR POWER TOOLS

Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rotating electrical machinery - AC, Rotating Electrical Machinery - DC, Rotating electrical machinery - Alternators and Generators, Protection devices, luminaries, current transformers, potential transformers and shunts according to the design specifications

PA0201 Use Tools, Materials and Equipment PA0202 Use Personal Protective Equipment AK0201 Methods and techniques for using personal protective equipment AK0202 Methods and techniques of using tools AK0203 Methods and techniques of using power tools AK0204 Methods and techniques of ensuring workshop / worksite safety Internal Assessment Criteria Tools, material and equipment were safely and correctly used while adhering to worksite and workshop safety procedures

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC AK0304: WORKSHOP SAFETY PROCEDURES AND PRACTICES

Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rotating electrical machinery - AC, Rotating Electrical Machinery - DC, Rotating electrical machinery - Alternators and Generators, Protection devices, luminaries, current transformers, potential transformers and shunts according to the design specifications

PA0201 Use Tools, Materials and Equipment PA0202 Use Personal Protective Equipment AK0201 Methods and techniques for using personal protective equipment AK0202 Methods and techniques of using tools AK0203 Methods and techniques of using power tools AK0204 Methods and techniques of ensuring workshop / worksite safety Internal Assessment Criteria Tools, material and equipment were safely and correctly used while adhering to worksite and workshop safety procedures

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC PA0301: USE INSTALLATION TOOLS.

Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rotating electrical machinery - AC, Rotating Electrical Machinery - DC, Rotating electrical machinery - Alternators and Generators, Protection devices, luminaries, current transformers, potential transformers and shunts according to the design specifications

671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to installing wire-ways

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC PA0302: USE TASK SPECIFIC PERSONAL PROTECTIVE EQUIPMENT

Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rotating electrical machinery - AC, Rotating Electrical Machinery - DC, Rotating electrical machinery - Alternators and Generators, Protection devices, luminaries, current transformers, potential transformers and shunts according to the design specifications

671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to installing wire-ways

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

IN SUMMARY

In this unit we have focussed on:

AK0301 Procedure for using personal protective equipment

AK0302 Techniques for identifying, inspecting, using, maintaining and caring for hand tools

AK0303 Techniques for identifying, selecting, using, maintaining and caring for power tools

AK0304 Workshop safety procedures and practices

PA0301 Use installation tools.

PA0302 Use task specific personal protective equipment

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 40

PM-05-PS04: INSTALL EQUIPMENT AND CONTROL SYSTEMS

In this Chapter we will discuss the following Topics:

AK0401 Relevant subsections of SANS 10142-1

AK0402 Statutory requirements regarding machine guards and lock-out systems.

AK0403 Protection principles

AK0404 Installation methods and techniques of electrical equipment and control systems

AK0405 Knowledge of relevant equipment and components

AK0406 Techniques for installation of basic electronic components

PA0401 Install equipment and control systems

PA0402 Select and install protection devices

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

The equipment and control systems were correctly selected as per job and manufacturer's specifications and the installation complies with SANS 10142-1. | The equipment and control systems were correctly installed as per job and manufacturer's specifications and the installation complies with SANS 10142-1. | The correct protection equipment and level of protection was selected as per job, manufacturer's specifications and relevant statutory requirements | The correct protection equipment and level of protection was installed as per job, manufacturer's specifications and relevant statutory requirements. | Electronic components were correctly installed, soldered and the circuitry was functional.

TOPIC AK0401: RELEVANT SUBSECTIONS OF SANS 10142-1

PA0401 - Install equipment and control systems

internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures 671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures 671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

AK0401 - Relevant subsections of SANS 10142-1

internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

PA0402 - Select and install protection devices

internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures 671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures 671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

AK0402 - Statutory requirements regarding machine guards and lock-out systems

internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations portable electrical equipment was checked for correct operation and functionality measuring instruments were correctly set up for application results were accurately interpreted against the specifications of the job requirements internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations portable electrical equipment was checked for correct operation and functionality

AK0403 - Protection principles

internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures

AK0404 - Installation methods and techniques of electrical equipment and control systems

internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures 671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the

learner an opportunity to: Practice electrical skills within a simulated work environment internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures 671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

AK0405 - Knowledge of relevant equipment and components

internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

AK0406 - Techniques for installation of basic electronic components

internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

TOPIC AK0402: STATUTORY REQUIREMENTS REGARDING MACHINE GUARDS AND LOCK-OUT SYSTEMS.

Internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures. Portable power tools were correctly used, as per workshop practices and safety The correct personal protective equipment was used and complied with the workshop PM-05-PS04: Install equipment and control systems Scope of Practical Skill Given equipment and control systems Includes: Batteries, Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rot

readings were recorded as per workshop procedures. Portable electrical equipment was checked for correct operation and functionality. Measuring instruments were correctly set up for application. Results were recorded on appropriate documentation. Results were accurately interpreted against the specifications of the job requirements. Electrical measuring instruments were used in accordance with their specifications

671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to installing wire-ways

TOPIC AK0403: PROTECTION PRINCIPLES

Internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures. Portable power tools were correctly used, as per workshop practices and safety The correct personal protective equipment was used and complied with the workshop PM-05-PS04: Install equipment and control systems Scope of Practical Skill Given equipment and control systems Includes: Batteries, Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rot

671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to installing wire-ways

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC AK0404: INSTALLATION METHODS AND TECHNIQUES OF ELECTRICAL EQUIPMENT AND CONTROL SYSTEMS

Internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures. Portable power tools were correctly used, as per workshop practices and safety The correct personal protective equipment was used and complied with the workshop PM-05-PS04: Install equipment and control systems Scope of Practical Skill Given equipment and control systems Includes: Batteries, Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rot

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to installing wire-ways

TOPIC AK0405: KNOWLEDGE OF RELEVANT EQUIPMENT AND COMPONENTS

Internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures. Portable power tools were correctly used, as per workshop practices and safety The correct personal protective equipment was used and complied with the workshop PM-05-PS04: Install equipment and control systems Scope of Practical Skill Given equipment and control systems Includes: Batteries, Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rot

671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to installing wire-ways

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC AK0406: TECHNIQUES FOR INSTALLALION OF BASIC ELECTRONIC COMPONENTS

Internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures. Portable power tools were correctly used, as per workshop practices and safety The correct personal protective equipment was used and complied with the workshop PM-05-PS04: Install equipment and control systems Scope of Practical Skill Given equipment and control systems Includes: Batteries, Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rot

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

TOPIC PA0401: INSTALL EQUIPMENT AND CONTROL SYSTEMS

Internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures. Portable power tools were correctly used, as per workshop practices and safety The correct personal protective equipment was used and complied with the workshop PM-05-PS04: Install equipment and control systems Scope of Practical Skill Given equipment and control systems Includes: Batteries, Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rot

671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to installing wire-ways

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC PA0402: SELECT AND INSTALL PROTECTION DEVICES

Internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures. Portable power tools were correctly used, as per workshop practices and safety The correct personal protective equipment was used and complied with the workshop PM-05-PS04: Install equipment and control systems Scope of Practical Skill Given equipment and control systems Includes: Batteries, Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rot

671101000-PM-04, Install wire-ways, L4, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to installing wire-ways

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

IN SUMMARY

In this unit we have focussed on:

AK0401 Relevant subsections of SANS 10142-1

AK0402 Statutory requirements regarding machine guards and lock-out systems.

AK0403 Protection principles

AK0404 Installation methods and techniques of electrical equipment and control systems

AK0405 Knowledge of relevant equipment and components

AK0406 Techniques for installation of basic electronic components

PA0401 Install equipment and control systems

PA0402 Select and install protection devices

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 41

PM-05-PS05: WIRE ELECTRICAL EQUIPMENT AND CONTROL SYSTEMS

In this Chapter we will discuss the following Topics:

AK0501 Wiring techniques

AK0502 Interpretation of electrical drawings

AK0503 Techniques for using relevant tools

PA0501 Use relevant types of tools and equipment

PA0502 Wire low voltage electrical systems, protection devices and electrical components.

PA0503 Clean the worksite

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Electrical equipment, control systems and protection devices were correctly wired to conform to safety, statutory and performance specifications.

Relevant tools and equipment were correctly used according to job and statutory requirements.

The wiring of electrical equipment and control systems were carried out in compliance with

SANS 10142-1.

The wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard.

The work site/area was left neat and tidy as per workshop procedures.

TOPIC AK0501: WIRING TECHNIQUES

AK0501 - Wiring techniques

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1 submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant

workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications the installation of wireways reflected the specific use of inspection points and accessories required according to SANS 10142-1

TOPIC AK0502: INTERPRETATION OF ELECTRICAL DRAWINGS

AK0502 - Interpretation of electrical drawings

manufacturer's specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer's specifications and relevant statutory requirements the wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard manufacturer's specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer's specifications and relevant statutory requirements

TOPIC AK0503: TECHNIQUES FOR USING RELEVANT TOOLS

AK0503 - Techniques for using relevant tools

the wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard the work site/area was left neat and tidy as per workshop procedures manufacturer's specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer's specifications and relevant statutory requirements manufacturer's specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer's specifications and relevant statutory requirements

TOPIC PA0501: USE RELEVANT TYPES OF TOOLS AND EQUIPMENT

PA0501 - Use relevant types of tools and equipment

manufacturer's specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer's specifications and relevant statutory requirements electronic components were correctly installed, soldered and the circuitry was functional internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures the wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard manufacturer's specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer's specifications and relevant statutory requirements electronic components were correctly installed, soldered and the circuitry was functional internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures

TOPIC PA0502: WIRE LOW VOLTAGE ELECTRICAL SYSTEMS, PROTECTION DEVICES AND ELECTRICAL COMPONENTS.

PA0502 - Wire low voltage electrical systems, protection devices and electrical components

manufacturer's specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer's specifications and relevant statutory requirements electronic components were correctly installed, soldered and the circuitry was functional internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables manufacturer's specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer's specifications and relevant statutory requirements electronic components were correctly installed, soldered and the circuitry was functional internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures

TOPIC PA0503: CLEAN THE WORKSITE

manufacturer's specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer's specifications and relevant statutory requirements. Electronic components were correctly installed, soldered and the circuitry was functional

PM-05-PS05: Wire electrical equipment and control systems PM-05-PS06: Terminate and connect cables and conductors 671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to inspecting (power off) and testing new and existing installations

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

IN SUMMARY

In this unit we have focussed on:

AK0501 Wiring techniques

AK0502 Interpretation of electrical drawings

AK0503 Techniques for using relevant tools

PA0501 Use relevant types of tools and equipment

PA0502 Wire low voltage electrical systems, protection devices and electrical components.

PA0503 Clean the worksite

Complete the workbook activities linked to this chapter, which are activities 1 – 6.

CHAPTER 42

PM-05-PS06: TERMINATE, JOINT AND CONNECT CABLES AND CONDUCTORS

In this Chapter we will discuss the following Topics:

AK0601 Termination and connecting techniques

AK0602 Cable and conductor jointing techniques

AK0603 Hazards associated with loose or defective connections

AK0604 Use of hand-and power tools

AK0605 Cable glanding procedures

AK0606 Safety, statutory and performance specifications

AK0607 Cable and conductor types, sizes and selection methods

PA0601 Make off and terminate cables to electrical equipment and control systems

PA0602 Join cables and conductors

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

The types and sizes of electrical cables and conductors were correctly identified and selected in accordance with the manufacturer's specifications and complies with SANS | 10142-1. | Electrical cables were correctly terminated according to safety, statutory and manufacturer's specifications | Electrical cables and conductors were correctly jointed, according to safety, statutory and manufacturer's specifications. | Electrical cables were correctly and neatly terminated according to safety, statutory requirements and comply with industry acceptable standards. | Electrical conductors were correctly connected and neatly secured in accordance with statutory requirements and industry acceptable standards. | Provider Accreditation Requirements for the Module

TOPIC AK0601: TERMINATION AND CONNECTING TECHNIQUES

PA0601 - Make off and terminate cables to electrical equipment and control systems

the wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard manufacturer's specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer's specifications

and relevant statutory requirements electronic components were correctly installed, soldered and the circuitry was functional manufacturer's specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer's specifications and relevant statutory requirements electronic components were correctly installed, soldered and the circuitry was functional

AK0601 - Termination and connecting techniques

the wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard manufacturer's specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer's specifications and relevant statutory requirements electronic components were correctly installed, soldered and the circuitry was functional manufacturer's specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer's specifications and relevant statutory requirements electronic components were correctly installed, soldered and the circuitry was functional

PA0602 - Join cables and conductors

the wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard manufacturer's specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer's specifications and relevant statutory requirements electronic components were correctly installed, soldered and the circuitry was functional there are many different types and sizes of cables in the industry, here we'll only cover a few that we may recognize manufacturer's specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer's specifications and relevant statutory requirements electronic components were correctly installed, soldered and the circuitry was functional

AK0602 - Cable and conductor jointing techniques

the wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard there are many different types and sizes of cables in the industry, here we'll only cover a few that we may recognize i have inserted a Cable to Current table that gives guidelines to the amount of Amps a cable can withstand as well as the voltage drop of the cable on Page 25 laying of wires / cables and fitting the wiring accessories in a building, according to the safety and electricity rules, keeping in view the architectural beauty of the building is called electrical wiring the purpose of electrical wiring is to provide electric power to the electric machines/appliances a wiring system uses wiring cables and different wiring accessories in order to implement a Single Phase Wiring Single Phase wiring installation is the most common wiring in residential buildings laying of wires / cables and fitting the wiring accessories in a building, according to the safety and electricity rules, keeping in view the architectural beauty of the building is called electrical wiring a wiring system uses wiring cables and different wiring accessories in order to implement a Single Phase Wiring Single Phase wiring installation is the most common wiring in residential buildings

AK0603 - Hazards associated with loose or defective connections

the wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard § The correct installation methods are selected in accordance with relevant regulations and codes, job instruction and environmental requirements § Tools, material and equipment are safely and correctly used while adhering to worksite and workshop safety procedures § Personal protective equipment are correctly selected and used according to § Power tools are selected and used correctly and safely while adhering fully to workplace safety regulations § The correct installation methods are selected in accordance with relevant regulations and codes, job instruction and environmental requirements § Tools, material and equipment are safely and correctly used while adhering to worksite and workshop safety procedures § Personal protective equipment are correctly selected and used according to § Power tools are selected and used correctly and safely while adhering fully to workplace safety regulations

AK0604 - Use of hand-and power tools

the wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures

AK0605 - Cable glanding procedures

the wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard the work site/area was left neat and tidy as per workshop procedures components were soldered according to soldering techniques pM-01-PS05: Carry out basic electric arc welding in an electrical environment Scope of Practical Skill Given work instructions, work area, documents, portable electric arc welding machines, welding cables the work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements the work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements

AK0606 - Safety, statutory and performance specifications

the wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard manufacturer’s specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer’s specifications and relevant statutory requirements electronic components were correctly installed, soldered and the circuitry was functional internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures manufacturer’s specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer’s specifications and relevant statutory requirements electronic components were correctly installed, soldered and the circuitry was functional internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures

AK0607 - Cable and conductor types, sizes and selection methods

the wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard there are many different types and sizes of cables in the industry, here we'll only cover a few that we may recognize i have inserted a Cable to Current table that gives guidelines to the amount of Amps a cable can withstand as well as the voltage drop of the cable on Page 25 laying of wires / cables and fitting the wiring accessories in a building, according to the safety and electricity rules, keeping in view the architectural beauty of the building is called electrical wiring the purpose of electrical wiring is to provide electric power to the electric machines/appliances a wiring system uses wiring cables and different wiring accessories in order to implement a Single Phase Wiring Single Phase wiring installation is the most common wiring in residential buildings laying of wires / cables and fitting the wiring accessories in a building, according to the safety and electricity rules, keeping in view the architectural beauty of the building is called electrical wiring a wiring system uses wiring cables and different wiring accessories in order to implement a Single Phase Wiring Single Phase wiring installation is the most common wiring in residential buildings

TOPIC AK0602: CABLE AND CONDUCTOR JOINTING TECHNIQUES

The wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard. The work site/area was left neat and tidy as per workshop procedures

wire joints, carry out soldering, 25. Practice on skinning, twisting electric current. and crimping. (15 Hrs) Conductors and insulators. measure insulation 26. Identify various types of cables Conducting materials and their resistance of and measure conductor size comparison. using SWG and micrometre. (8 9-10 27. Make simple twist, married, Joints in electrical conductors. wire joints, carry Tee and western union joints. Techniques of soldering. Hrs) Types of solders and flux

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC AK0603: HAZARDS ASSOCIATED WITH LOOSE OR DEFECTIVE CONNECTIONS

The wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard. The work site/area was left neat and tidy as per workshop procedures

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

and you must be sure you get the conductors for 240- finding faults in circuits is inefficient and time Volt circuits connected to the proper terminals. consuming. Loads that are connected to a panel board should be The first step to take in troubleshooting circuits is to divided as evenly as possible between the supply inspect the circuit visually. Check for loose conductors. This process of equalizing the load is connections, loose wires, abraded wires, and loose commonly referred to as load balancing

TOPIC AK0604: USE OF HAND-AND POWER TOOLS

The wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard. The work site/area was left neat and tidy as per workshop procedures

PM-05-PS05: Wire electrical equipment and control systems PM-05-PS06: Terminate and connect cables and conductors 671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to inspecting (power off) and testing new and existing installations

Occupational Certificate: Qualification: Name E-mail Phone Logo Planning and preparing work site, equipment, tools, consumables and materials Maintaining and repairing electrical equipment, control systems and installations20 671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, L3, 671101000-PM-02, Plan worksite set up for installing, wiring and connecting 671101000-PM-03, Prepare worksite set up for installing, wiring and connecting 671101000-PM-05, Install, wire and connect electrical equipment

TOPIC AK0605: CABLE GLANDING PROCEDURES

The wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard. The work site/area was left neat and tidy as per workshop procedures

applicable safety standards. The work area was cleaned and waste materials disposed of, according to local Safety equipment was cleaned and stored according to statutory requirements

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC AK0606: SAFETY, STATUTORY AND PERFORMANCE SPECIFICATIONS

The wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard. The work site/area was left neat and tidy as per workshop procedures

manufacturer's specifications and relevant statutory requirements The correct protection equipment and level of protection was installed as per job, manufacturer's specifications and relevant statutory requirements. Electronic components were correctly installed, soldered and the circuitry was functional

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

TOPIC AK0607: CABLE AND CONDUCTOR TYPES, SIZES AND SELECTION METHODS

The wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard. The work site/area was left neat and tidy as per workshop procedures

Prepare western union joint in bare conductor. Solder the finished copper conductor joints with precaution. Prepare termination of cable lugs by using crimping tool. Make straight joint in different types of underground cables. Measure insulation resistance of underground cable. Verify characteristics of Identify types of wires, cables and verify their specifications

all the cables in a conduit or trunking to the whole space enclosed by the ratio of the space occupied by all the cables in a conduit or trunking to conduit or trunking is known as the whole space enclosed by the conduit or trunking is known as the space factor. Where sizes and types of cable and trunking are not covered by the

Building structure made good after installation of trunking: (a) fi re barriers in trunking and (b) cable supports in vertical trunking

TOPIC PA0601: MAKE OFF AND TERMINATE CABLES TO ELECTRICAL EQUIPMENT AND CONTROL SYSTEMS

The wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard. The work site/area was left neat and tidy as per workshop procedures

PM-05-PS05: Wire electrical equipment and control systems PM-05-PS06: Terminate and connect cables and conductors 671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to inspecting (power off) and testing new and existing installations

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC PA0602: JOIN CABLES AND CONDUCTORS

The wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard. The work site/area was left neat and tidy as per workshop procedures

PM-05-PS05: Wire electrical equipment and control systems PM-05-PS06: Terminate and connect cables and conductors 671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to inspecting (power off) and testing new and existing installations

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

IN SUMMARY

In this unit we have focussed on:

AK0601 Termination and connecting techniques

AK0602 Cable and conductor jointing techniques

AK0603 Hazards associated with loose or defective connections

AK0604 Use of hand-and power tools

AK0605 Cable glanding procedures

AK0606 Safety, statutory and performance specifications

AK0607 Cable and conductor types, sizes and selection methods

PA0601 Make off and terminate cables to electrical equipment and control systems

PA0602 Join cables and conductors

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 43

PM-06-PS01: INSPECT ELECTRICAL EQUIPMENT, CONTROL SYSTEMS AND INSTALLATIONS UNDER POWER OFF CONDITIONS

In this Chapter we will discuss the following Topics:

AK0101 Installation specifications equipment, control systems and installations

AK0102 Relevant subsections of SANS 10142-1

AK0103 Operational procedures of equipment, control systems and installations

AK0104 Methods for the termination of wiring and cabling

PA0101 Visually inspect installed equipment, control systems and installations

PA0102 Confirm conductor and cable selection

PA0103 Inspect all connections and terminations

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Operational procedures were correctly interpreted and applied in accordance with installation specifications | Inspection report that confirms the correctness of the installation was completed and | submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. | All terminations and connections were inspected and comply with SANS 10142-1. | The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications.

TOPIC AK0101: INSTALLATION SPECIFICATIONS EQUIPMENT, CONTROL SYSTEMS AND INSTALLATIONS

PA0101 - Visually inspect installed equipment, control systems and installations

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and

document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

AK0101 - Installation specifications equipment, control systems and installations

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

PA0102 - Confirm conductor and cable selection

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment the wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard the work site/area was left neat and tidy as per workshop procedures 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

AK0102 - Relevant subsections of SANS 10142-1

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures

PA0103 - Inspect all connections and terminations

§ The correct installation methods are selected in accordance with relevant regulations and codes, job instruction and environmental requirements § Tools, material and equipment are safely and correctly used while adhering to worksite and workshop safety procedures § Personal protective equipment are correctly selected and used according to § Power tools are selected and used correctly and safely while adhering fully to workplace safety regulations submitted in accordance with the visual inspection subsection of SANS 10142-1

and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications § The correct installation methods are selected in accordance with relevant regulations and codes, job instruction and environmental requirements § Tools, material and equipment are safely and correctly used while adhering to worksite and workshop safety procedures § Personal protective equipment are correctly selected and used according to § Power tools are selected and used correctly and safely while adhering fully to workplace safety regulations submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

AK0103 - Operational procedures of equipment, control systems and installations

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

AK0104 - Methods for the termination of wiring and cabling

the wiring of electrical equipment and control systems was neatly loomed (vertically and horizontally) and complies with industry acceptable standard the work site/area was left neat and tidy as per workshop procedures § The correct installation methods are selected in accordance with relevant regulations and codes, job instruction and environmental requirements § Tools, material and equipment are safely and correctly used while adhering to worksite and workshop safety procedures § Personal protective equipment are correctly selected and used according to § Power tools are selected and used correctly and safely while adhering fully to workplace safety regulations § The correct installation methods are selected in accordance with relevant regulations and codes, job instruction and environmental requirements § Tools, material and equipment are safely and correctly used while adhering to worksite and workshop safety procedures § Personal protective equipment are correctly selected and used according to § Power tools are selected and used correctly and safely while adhering fully to workplace safety regulations

TOPIC AK0102: RELEVANT SUBSECTIONS OF SANS 10142-1

671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

TOPIC AK0103: OPERATIONAL PROCEDURES OF EQUIPMENT, CONTROL SYSTEMS AND INSTALLATIONS

671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

671101000-PM-03, Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems, L3, Cr30 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC AK0104: METHODS FOR THE TERMINATION OF WIRING AND CABLING

671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and

PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

§ The correct installation methods are selected in accordance with relevant regulations and codes, job instruction and environmental requirements. § Tools, material and equipment are safely and correctly used while adhering to worksite and workshop safety procedures. § Personal protective equipment are correctly selected and used according to § Power tools are selected and used correctly and safely while adhering fully to workplace safety regulations

TOPIC PA0101: VISUALLY INSPECT INSTALLED EQUIPMENT, CONTROL SYSTEMS AND INSTALLATIONS

671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

TOPIC PA0102: CONFIRM CONDUCTOR AND CABLE SELECTION

671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-05, Install, wire and connect electrical equipment and control systems, L3, Cr38 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC PA0103: INSPECT ALL CONNECTIONS AND TERMINATIONS

671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

§ The correct installation methods are selected in accordance with relevant regulations and codes, job instruction and environmental requirements. § Tools, material and equipment are safely and correctly used while adhering to worksite and workshop safety procedures. § Personal protective equipment are correctly selected and used according to § Power tools are selected and used correctly and safely while adhering fully to workplace safety regulations

IN SUMMARY

In this unit we have focussed on:

AK0101 Installation specifications equipment, control systems and installations

AK0102 Relevant subsections of SANS 10142-1

AK0103 Operational procedures of equipment, control systems and installations

AK0104 Methods for the termination of wiring and cabling

PA0101 Visually inspect installed equipment, control systems and installations

PA0102 Confirm conductor and cable selection

PA0103 Inspect all connections and terminations

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 44

PM-06-PS02: TEST ELECTRICAL EQUIPMENT, CONTROL SYSTEMS AND INSTALLATIONS UNDER POWER OFF CONDITIONS

In this Chapter we will discuss the following Topics:

AK0201 Relevant subsections of SANS 10142-1

AK0202 Testing techniques for electrical components and protective devices

AK0203 Test parameters for protective devices and electrical components

AK0204 Techniques for the safe use of electrical testing equipment

PA0201 Interpret task instructions

PA0202 Select and use of the correct test instruments

PA0203 Select and use of relevant personal protective equipment

PA0204 Isolate and lock-out of identified equipment.

PA0206 Test electrical equipment, control systems and installations

PA0207 Interpret, record and report test results

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Work Instructions and testing procedures were correctly interpreted and applied. | Test results were correctly interpreted and recorded in accordance with relevant subsections of SANS 10142-1 and SANS 10242. | A Test report that confirms the correctness of the installation was completed in accordance with the testing subsections of SANS 10142-1, SANS10242 and relevant workshop practices. | Personal protective equipment was selected and correctly used in accordance with the | relevant subsection of the OHS Act and the Mines Health and Safety Act. | The selected equipment was isolated and locked out. | Provider Accreditation Requirements for the Module

TOPIC AK0201: RELEVANT SUBSECTIONS OF SANS 10142-1

PA0201 - Interpret task instructions

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's

functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

AK0201 - Relevant subsections of SANS 10142-1

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications 671101000-PM-04, Install wire-ways, L4, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

PA0202 - Select and use of the correct test instruments

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

AK0202 - Testing techniques for electrical components and protective devices

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures

PA0203 - Select and use of relevant personal protective equipment

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures

AK0203 - Test parameters for protective devices and electrical components

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures

PA0204 - Isolate and lock-out of identified equipment

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's

functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified

AK0204 - Techniques for the safe use of electrical testing equipment

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures

PA0206 - Test electrical equipment, control systems and installations

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications the you will be practicing skills related to inspecting (power off) and testing new and existing installations submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's

functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications the you will be practicing skills related to inspecting (power off) and testing new and existing installations

PA0207 - Interpret, record and report test results

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

TOPIC AK0202: TESTING TECHNIQUES FOR ELECTRICAL COMPONENTS AND PROTECTIVE DEVICES

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

TOPIC AK0203: TEST PARAMETERS FOR PROTECTIVE DEVICES AND ELECTRICAL COMPONENTS

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

TOPIC AK0204: TECHNIQUES FOR THE SAFE USE OF ELECTRICAL TESTING EQUIPMENT

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

TOPIC PA0201: INTERPRET TASK INSTRUCTIONS

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

Approved work permit or other documentation is completed to reflect risk assessment conducted in accordance with approval procedures PM-02-PS02: Read and interpret electrical drawings Scope of Practical Skill Given work instructions, work area, case study, documents, templates, you should be able to: PA0201 Identify signs

and symbols PA0202 Read and interpret electrical drawings AK0201 Methods of interpreting scales AK0202 Techniques of interpreting diagrams AK0203 Procedures for evaluating the work site

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

TOPIC PA0202: SELECT AND USE OF THE CORRECT TEST INSTRUMENTS

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

TOPIC PA0203: SELECT AND USE OF RELEVANT PERSONAL PROTECTIVE EQUIPMENT

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

Scribers, dividers and centre punches were sharpened to ensure serviceability. PM-01-PS02: Select, use and care for power tools Scope of Practical Skill Given work instructions, work area, documents, templates, drilling machines, grinders, and saws standard operating procedures and, statutory requirements, you should be PA0201 Select a power tool pertaining to specific job requirements. PA0202 Use fixed power tools. PA0203 Use portable power tools

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified

through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

TOPIC PA0204: ISOLATE AND LOCK-OUT OF IDENTIFIED EQUIPMENT.

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

Scribers, dividers and centre punches were sharpened to ensure serviceability. PM-01-PS02: Select, use and care for power tools Scope of Practical Skill Given work instructions, work area, documents, templates, drilling machines, grinders, and saws standard operating procedures and, statutory requirements, you should be PA0201 Select a power tool pertaining to specific job requirements. PA0202 Use fixed power tools. PA0203 Use portable power tools

TOPIC PA0206: TEST ELECTRICAL EQUIPMENT, CONTROL SYSTEMS AND INSTALLATIONS

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

PM-05-PS05: Wire electrical equipment and control systems PM-05-PS06: Terminate and connect cables and conductors 671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to inspecting (power off) and testing new and existing installations

671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC PA0207: INTERPRET, RECORD AND REPORT TEST RESULTS

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment. WA0202 Inspect protection devises used in electrical equipment and settings verified. WA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations. WA0206 Complete and record electrical tests

WA0209 document findings and test results SE0201 Completed observation checklist SE0202 Completed section in logbook SE0203 Completed electrical test report WM-03- Undertake all activities without assistance, but Date Signature WE03 under supervision, in the processes of testing and inspecting of electrical equipment, control systems WA0201 Complete a visual inspection of electrical WA0202 Inspect protection devises used in electrical equipment and settings verified

IN SUMMARY

In this unit we have focussed on:

AK0201 Relevant subsections of SANS 10142-1

AK0202 Testing techniques for electrical components and protective devices

AK0203 Test parameters for protective devices and electrical components

AK0204 Techniques for the safe use of electrical testing equipment

PA0201 Interpret task instructions

PA0202 Select and use of the correct test instruments

PA0203 Select and use of relevant personal protective equipment

PA0204 Isolate and lock-out of identified equipment.

PA0206 Test electrical equipment, control systems and installations

PA0207 Interpret, record and report test results

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 45

PM-07-PS01: CARRY OUT COMMISSIONING TESTS (WITH POWER ON) ON ELECTRICAL INSTALLATION. AND CONTROL SYSTEMS

In this Chapter we will discuss the following Topics:

AK0101 Techniques to carry out commissioning tests

AK0102 Fault finding techniques

AK0103 Isolating and energizing processes

AK0104 Risks associated with live testing

AK0105 Correct use of personal protective equipment

PA0101 Carry out risk assessment

PA0102 Use the relevant tools and testing equipment

PA0103 Carry out relevant commissioning tests

PA0104 Interpret commissioning tests results

PA0105 Report commissioning results

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Work Instructions and testing procedures were correctly interpreted and applied. | Risks were identified and mitigated in accordance with risk management procedures | A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements. | Isolating and energizing processes were correctly followed | Tools and testing equipment including the relevant hand tools, mustimeters, insulation tester and tong tester is correctly used | The relevant commissioning tests are correctly completed according to relevant regulations

TOPIC AK0101: TECHNIQUES TO CARRY OUT COMMISIONING TESTS

PA0101 - Carry out risk assessment

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under

the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

PA0102 - Use the relevant tools and testing equipment

subsections of SANS 10142-1 and SANS 10242 the selected equipment was isolated and locked out

PA0103 - Carry out relevant commissioning tests

the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

AK0103 - Isolating and energizing processes

risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

PA0104 - Interpret commissioning tests results

risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures

PA0105 - Report commissioning results

the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a

visual inspection of installations the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

AK0105 - Correct use of personal protective equipment

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices

TOPIC AK0102: FAULT FINDING TECHNIQUES

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

§ Measurements and observations are interpreted and captured accurately according to manufacturers specifications § All the relevant units/variables are recorded accurately § Hand-over documents are completed as per hand-over procedures § Demonstrate knowledge and understanding of commissioning control systems The ability to maintain and repair electrical equipment, control systems and installations Associated Assessment Criteria § Correct work instructions are followed and correct documentation including checklist

TOPIC AK0103: ISOLATING AND ENERGIZING PROCESSES

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

TOPIC AK0104: RISKS ASSOCIATED WITH LIVE TESTING

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

§ Measurements and observations are interpreted and captured accurately according to manufacturers specifications § All the relevant units/variables are recorded accurately § Hand-over documents are completed as per hand-over procedures § Demonstrate knowledge and understanding of commissioning control systems The ability to maintain and repair electrical equipment, control systems and installations Associated Assessment Criteria § Correct work instructions are followed and correct documentation including checklist

TOPIC AK0105: CORRECT USE OF PERSONAL PROTECTIVE EQUIPMENT

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted

PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

TOPIC PA0101: CARRY OUT RISK ASSESSMENT

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

TOPIC PA0102: USE THE RELEVANT TOOLS AND TESTING EQUIPMENT

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

TOPIC PA0103: CARRY OUT RELEVANT COMMISSIONING TESTS

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

manufacturer’s specifications. The correct electrical testing and fault finding procedures were applied. Defects were rectified as per installation requirements. Repairs were recorded and reported as per reporting procedures

TOPIC PA0104: INTERPRET COMMISSIONING TESTS RESULTS

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

manufacturer’s specifications. The correct electrical testing and fault finding procedures were applied. Defects were rectified as per installation requirements. Repairs were recorded and reported as per reporting procedures

TOPIC PA0105: REPORT COMMISSIONING RESULTS

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

manufacturer's specifications. The correct electrical testing and fault finding procedures were applied. Defects were rectified as per installation requirements. Repairs were recorded and reported as per reporting procedures

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

IN SUMMARY

In this unit we have focussed on:

AK0101 Techniques to carry out commissioning tests

AK0102 Fault finding techniques

AK0103 Isolating and energizing processes

AK0104 Risks associated with live testing

AK0105 Correct use of personal protective equipment

PA0101 Carry out risk assessment

PA0102 Use the relevant tools and testing equipment

PA0103 Carry out relevant commissioning tests

PA0104 Interpret commissioning tests results

PA0105 Report commissioning results

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 46

PM-07-PS02: RECTIFY DEFECTS FOUND ON ELECTRICAL INSTALLATION. AND CONTROL SYSTEMS

In this Chapter we will discuss the following Topics:

AK0201 Isolating procedures

AK0202 Reading and interpretation of electrical drawings and operating procedure of electrical installation. Or control system

AK0203 Fault finding procedures and techniques for electrical installation.or control system

AK0204 Electrical testing equipment and testing procedures

AK0205 Relevant hand tools required for repairs

AK0206 Procedures for repairing and replacing electrical installation.or control system components

PA0201 Isolate the defective electrical installation.or control system

PA0202 Interpret the applicable operating procedure

PA0203 Trace and identify the defect on electrical installation.or control system

PA0204 Repair defects on electrical installation.or control system

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Electrical installation or control system was correctly and safely isolated according to isolating procedures | Operating procedures and electrical diagrams were correctly interpreted according to manufacturer's specifications. | The correct electrical testing and fault finding procedures were applied. | Defects were rectified as per installation requirements. | Repairs were recorded and reported as per reporting procedures.

TOPIC AK0201: ISOLATING PROCEDURES

PA0201 - Isolate the defective electrical installation.or control system

risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications risks were identified and mitigated in accordance with risk management procedures A test report

that confirms the correctness of the installation was completed in accordance with relevant statutory requirements the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

AK0201 - Isolating procedures

risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

PA0202 - Interpret the applicable operating procedure

risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

AK0202 - Reading and interpretation of electrical drawings and operating procedure of electrical installation.or control system

risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

PA0203 - Trace and identify the defect on electrical installation.or control system

risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

PA0204 - Repair defects on electrical installation.or control system

risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements the you will be practicing skills related to inspecting (power off) and testing new and existing installations risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements the you will be practicing skills related to inspecting (power off) and testing new and existing installations

AK0204 - Electrical testing equipment and testing procedures

risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

AK0205 - Relevant hand tools required for repairs

risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements the correct electrical testing and fault finding procedures were applied defects were rectified as per installation requirements repairs were recorded and reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements the correct electrical testing and fault finding procedures were applied

DEFECTS WERE RECTIFIED AS PER INSTALLATION REQUIREMENTS

repairs were recorded and reported as per reporting procedures

TOPIC AK0202: READING AND INTERPRETATION OF ELECTRICAL DRAWINGS AND OPERATING PROCEDURE OF ELECTRICAL INSTALLATION.OR CONTROL SYSTEM

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

TOPIC AK0203: FAULT FINDING PROCEDURES AND TECHNIQUES FOR ELECTRICAL INSTALLATION.OR CONTROL SYSTEM

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and

PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

TOPIC AK0204: ELECTRICAL TESTING EQUIPMENT AND TESTING PROCEDURES

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

TOPIC AK0205: RELEVANT HAND TOOLS REQUIRED FOR REPAIRS

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

PM-05-PS05: Wire electrical equipment and control systems PM-05-PS06: Terminate and connect cables and conductors 671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to:

Practice electrical skills within a simulated work environment. The you will be practicing skills related to inspecting (power off) and testing new and existing installations

TOPIC AK0206: PROCEDURES FOR REPAIRING AND REPLACING ELECTRICAL INSTALLATION.OR CONTROL SYSTEM COMPONENTS

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

TOPIC PA0201: ISOLATE THE DEFECTIVE ELECTRICAL INSTALLATION.OR CONTROL SYSTEM

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation’s functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

TOPIC PA0202: INTERPRET THE APPLICABLE OPERATING PROCEDURE

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

TOPIC PA0203: TRACE AND IDENTIFY THE DEFECT ON ELECTRICAL INSTALLATION.OR CONTROL SYSTEM

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices. All terminations and connections were inspected and comply with SANS 10142-1. The installation’s functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

TOPIC PA0204: REPAIR DEFECTS ON ELECTRICAL INSTALLATION.OR CONTROL SYSTEM

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

PM-05-PS05: Wire electrical equipment and control systems PM-05-PS06: Terminate and connect cables and conductors 671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to inspecting (power off) and testing new and existing installations

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

IN SUMMARY

In this unit we have focussed on:

AK0201 Isolating procedures

AK0202 Reading and interpretation of electrical drawings and operating procedure of electrical installation.or control system

AK0203 Fault finding procedures and techniques for electrical installation.or control system

AK0204 Electrical testing equipment and testing procedures

AK0205 Relevant hand tools required for repairs

AK0206 Procedures for repairing and replacing electrical installation.or control system components

PA0201 Isolate the defective electrical installation.or control system

PA0202 Interpret the applicable operating procedure

PA0203 Trace and identify the defect on electrical installation.or control system

PA0204 Repair defects on electrical installation.or control system

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 47

PM-07-PS03: RECORD AND REPORT COMMISSIONING RESULTS AND HAND OVER THE INSTALLATIONS AND CONTROL SYSTEMS

In this Chapter we will discuss the following Topics:

AK0301 Relevant statutory recording requirements

AK0302 Data capturing and information procedure

AK0303 Hand-over procedure

PA0301 Record and report commissioning test results of installations and control systems

PA0302 Complete hand-over documentation

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Measurements and observations were interpreted and captured accurately according to manufacturers specifications | All the relevant units/variables were recorded accurately | Hand-over documents were completed as per hand-over procedures | Provider Accreditation Requirements for the Module

TOPIC AK0301: RELEVANT STATUTORY RECORDING REQUIREMENTS

PA0301 - Record and report commissioning test results of installations and control systems

the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements § The correct installation methods are selected in accordance with relevant regulations and codes, job instruction and environmental requirements § Tools, material and equipment are safely and correctly used while adhering to worksite and workshop safety procedures § Personal protective equipment are correctly selected and used according to § Power tools are selected and used correctly and safely while adhering fully to workplace safety regulations the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements § The correct installation methods are selected in accordance with relevant regulations and codes, job instruction and environmental requirements § Tools, material and equipment are safely and correctly used while adhering to worksite and workshop safety procedures § Personal

protective equipment are correctly selected and used according to § Power tools are selected and used correctly and safely while adhering fully to workplace safety regulations

AK0301 - Relevant statutory recording requirements

the correct electrical testing and fault finding procedures were applied defects were rectified as per installation requirements repairs were recorded and reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements the correct electrical testing and fault finding procedures were applied

---

DEFECTS WERE RECTIFIED AS PER INSTALLATION REQUIREMENTS

repairs were recorded and reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

PA0302 - Complete hand-over documentation

the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

AK0302 - Data capturing and information procedure

the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

AK0303 - Hand-over procedure

the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements the correct electrical testing and fault finding procedures were applied repairs were recorded and

reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

TOPIC AK0302: DATA CAPTURING AND INFORMATION PROCEDURE

manufacturer's specifications. The correct electrical testing and fault finding procedures were applied. Defects were rectified as per installation requirements. Repairs were recorded and reported as per reporting procedures

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rotating electrical machinery - AC, Rotating Electrical Machinery - DC, Rotating electrical machinery - Alternators and Generators, Protection devices, luminaries, current transformers, potential transformers and shunts according to the design specifications

TOPIC AK0303: HAND-OVER PROCEDURE

manufacturer's specifications. The correct electrical testing and fault finding procedures were applied. Defects were rectified as per installation requirements. Repairs were recorded and reported as per reporting procedures

Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rotating electrical machinery - AC, Rotating Electrical Machinery - DC, Rotating electrical machinery - Alternators and Generators, Protection devices, luminaries, current transformers, potential transformers and shunts according to the design specifications

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

TOPIC PA0301: RECORD AND REPORT COMMISSIONING TEST RESULTS OF INSTALLATIONS AND CONTROL SYSTEMS

manufacturer's specifications. The correct electrical testing and fault finding procedures were applied. Defects were rectified as per installation requirements. Repairs were recorded and reported as per reporting procedures

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment. WA0202 Inspect protection devises used in electrical equipment and settings verified. WA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations. WA0206 Complete and record electrical tests

WA0209 document findings and test results SE0201 Completed observation checklist SE0202 Completed section in logbook SE0203 Completed electrical test report WM-03- Undertake all activities without assistance, but Date Signature WE03 under supervision, in the processes of testing and inspecting of electrical equipment, control systems WA0201 Complete a visual inspection of electrical WA0202 Inspect protection devises used in electrical equipment and settings verified

TOPIC PA0302: COMPLETE HAND-OVER DOCUMENTATION

manufacturer's specifications. The correct electrical testing and fault finding procedures were applied. Defects were rectified as per installation requirements. Repairs were recorded and reported as per reporting procedures

Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rotating electrical machinery - AC, Rotating Electrical Machinery - DC, Rotating electrical machinery - Alternators and Generators, Protection devices, luminaries, current transformers, potential transformers and shunts according to the design specifications

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

IN SUMMARY

In this unit we have focussed on:

AK0301 Relevant statutory recording requirements

AK0302 Data capturing and information procedure

AK0303 Hand-over procedure

PA0301 Record and report commissioning test results of installations and control systems

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 48

PM-08-PS01: OBTAIN WORK INSTRUCTIONS AND GATHER DRAWINGS AND DOCUMENTATION

In this Chapter we will discuss the following Topics:

AK0101 Plan and prepare procedures

AK0102 Risk assessment procedure

AK0103 Lock out procedures

AK0104 Relevant SANS 10142-1 requirements

AK0105 Safety procedures

PA0101 Interpret work instructions

PA0102 Gather documentation (work permits, drawings, work file)

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Correct work instructions were followed in accordance with relevant workshop procedures | Correct documentation including checklists were obtained

TOPIC AK0101: PLAN AND PREPARE PROCEDURES

PA0101 - Interpret work instructions

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

AK0101 - Plan and prepare procedures

test and inspect electrical equipment, control systems and installations

AK0102 - Risk assessment procedure

pA0103 Relate the risk to the hazard risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with

relevant statutory requirements risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

AK0104 - Relevant SANS 10142-1 requirements

submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications submitted in accordance with the visual inspection subsection of SANS 10142-1 and relevant workshop practices all terminations and connections were inspected and comply with SANS 10142-1 the installation's functionality was inspected and complies with SANS 10142-1 and/or the relevant manufacturers specifications

AK0105 - Safety procedures

pA0103 Relate the risk to the hazard risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

TOPIC AK0102: RISK ASSESSMENT PROCEDURE

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

PM-02-PS01: Undertake a preliminary risk assessment Scope of Practical Skill Given work instructions, checklists, work area, case study, drawings, documents, templates, forms, safety and quality principles, standard operating procedures and, statutory requirements, you should be able to: PA0101 Conduct a preliminary risk assessment. PA0102 Identify the hazards. PA0103 Relate the risk to the hazard. PA0104 Conduct mitigation of the risk

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

TOPIC AK0103: LOCK OUT PROCEDURES

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

Equipment were procured to meet the specific needs of the activity and are fit for purpose Material requirements were procured to meet the specific needs of the activity and are fit Consumable requirements were procured to meet the specific needs of the activity and are fit for purpose PM-03-PS02: Prepare and move resources Scope of Practical Skill Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, you should be able to: PA0201

SE0301 Obtained and signed requisition/order documentation as verified by a signature from a qualified person SE0302 Signed job card as verified by a signature from a qualified person SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a Contextualised Workplace Knowledge Company specific policies and procedures (LV operating procedures, l

TOPIC AK0104: RELEVANT SANS 10142-1 REQUIREMENTS

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment

TOPIC AK0105: SAFETY PROCEDURES

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

PM-02-PS01: Undertake a preliminary risk assessment Scope of Practical Skill Given work instructions, checklists, work area, case study, drawings, documents, templates, forms, safety and quality principles, standard operating procedures and, statutory requirements, you should be able to: PA0101 Conduct a preliminary risk assessment. PA0102 Identify the hazards. PA0103 Relate the risk to the hazard. PA0104 Conduct mitigation of the risk

AK0104 Identification of hazardous substances AK0105 Electrical safety procedures for working in close proximity to live equipment Internal Assessment Criteria Risk assessment techniques were used to identify risks within the installation in accordance with workshop practices. Risks were mitigated and or rectified through the implementation of the correct risk A risk assessment report was completed that included health and safety considerations in accordance with risk management procedures

TOPIC PA0101: INTERPRET WORK INSTRUCTIONS

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

skills related to fault find and repair electrical control systems and electrical installation. you will be required to: PM-08-PS01: Obtain work instructions and gather drawings and documentation PM-08-PS02: Select tools and testing instruments PM-08-PS03: Carry out fault finding PM-08-PS04: Rectify faults where applicable PM-08-PS05: Record findings and provide feedback Details of Work Experience Modules 671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

TOPIC PA0102: GATHER DOCUMENTATION (WORK PERMITS, DRAWINGS, WORK FILE)

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

skills related to fault find and repair electrical control systems and electrical installation. you will be required to: PM-08-PS01: Obtain work instructions and gather drawings and documentation PM-08-PS02: Select tools and testing instruments PM-08-PS03: Carry out fault finding PM-08-PS04: Rectify faults where applicable PM-08-PS05: Record findings and provide feedback Details of Work Experience Modules 671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.

671101000-PM-01, 671101000-PM-01, Use hand, power and measuring tools, Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to preparing and using of hand, power and measuring tools

IN SUMMARY

In this unit we have focussed on:

AK0101 Plan and prepare procedures

AK0102 Risk assessment procedure

AK0103 Lock out procedures

AK0104 Relevant SANS 10142-1 requirements

AK0105 Safety procedures

PA0101 Interpret work instructions

PA0102 Gather documentation (work permits, drawings, work file)

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 49

PM-08-PS02: SELECT TOOLS AND TESTING INSTRUMENTS

In this Chapter we will discuss the following Topics:

AK0201 Hand tools and their uses

AK0202 Power tools and their uses

AK0203 Testing instruments and their uses

AK0204 Safety and standard operating procedures.

PA0201 Select hand and power tools

PA0202 Select testing instruments

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Correct hand tools were selected | Correct power were selected | Correct testing equipment was selected

TOPIC AK0201: HAND TOOLS AND THEIR USES

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

skills related to fault find and repair electrical control systems and electrical installation. you will be required to: PM-08-PS01: Obtain work instructions and gather drawings and documentation PM-08-PS02: Select tools and testing instruments PM-08-PS03: Carry out fault finding PM-08-PS04: Rectify faults where applicable PM-08-PS05: Record findings and provide feedback Details of Work Experience Modules 671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

TOPIC AK0202: POWER TOOLS AND THEIR USES

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

Scribers, dividers and centre punches were sharpened to ensure serviceability. PM-01-PS02: Select, use and care for power tools Scope of Practical Skill Given work instructions, work area, documents, templates, drilling machines, grinders, and saws standard operating procedures and, statutory requirements, you should be PA0201 Select a power tool pertaining to specific job requirements. PA0202 Use fixed power tools. PA0203 Use portable power tools

skills related to fault find and repair electrical control systems and electrical installation. you will be required to: PM-08-PS01: Obtain work instructions and gather drawings and documentation PM-08-PS02: Select tools and testing instruments PM-08-PS03: Carry out fault finding PM-08-PS04: Rectify faults where applicable PM-08-PS05: Record findings and provide feedback Details of Work Experience Modules 671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.

TOPIC AK0203: TESTING INSTRUMENTS AND THEIR USES

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

skills related to fault find and repair electrical control systems and electrical installation. you will be required to: PM-08-PS01: Obtain work instructions and gather drawings and documentation PM-08-PS02: Select tools and testing instruments PM-08-PS03: Carry out fault finding PM-08-PS04: Rectify faults where applicable PM-08-PS05: Record findings and provide feedback Details of Work Experience Modules 671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

TOPIC AK0204: SAFETY AND STANDARD OPERATING PROCEDURES.

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

TOPIC PA0201: SELECT HAND AND POWER TOOLS

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

skills related to fault find and repair electrical control systems and electrical installation. you will be required to: PM-08-PS01: Obtain work instructions and gather drawings and documentation PM-08-PS02: Select tools and testing instruments PM-08-PS03: Carry out fault finding PM-08-PS04: Rectify faults where applicable PM-08-PS05: Record findings and provide feedback Details of Work Experience Modules 671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22

Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

TOPIC PA0202: SELECT TESTING INSTRUMENTS

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

skills related to fault find and repair electrical control systems and electrical installation. you will be required to: PM-08-PS01: Obtain work instructions and gather drawings and documentation PM-08-PS02: Select tools and testing instruments PM-08-PS03: Carry out fault finding PM-08-PS04: Rectify faults where applicable PM-08-PS05: Record findings and provide feedback Details of Work Experience Modules 671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

IN SUMMARY

In this unit we have focussed on:

AK0201 Hand tools and their uses

AK0202 Power tools and their uses

AK0203 Testing instruments and their uses

AK0204 Safety and standard operating procedures.

PA0201 Select hand and power tools

PA0202 Select testing instruments

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 50

PM-08-PS03: CARRY OUT FAULTFINDING

In this Chapter we will discuss the following Topics:

AK0301 Fault finding procedures

AK0302 Fault finding techniques

AK0303 Use of testing instruments

AK0304 Interpretation of electrical schematics and drawings

AK0305 Functionality of components

PA0301 Carry out tests that include but not limited to: visual inspections, functional, continuity, insulation resistance and polarity tests

PA0302 Conduct fault finding

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Fault finding procedures were correctly followed | Fault finding techniques were correctly applied | Electrical schematics and drawings were correctly interpreted | Test instruments were correctly and safely used | Safety and standard operating procedures were correctly followed | Functionality of components inspected in accordance with manufacturers specifications | Faults were correctly identified | Causes were correctly interpreted

TOPIC AK0301: FAULT FINDING PROCEDURES

PA0301 - Carry out tests that include but not limited to: visual inspections, functional, continuity, insulation resistance and polarity tests

the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures

AK0301 - Fault finding procedures

the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory

requirements the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

PA0302 - Conduct fault finding

the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures the correct electrical testing and fault finding procedures were applied repairs were recorded and reported as per reporting procedures

TOPIC AK0302: FAULT FINDING TECHNIQUES

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

TOPIC AK0303: USE OF TESTING INSTRUMENTS

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rotating electrical machinery - AC, Rotating Electrical Machinery - DC, Rotating electrical machinery - Alternators and Generators, Protection devices, luminaries, current transformers, potential transformers and shunts according to the design specifications

TOPIC AK0304: INTERPRETATION OF ELECTRICAL SCHEMATICS AND DRAWINGS

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

Equipment were procured to meet the specific needs of the activity and are fit for purpose Material requirements were procured to meet the specific needs of the activity and are fit Consumable requirements were procured to meet the specific needs of the activity and are fit for purpose PM-03-PS02: Prepare and move resources Scope of Practical Skill Given work instructions, checklists, work area, activity documents, safety and quality principles and standard operating procedures, you should be able to: PA0201

TOPIC AK0305: FUNCTIONALITY OF COMPONENTS

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses

AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

§ Measurements and observations are interpreted and captured accurately according to manufacturers specifications § All the relevant units/variables are recorded accurately § Hand-over documents are completed as per hand-over procedures § Demonstrate knowledge and understanding of commissioning control systems The ability to maintain and repair electrical equipment, control systems and installations Associated Assessment Criteria § Correct work instructions are followed and correct documentation including checklist

TOPIC PA0301: CARRY OUT TESTS THAT INCLUDE BUT NOT LIMITED TO: VISUAL INSPECTIONS, FUNCTIONAL, CONTINUITY, INSULATION RESISTANCE AND POLARITY TESTS

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

manufacturer’s specifications. The correct electrical testing and fault finding procedures were applied. Defects were rectified as per installation requirements. Repairs were recorded and reported as per reporting procedures

TOPIC PA0302: CODUCT FAULT FINDING

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must:

PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

manufacturer's specifications. The correct electrical testing and fault finding procedures were applied. Defects were rectified as per installation requirements. Repairs were recorded and reported as per reporting procedures

skills related to fault find and repair electrical control systems and electrical installation. you will be required to: PM-08-PS01: Obtain work instructions and gather drawings and documentation PM-08-PS02: Select tools and testing instruments PM-08-PS03: Carry out fault finding PM-08-PS04: Rectify faults where applicable PM-08-PS05: Record findings and provide feedback Details of Work Experience Modules 671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.

IN SUMMARY

In this unit we have focussed on:

AK0301 Fault finding procedures

AK0302 Fault finding techniques

AK0303 Use of testing instruments

AK0304 Interpretation of electrical schematics and drawings

AK0305 Functionality of components

PA0301 Carry out tests that include but not limited to: visual inspections, functional, continuity, insulation resistance and polarity tests

PA0302 Conduct fault finding

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 51

PM-08-PS04: RECTIFY FAULTS AND RE-COMMISSION

In this Chapter we will discuss the following Topics:

AK0401 Functionality of components

AK04010 Risks associated with live testing

AK04011 Correct use of personal protective equipment

AK04012 Knowledge of recording procedures and documentation

AK04013 Feedback procedures

AK0402 Various fault conditions

AK0403 Various control systems

AK0405 Use of testing instruments

AK0406 Interpretation of electrical schematics and drawings

AK0407 Techniques to carry out commissioning tests

AK0408 Fault finding techniques

AK0409 Isolating and energizing processes

PA0401 Rectify the faults identified

PA0402 Complete a re-commissioning test

PA0403 Report findings and provide feedback

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

Electrical installation or control system was correctly and safely isolated according to isolating procedures | Operating procedures and electrical diagrams were correctly interpreted according to manufacturer's specifications. | Identified faults were correctly rectified to the original operating status | Risks associated with live testing were correctly identified and safety precautions taken. | The correct electrical testing procedures were applied. | Re-commissioning tests were done and complied with standard operating procedures | Documentation was correctly completed according to recording procedures | Feedback was correctly given

within the prescribed time frame | Causes were correctly interpreted | Provider Accreditation Requirements for the Module

TOPIC AK0401: FUNCTIONALITY OF COMPONENTS

PA0401 - Rectify the faults identified

risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

PA0402 - Complete a re-commissioning test

the correct electrical testing and fault finding procedures were applied defects were rectified as per installation requirements repairs were recorded and reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements the correct electrical testing and fault finding procedures were applied

DEFECTS WERE RECTIFIED AS PER INSTALLATION REQUIREMENTS

repairs were recorded and reported as per reporting procedures risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

AK0402 - Various fault conditions

readings were recorded as per workshop procedures portable electrical equipment was checked for correct operation and functionality measuring instruments were correctly set up for application results were accurately interpreted against the specifications of the job requirements

READINGS WERE RECORDED AS PER WORKSHOP PROCEDURES

portable electrical equipment was checked for correct operation and functionality

AK0403 - Various control systems

readings were recorded as per workshop procedures portable electrical equipment was checked for correct operation and functionality measuring instruments were correctly set up for application results were accurately interpreted against the specifications of the job requirements

READINGS WERE RECORDED AS PER WORKSHOP PROCEDURES

portable electrical equipment was checked for correct operation and functionality

AK0405 - Use of testing instruments

internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures

AK0406 - Interpretation of electrical schematics and drawings

risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

AK0409 - Isolating and energizing processes

risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

AK04011 - Correct use of personal protective equipment

readings were recorded as per workshop procedures portable electrical equipment was checked for correct operation and functionality measuring instruments were correctly set up for application results were accurately interpreted against the specifications of the job requirements

---

READINGS WERE RECORDED AS PER WORKSHOP PROCEDURES

portable electrical equipment was checked for correct operation and functionality

AK04012 - Knowledge of recording procedures and documentation

the correct electrical testing and fault finding procedures were applied defects were rectified as per installation requirements repairs were recorded and reported as per reporting procedures the correct electrical testing and fault finding procedures were applied

---

DEFECTS WERE RECTIFIED AS PER INSTALLATION REQUIREMENTS

repairs were recorded and reported as per reporting procedures

TOPIC AK04010: RISKS ASSOCIATED WITH LIVE TESTING

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and

PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

§ Measurements and observations are interpreted and captured accurately according to manufacturers specifications § All the relevant units/variables are recorded accurately § Hand-over documents are completed as per hand-over procedures § Demonstrate knowledge and understanding of commissioning control systems The ability to maintain and repair electrical equipment, control systems and installations Associated Assessment Criteria § Correct work instructions are followed and correct documentation including checklist

TOPIC AK04011: CORRECT USE OF PERSONAL PROTECTIVE EQUIPMENT

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

readings were recorded as per workshop procedures. Portable electrical equipment was checked for correct operation and functionality. Measuring instruments were correctly set up for application. Results were recorded on appropriate documentation. Results were accurately interpreted against the specifications of the job requirements. Electrical measuring instruments were used in accordance with their specifications

TOPIC AK04012: KNOWLEDGE OF RECORDING PROCEDURES AND DOCUMENTATION

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

§ Measurements and observations are interpreted and captured accurately according to manufacturers specifications § All the relevant units/variables are recorded accurately § Hand-over documents are completed as per hand-over procedures § Demonstrate knowledge and understanding of commissioning control systems The ability to maintain and repair electrical equipment, control systems and installations Associated Assessment Criteria § Correct work instructions are followed and correct documentation including checklist

TOPIC AK04013: FEEDBACK PROCEDURES

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

skills related to fault find and repair electrical control systems and electrical installation. you will be required to: PM-08-PS01: Obtain work instructions and gather drawings and documentation PM-08-PS02: Select tools and testing instruments PM-08-PS03: Carry out fault finding PM-08-PS04: Rectify faults where applicable PM-08-PS05: Record findings and provide feedback Details of Work Experience Modules 671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.

TOPIC AK0402: VARIOUS FAULT CONDITIONS

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the

fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

readings were recorded as per workshop procedures. Portable electrical equipment was checked for correct operation and functionality. Measuring instruments were correctly set up for application. Results were recorded on appropriate documentation. Results were accurately interpreted against the specifications of the job requirements. Electrical measuring instruments were used in accordance with their specifications

Internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures. Portable power tools were correctly used, as per workshop practices and safety The correct personal protective equipment was used and complied with the workshop PM-05-PS04: Install equipment and control systems Scope of Practical Skill Given equipment and control systems Includes: Batteries, Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rot

TOPIC AK0403: VARIOUS CONTROL SYSTEMS

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

Regulations and statutory requirements for wiring of premises Electrical diagrams and symbols Electrical components and their applications Internal Assessment Criteria and Weight List, identify and explain the meaning of all standard International Electrotechnical Commission (IEC) wiring symbols given on work drawings Identify electrical components and draw schematic diagrams of installations State and explain the safety purpose of earthing, fuse, circuit breakers and earth leakage Describe the

readings were recorded as per workshop procedures. Portable electrical equipment was checked for correct operation and functionality. Measuring instruments were correctly set up for application. Results were recorded on appropriate documentation. Results were accurately interpreted against the specifications of the job requirements. Electrical measuring instruments were used in accordance with their specifications

TOPIC AK0405: USE OF TESTING INSTRUMENTS

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the

fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

Internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures. Portable power tools were correctly used, as per workshop practices and safety The correct personal protective equipment was used and complied with the workshop PM-05-PS04: Install equipment and control systems Scope of Practical Skill Given equipment and control systems Includes: Batteries, Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rot

TOPIC AK0406: INTERPRETATION OF ELECTRICAL SCHEMATICS AND DRAWINGS

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

Internal Assessment Criteria Hand tools were correctly used, as per workshop practices and safety procedures. Portable power tools were correctly used, as per workshop practices and safety The correct personal protective equipment was used and complied with the workshop PM-05-PS04: Install equipment and control systems Scope of Practical Skill Given equipment and control systems Includes: Batteries, Transformers, Measuring instruments (portable & fixed), Testing instruments, Cables, Switchgear and control gear, Rot

TOPIC AK0407: TECHNIQUES TO CARRY OUT COMMISIONING TESTS

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

Explain fault finding principles and techniques for alternating and direct current control Provider Accreditation Requirements for the Knowledge Module Classroom furniture (chairs and tables, audio equipment and all other equipment conducive to a learning environment) Handouts and stationery (electronic consumables, pencils/paper) Electrical learning material Human Resource Requirements: Relevant qualifications/experience Accredited with National Artisan Moderation Body (NAMB)/Assessment Quality Compliant with Safe

TOPIC AK0408: FAULT FINDING TECHNIQUES

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

Scope of Practical Skill Given work instructions, checklists, work area, case study, testing instruments, activity documents, any templates, form, safety and quality principles and standard operating procedures you must: PA0201 Select hand and power tools PA0202 Select testing instruments AK0201 Hand tools and their uses AK0202 Power tools and their uses AK0203 Testing instruments and their uses AK0204 Safety and standard operating procedures

TOPIC AK0409: ISOLATING AND ENERGIZING PROCESSES

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

671101000-PM-07, Carry out commissioning tests, L3, Cr13 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Carry out commissioning tests you will be required to: PM-07-PS01: Carry out commissioning tests (with power on) on electrical installation. and PM-07-PS02: Rectify defects found on electrical installation. and control systems PM-07-PS03: Report product deficiencies and rectify workmanship Guidelines for Practical Skills PM-07

Risks were identified and mitigated in accordance with risk management procedures A test report that confirms the correctness of the installation was completed in accordance with relevant statutory requirements

TOPIC PA0401: RECTIFY THE FAULTS IDENTIFIED

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

skills related to fault find and repair electrical control systems and electrical installation. you will be required to: PM-08-PS01: Obtain work instructions and gather drawings and documentation PM-08-PS02: Select tools and testing instruments PM-08-PS03: Carry out fault finding PM-08-PS04: Rectify faults where applicable PM-08-PS05: Record findings and provide feedback Details of Work Experience Modules 671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

TOPIC PA0402: COMPLETE A RE-COMMISSIONING TEST

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

skills related to fault find and repair electrical control systems and electrical installation. you will be required to: PM-08-PS01: Obtain work instructions and gather drawings and documentation PM-08-PS02: Select tools and testing instruments PM-08-PS03: Carry out fault finding PM-08-PS04: Rectify faults where applicable PM-08-PS05: Record findings and provide feedback Details of Work Experience Modules 671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.

PM-05-PS05: Wire electrical equipment and control systems PM-05-PS06: Terminate and connect cables and conductors 671101000-PM-06, Conduct pre-commission inspection (power off) and test new and existing installations, L3, Cr5 The focus of the learning in this module is on providing the learner an opportunity to: Practice electrical skills within a simulated work environment. The you will be practicing skills related to inspecting (power off) and testing new and existing installations

TOPIC PA0403: REPORT FINDINGS AND PROVIDE FEEDBACK

Safety and standard operating procedures were correctly followed Functionality of components inspected in accordance with manufacturers specifications Faults were correctly identified Causes were correctly interpreted PM-08-PS04: Rectify faults and re-commission Scope of Practical Skill Given the location and cause of the fault, you should be able to: PA0401 Rectify the faults identified PA0402 Complete a re-commissioning test PA0403 Report findings and provide feedback AK0401 Functionality of components AK

skills related to fault find and repair electrical control systems and electrical installation. you will be required to: PM-08-PS01: Obtain work instructions and gather drawings and documentation PM-08-PS02: Select tools and testing instruments PM-08-PS03: Carry out fault finding PM-08-PS04: Rectify faults where applicable PM-08-PS05: Record findings and provide feedback Details of Work Experience Modules 671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.

Compliant with Safety, Health, Environmental, Risk and Quality (SHERQ) Compliant with Compensation for Occupational Injuries and Diseases Act (COIDA) Critical Practical Activities to be Assessed Externally for the Module 671101000-PM-08, Fault find and repair electrical control systems and electrical installation., L4, Cr22 Purpose of the Practical Skill Module The focus of the learning in this module is on providing the learner an opportunity to: Fault find and repair electrical control systems and electrical installation.

IN SUMMARY

In this unit we have focussed on:

AK0401 Functionality of components

AK04010 Risks associated with live testing

AK04011 Correct use of personal protective equipment

AK04012 Knowledge of recording procedures and documentation

AK04013 Feedback procedures

AK0402 Various fault conditions

AK0403 Various control systems

AK0405 Use of testing instruments

AK0406 Interpretation of electrical schematics and drawings

AK0407 Techniques to carry out commissioning tests

AK0408 Fault finding techniques

AK0409 Isolating and energizing processes

PA0401 Rectify the faults identified

PA0402 Complete a re-commissioning test

PA0403 Report findings and provide feedback

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 52

WM-01-WE01: OBSERVE AND ASSIST A QUALIFIED PERSON IN THE PLANNING AND PREPARATION PROCESS FOR ELECTRICAL INSTALLATION. AND CONTROL SYSTEMS

In this Chapter we will discuss the following Topics:

WA0101 Obtaining relevant documentation and completing requisition/order

WA0102 Obtaining drawings in order to determine job requirements

WA0103 Obtaining tools and equipment from stores and to ensure transportation to worksite

WA0104 Checking worksite for safety and risk analysis requirements

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

SE0101 Obtained and signed requisition/order documentation by both learner and a qualified person

SE0102 Signed job card by both learner and a qualified person

SE0103 Signed workplace log or other relevant documentation to indicate number of hours spent by both learner and a qualified person

SE0104 Signed off completed risk assessment checklist by both learner and a qualified person

TOPIC WA0101: OBTAINING RELEVANT DOCUMENTATION AND COMPLETING REQUISITION/ORDER

and prepare for electrical installation. Guidelines for Work Experiences WM-01-WE01: Observe and assist a qualified person in the planning and preparation process for electrical installation. and control systems Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Obtaining relevant documentation and completing requisition/order WA0102 Obtaining drawings in order to determine job requirements WA0103 Obtaining tools and equipment from stores and to ensure transport

671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting and maintaining of electrical installation. and control systems, L4, Cr8 WM-01- Observe and assist a qualified person in the Date Signature WE01 planning and preparation process for electrical installation. and control

systems WA0101 Obtaining relevant documentation and completing WA0102 Obtaining drawings in order to determine job WA0103 Obtaining tools and equipment from stores and to ensure transport.

WA0304 Checking worksite for safety and risk analysis SE0301 Obtained and signed requisition/order documentation as verified by a signature from a SE0302 Signed job card as verified by a signature from a SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person Contextualised Workplace Knowledge Company specific policies and p

TOPIC WA0102: OBTAINING DRAWINGS IN ORDER TO DETERMINE JOB REQUIREMENTS

and prepare for electrical installation. Guidelines for Work Experiences WM-01-WE01: Observe and assist a qualified person in the planning and preparation process for electrical installation. and control systems Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Obtaining relevant documentation and completing requisition/order WA0102 Obtaining drawings in order to determine job requirements WA0103 Obtaining tools and equipment from stores and to ensure transport.

671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting. ssioning and maintaining of electrical installation. and control systems, L4, Cr8 WM-01- Observe and assist a qualified person in the Date Signature WE01 planning and preparation process for electrical installation. and control systems WA0101 Obtaining relevant documentation and completing WA0102 Obtaining drawings in order to determine job WA0103 Obtaining tools and equipment from stores and to ensure transport.

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the processes of commissioning electrical installation. and control systems in a real life electrical environment

TOPIC WA0103: OBTAINING TOOLS AND EQUIPMENT FROM STORES AND TO ENSURE TRANSPORTATION TO WORKSITE

671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.ssioning and maintaining of electrical installation. and control systems, L4, Cr8 WM-01- Observe and assist a qualified person in the Date Signature WE01 planning and preparation process for electrical installation. and control systems WA0101 Obtaining relevant documentation and completing WA0102 Obtaining drawings in order to determine job WA0103 Obtaining tools and equipment from stores and to ensure transport

and prepare for electrical installation. Guidelines for Work Experiences WM-01-WE01: Observe and assist a qualified person in the planning and preparation process for electrical installation. and control systems Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Obtaining relevant documentation and completing requisition/order WA0102 Obtaining drawings in order to determine job requirements WA0103 Obtaining tools and equipment from stores and to ensure transport

SE0104 Signed off completed risk assessment checklist by both learner and a qualified WM-01-WE02: Plan and prepare for electrical installation. and control systems under the direct supervision of a qualified person Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Obtain relevant documentation and complete requisition WA0202 Obtain drawings in order to determine job requirements under WA0203 Obtain tools and equipment from stores and ensure transportation to work

TOPIC WA0104: CHECKING WORKSITE FOR SAFETY AND RISK ANALYSIS REQUIREMENTS

and prepare for electrical installation. Guidelines for Work Experiences WM-01-WE01: Observe and assist a qualified person in the planning and preparation process for electrical installation. and control systems Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Obtaining relevant documentation and completing requisition/order WA0102 Obtaining drawings in order to determine job requirements WA0103 Obtaining tools and equipment from stores and to ensure transport

SE0104 Signed off completed risk assessment checklist by both learner and a qualified WM-01-WE02: Plan and prepare for electrical installation. and control systems under the direct supervision of a qualified person Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Obtain relevant documentation and complete requisition WA0202 Obtain drawings in order to determine job requirements under WA0203 Obtain tools and equipment from stores and ensure transportation to work

671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.ssioning and maintaining of electrical installation. and control systems, L4, Cr8 WM-01- Observe and assist a qualified person in the Date Signature WE01 planning and preparation process for electrical installation. and control systems WA0101 Obtaining relevant documentation and completing WA0102 Obtaining drawings in order to determine job WA0103 Obtaining tools and equipment from stores and to ensure transport

IN SUMMARY

In this unit we have focussed on:

WA0101 Obtaining relevant documentation and completing requisition/order

WA0102 Obtaining drawings in order to determine job requirements

WA0103 Obtaining tools and equipment from stores and to ensure transportation to worksite

Complete the workbook activities linked to this chapter, which are activities 1 – 4.

CHAPTER 53

WM-01-WE02: PLAN AND PREPARE FOR ELECTRICAL INSTALLATION. AND CONTROL SYSTEMS UNDER THE DIRECT SUPERVISION OF A QUALIFIED PERSON

In this Chapter we will discuss the following Topics:

WA0201 Obtain relevant documentation and complete requisition

WA0202 Obtain drawings in order to determine job requirements under

WA0203 Obtain tools and equipment from stores and ensure transportation to worksite

WA0204 Checking worksite for safety and risk analysis requirements

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

SE0201 Obtained and signed requisition/order documentation by both learner and a qualified person

SE0202 Signed job card by both learner and a qualified person

SE0203 Signed workplace log or other relevant documentation to indicate number of hours spent by both learner and a qualified person

SE0204 Signed off completed risk assessment checklist by both learner and a qualified person

TOPIC WA0201: OBTAIN RELEVANT DOCUMENTATION AND COMPLETE REQUISITION

SE0104 Signed off completed risk assessment checklist by both learner and a qualified WM-01-WE02: Plan and prepare for electrical installation. and control systems under the direct supervision of a qualified person Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Obtain relevant documentation and complete requisition WA0202 Obtain drawings in order to determine job requirements under WA0203 Obtain tools and equipment from stores and ensure transportation to work

WA0201 Obtain relevant documentation and complete WA0202 Obtain drawings in order to determine job WA00203 Obtain tools and equipment from stores and ensure transportation to worksite WA0204 Checking worksite for safety and risk analysis SE0201 Obtained and signed requisition/order documentation by both

learner and a qualified SE0202 Signed job card by both learner and a qualified SE0203 Signed workplace log or other relevant documentation to indicate number of hours spent by both learner and a qualified person SE0

skills related to fault find and repair electrical control systems and electrical installation. you will be required to: PM-08-PS01: Obtain work instructions and gather drawings and documentation PM-08-PS02: Select tools and testing instruments PM-08-PS03: Carry out fault finding PM-08-PS04: Rectify faults where applicable PM-08-PS05: Record findings and provide feedback Details of Work Experience Modules 671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.

TOPIC WA0202: OBTAIN DRAWINGS IN ORDER TO DETERMINE JOB REQUIREMENTS UNDER

SE0104 Signed off completed risk assessment checklist by both learner and a qualified WM-01-WE02: Plan and prepare for electrical installation. and control systems under the direct supervision of a qualified person Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Obtain relevant documentation and complete requisition WA0202 Obtain drawings in order to determine job requirements under WA0203 Obtain tools and equipment from stores and ensure transportation to work

WA0201 Obtain relevant documentation and complete WA0202 Obtain drawings in order to determine job WA00203 Obtain tools and equipment from stores and ensure transportation to worksite WA0204 Checking worksite for safety and risk analysis SE0201 Obtained and signed requisition/order documentation by both learner and a qualified SE0202 Signed job card by both learner and a qualified SE0203 Signed workplace log or other relevant documentation to indicate number of hours spent by both learner and a qualified person SE0

skills related to fault find and repair electrical control systems and electrical installation. you will be required to: PM-08-PS01: Obtain work instructions and gather drawings and documentation PM-08-PS02: Select tools and testing instruments PM-08-PS03: Carry out fault finding PM-08-PS04: Rectify faults where applicable PM-08-PS05: Record findings and provide feedback Details of Work Experience Modules 671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.

TOPIC WA0203: OBTAIN TOOLS AND EQUIPMENT FROM STORES AND ENSURE TRANSPORTATION TO WORKSITE

SE0104 Signed off completed risk assessment checklist by both learner and a qualified WM-01-WE02: Plan and prepare for electrical installation. and control systems under the direct supervision of a qualified person Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Obtain relevant documentation and complete requisition WA0202 Obtain drawings in order to determine job requirements under WA0203 Obtain tools and equipment from stores and ensure transportation to work

WA0201 Obtain relevant documentation and complete WA0202 Obtain drawings in order to determine job WA00203 Obtain tools and equipment from stores and ensure transportation to worksite WA0204 Checking worksite for safety and risk analysis SE0201 Obtained and signed requisition/order documentation by both learner and a qualified SE0202 Signed job card by both learner and a qualified SE0203 Signed workplace log or other relevant documentation to indicate number of hours spent by both learner and a qualified person SE0

671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.ssioning and maintaining of electrical installation. and control systems, L4, Cr8 WM-01- Observe and assist a qualified person in the Date Signature WE01 planning and preparation process for electrical installation. and control systems WA0101 Obtaining relevant documentation and completing WA0102 Obtaining drawings in order to determine job WA0103 Obtaining tools and equipment from stores and to ensure transport

TOPIC WA0204: CHECKING WORKSITE FOR SAFETY AND RISK ANALYSIS REQUIREMENTS

SE0104 Signed off completed risk assessment checklist by both learner and a qualified WM-01-WE02: Plan and prepare for electrical installation. and control systems under the direct supervision of a qualified person Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Obtain relevant documentation and complete requisition WA0202 Obtain drawings in order to determine job requirements under WA0203 Obtain tools and equipment from stores and ensure transportation to work

and prepare for electrical installation. Guidelines for Work Experiences WM-01-WE01: Observe and assist a qualified person in the planning and preparation process for electrical installation. and control systems Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Obtaining relevant documentation and completing requisition/order WA0102 Obtaining drawings in order to determine job requirements WA0103 Obtaining tools and equipment from stores and to ensure transport

WA0201 Obtain relevant documentation and complete WA0202 Obtain drawings in order to determine job WA00203 Obtain tools and equipment from stores and ensure transportation to worksite WA0204 Checking worksite for safety and risk analysis SE0201 Obtained and signed requisition/order documentation by both learner and a qualified SE0202 Signed job card by both learner and a qualified SE0203 Signed workplace log or other relevant documentation to indicate number of hours spent by both learner and a qualified person SE0

IN SUMMARY

In this unit we have focussed on:

WA0201 Obtain relevant documentation and complete requisition

WA0202 Obtain drawings in order to determine job requirements under

WA0203 Obtain tools and equipment from stores and ensure transportation to worksite

Complete the workbook activities linked to this chapter, which are activities 1 – 4.

CHAPTER 54

WM-01-WE03: UNDERTAKE ALL ACTIVITIES WITHOUT ASSISTANCE, BUT UNDER SUPERVISION, TO PLAN AND PREPARE FOR ELECTRICAL INSTALLATION..

In this Chapter we will discuss the following Topics:

WA0301 Obtain relevant documentation and complete requisition/order

WA0302 Obtain drawings in order to determine job requirements

WA0303 Obtain tools and equipment from stores and ensure transportation to worksite

WA0304 Checking worksite for safety and risk analysis requirements

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

SE0301 Obtained and signed requisition/order documentation as verified by a signature from a qualified person

SE0302 Signed job card as verified by a signature from a qualified person

SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person

SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person

TOPIC WA0301: OBTAIN RELEVANT DOCUMENTATION AND COMPLETE REQUISITION/ORDER

SE0104 Signed off completed risk assessment checklist by both learner and a qualified WM-01-WE02: Plan and prepare for electrical installation. and control systems under the direct supervision of a qualified person Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Obtain relevant documentation and complete requisition WA0202 Obtain drawings in order to determine job requirements under WA0203 Obtain tools and equipment from stores and ensure transportation to work

WA0201 Obtain relevant documentation and complete WA0202 Obtain drawings in order to determine job WA00203 Obtain tools and equipment from stores and ensure transportation to worksite WA0204 Checking

worksite for safety and risk analysis SE0201 Obtained and signed requisition/order documentation by both learner and a qualified SE0202 Signed job card by both learner and a qualified SE0203 Signed workplace log or other relevant documentation to indicate number of hours spent by both learner and a qualified person SE0

skills related to fault find and repair electrical control systems and electrical installation. you will be required to: PM-08-PS01: Obtain work instructions and gather drawings and documentation PM-08-PS02: Select tools and testing instruments PM-08-PS03: Carry out fault finding PM-08-PS04: Rectify faults where applicable PM-08-PS05: Record findings and provide feedback Details of Work Experience Modules 671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.

TOPIC WA0302: OBTAIN DRAWINGS IN ORDER TO DETERMINE JOB REQUIREMENTS

SE0104 Signed off completed risk assessment checklist by both learner and a qualified WM-01-WE02: Plan and prepare for electrical installation. and control systems under the direct supervision of a qualified person Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Obtain relevant documentation and complete requisition WA0202 Obtain drawings in order to determine job requirements under WA0203 Obtain tools and equipment from stores and ensure transportation to work

WA0201 Obtain relevant documentation and complete WA0202 Obtain drawings in order to determine job WA00203 Obtain tools and equipment from stores and ensure transportation to worksite WA0204 Checking worksite for safety and risk analysis SE0201 Obtained and signed requisition/order documentation by both learner and a qualified SE0202 Signed job card by both learner and a qualified SE0203 Signed workplace log or other relevant documentation to indicate number of hours spent by both learner and a qualified person SE0

skills related to fault find and repair electrical control systems and electrical installation. you will be required to: PM-08-PS01: Obtain work instructions and gather drawings and documentation PM-08-PS02: Select tools and testing instruments PM-08-PS03: Carry out fault finding PM-08-PS04: Rectify faults where applicable PM-08-PS05: Record findings and provide feedback Details of Work Experience Modules 671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting.

TOPIC WA0303: OBTAIN TOOLS AND EQUIPMENT FROM STORES AND ENSURE TRANSPORTATION TO WORKSITE

SE0104 Signed off completed risk assessment checklist by both learner and a qualified WM-01-WE02: Plan and prepare for electrical installation. and control systems under the direct supervision of a qualified person Scope of Work Experience The person will be expected to engage in the following work activities: WA0201

Obtain relevant documentation and complete requisition WA0202 Obtain drawings in order to determine job requirements under WA0203 Obtain tools and equipment from stores and ensure transportation to work

WA0201 Obtain relevant documentation and complete WA0202 Obtain drawings in order to determine job WA00203 Obtain tools and equipment from stores and ensure transportation to worksite WA0204 Checking worksite for safety and risk analysis SE0201 Obtained and signed requisition/order documentation by both learner and a qualified SE0202 Signed job card by both learner and a qualified SE0203 Signed workplace log or other relevant documentation to indicate number of hours spent by both learner and a qualified person SE0

671101000-WM-01, Planning and preparation process for the wiring, connection, testing, inspecting and maintaining of electrical installation. and control systems, L4, Cr8 WM-01- Observe and assist a qualified person in the Date Signature WE01 planning and preparation process for electrical installation. and control systems WA0101 Obtaining relevant documentation and completing WA0102 Obtaining drawings in order to determine job WA0103 Obtaining tools and equipment from stores and to ensure transport.

TOPIC WA0304: CHECKING WORKSITE FOR SAFETY AND RISK ANALYSIS REQUIREMENTS

SE0104 Signed off completed risk assessment checklist by both learner and a qualified WM-01-WE02: Plan and prepare for electrical installation. and control systems under the direct supervision of a qualified person Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Obtain relevant documentation and complete requisition WA0202 Obtain drawings in order to determine job requirements under WA0203 Obtain tools and equipment from stores and ensure transportation to work

and prepare for electrical installation. Guidelines for Work Experiences WM-01-WE01: Observe and assist a qualified person in the planning and preparation process for electrical installation. and control systems Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Obtaining relevant documentation and completing requisition/order WA0102 Obtaining drawings in order to determine job requirements WA0103 Obtaining tools and equipment from stores and to ensure transport.

WA0304 Checking worksite for safety and risk analysis SE0301 Obtained and signed requisition/order documentation as verified by a signature from a SE0302 Signed job card as verified by a signature from a SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person Contextualised Workplace Knowledge Company specific policies and p

IN SUMMARY

In this unit we have focussed on:

WA0301 Obtain relevant documentation and complete requisition/order.

WA0302 Obtain drawings in order to determine job requirements.

WA0303 Obtain tools and equipment from stores and ensure transportation to worksite.

WA0304 Checking worksite for safety and risk analysis requirements.

Complete the workbook activities linked to this chapter, which are activities 1 – 4.

CHAPTER 55

WM-02-WE01: OBSERVE AND ASSIST A QUALIFIED ELECTRICIAN IN THE PROCESSES OF INSTALLATION, WIRING AND CONNECTION OF ELECTRICAL EQUIPMENT AND CONTROL SYSTEMS

In this Chapter we will discuss the following Topics:

WA0101 Apply drawing specifications.

WA0102 Use access to height equipment.

WA0103 Mark out and do measurements of the installation.

WA0104 Drill and/ tap and/or chase walls and/or trenching

WA0105 Install and secure wire ways.

WA0106 Mount and fasten equipment.

WA0107 Draw in conductors and cables.

WA0108 Make off cables.

WA0109 Terminate cables and conductors.

WA0110 Label and number cables and conductors

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

SE0101 Completed observation checklist | SE0102 Completed section in logbook | SE0103 List of equipment installed.

TOPIC WA0101: APPLY DRAWING SPECIFICATIONS

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-02, Processes of installing, wiring and connecting of electrical equipment and control systems, L4, Cr74 Purpose of the Work Experience Module The focus of the work experience is on

providing the learner an opportunity to: Gain exposure to work as part of a team in the process of wiring, connection of electrical equipment and control systems in a real life electrical environment

WA0304 Checking worksite for safety and risk analysis SE0301 Obtained and signed requisition/order documentation as verified by a signature from a SE0302 Signed job card as verified by a signature from a SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person Contextualised Workplace Knowledge Company specific policies and p

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

TOPIC WA0102: USE ACCESS TO HEIGHT EQUIPMENT

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-02, Processes of installing, wiring and connecting of electrical equipment and control systems, L4, Cr74 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the process of wiring, connection of electrical equipment and control systems in a real life electrical environment

WA0304 Checking worksite for safety and risk analysis SE0301 Obtained and signed requisition/order documentation as verified by a signature from a SE0302 Signed job card as verified by a signature from a SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person Contextualised Workplace Knowledge Company specific policies and p

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

TOPIC WA0103: MARK OUT AND DO MEASUREMENTS OF THE INSTALLATION

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-02, Processes of installing, wiring and connecting of electrical equipment and control systems, L4, Cr74 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the process of wiring, connection of electrical equipment and control systems in a real life electrical environment

WA0304 Checking worksite for safety and risk analysis SE0301 Obtained and signed requisition/order documentation as verified by a signature from a SE0302 Signed job card as verified by a signature from a SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person Contextualised Workplace Knowledge Company specific policies and p

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

TOPIC WA0104: DRILL AND/ TAP AND/OR CHASE WALLS AND/OR TRENCHING

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02-WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

WA0303 Mark out and do measurements of the site WA0304 Drill and/ tap and/or chase walls and/or trenching WA0305 Install and secure wire ways WA0305 Mount and fasten equipment WA0306 Draw in conductors and cables WA0307 Make off cables WA0308 Terminate cables and conductors WA0309 Label and number cables and conductors SE0301 Completed observation checklist SE0302 Completed section in logbook SE0303 List

of equipment installed Contextualised Workplace Knowledge Company specific policies and procedures (LV operating

TOPIC WA0105: INSTALL AND SECURE WIRE WAYS

WA0304 Checking worksite for safety and risk analysis SE0301 Obtained and signed requisition/order documentation as verified by a signature from a SE0302 Signed job card as verified by a signature from a SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person Contextualised Workplace Knowledge Company specific policies and p

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

TOPIC WA0106: MOUNT AND FASTEN EQUIPMENT

WA0304 Checking worksite for safety and risk analysis SE0301 Obtained and signed requisition/order documentation as verified by a signature from a SE0302 Signed job card as verified by a signature from a SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person Contextualised Workplace Knowledge Company specific policies and p

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02-

WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

TOPIC WA0107: DRAW IN CONDUCTORS AND CABLES

WA0304 Checking worksite for safety and risk analysis SE0301 Obtained and signed requisition/order documentation as verified by a signature from a SE0302 Signed job card as verified by a signature from a SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person Contextualised Workplace Knowledge Company specific policies and p

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

TOPIC WA0108: MAKE OFF CABLES

WA0304 Checking worksite for safety and risk analysis SE0301 Obtained and signed requisition/order documentation as verified by a signature from a SE0302 Signed job card as verified by a signature from a SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by

a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person Contextualised Workplace Knowledge Company specific policies and p

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

TOPIC WA0109: TERMINATE CABLES AND CONDUCTORS

WA0304 Checking worksite for safety and risk analysis SE0301 Obtained and signed requisition/order documentation as verified by a signature from a SE0302 Signed job card as verified by a signature from a SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person Contextualised Workplace Knowledge Company specific policies and p

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

TOPIC WA0110: LABEL AND NUMBER CABLES AND CONDUCTORS

WA0304 Checking worksite for safety and risk analysis SE0301 Obtained and signed requisition/order documentation as verified by a signature from a SE0302 Signed job card as verified by a signature from a SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person Contextualised Workplace Knowledge Company specific policies and p

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02-WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

IN SUMMARY

In this unit we have focussed on:

WA0101 Apply drawing specifications.

WA0102 Use access to height equipment.

WA0103 Mark out and do measurements of the installation.

WA0104 Drill and/ tap and/or chase walls and/or trenching

WA0105 Install and secure wire ways.

WA0106 Mount and fasten equipment.

WA0107 Draw in conductors and cables.

WA0108 Make off cables.

WA0109 Terminate cables and conductors.

WA0110 Label and number cables and conductors

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 56

WM-02-WE02: INSTALL, WIRE AND CONNECT ELECTRICAL EQUIPMENT AND CONTROL SYSTEMS UNDER THE DIRECT SUPERVISION OF A QUALIFIED ELECTRICIAN

In this Chapter we will discuss the following Topics:

WA0201 Application of drawing specifications

WA0202 Use access equipment.

WA0203 Mark out and do measurements of the site.

WA0204 Drill and/ tap and/chase walls and/complete trenches

WA0205 Install and secure wire ways.

WA0206 Draw in conductors and cables.

WA0207 Make off cables.

WA0208 Terminate cables and conductors.

WA0209 Label and number cables and conductors

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

SE0201 Completed observation checklist | SE0202 Completed section in logbook | SE0203 List of equipment installed.

TOPIC WA0201: APPLICATION OF DRAWING SPECIFICATIONS

WA0202 - Use access equipment

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and

document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

TOPIC WA0202: USE ACCESS EQUIPMENT

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

rotation, fault location test, earth resistance tests The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment. WA0202 Inspect protection devises used in electrical equipment and settings verified. WA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations. WA0206 Complete and record electrical tests

TOPIC WA0203: MARK OUT AND DO MEASUREMENTS OF THE SITE

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206

Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

WA0303 Mark out and do measurements of the site WA0304 Drill and/ tap and/or chase walls and/or trenching WA0305 Install and secure wire ways WA0305 Mount and fasten equipment WA0306 Draw in conductors and cables WA0307 Make off cables WA0308 Terminate cables and conductors WA0309 Label and number cables and conductors SE0301 Completed observation checklist SE0302 Completed section in logbook SE0303 List of equipment installed Contextualised Workplace Knowledge Company specific policies and procedures (LV operating

TOPIC WA0204: DRILL AND/ TAP AND/CHASE WALLS AND/COMPLETE TRENCHES

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02-WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment. WA0202 Inspect protection devises used in electrical equipment and settings verified. WA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations. WA0206 Complete and record electrical tests

TOPIC WA0205: INSTALL AND SECURE WIRE WAYS

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed

observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02-WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

WA0304 Checking worksite for safety and risk analysis SE0301 Obtained and signed requisition/order documentation as verified by a signature from a SE0302 Signed job card as verified by a signature from a SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person Contextualised Workplace Knowledge Company specific policies and p

TOPIC WA0206: DRAW IN CONDUCTORS AND CABLES

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02-WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

WA0304 Checking worksite for safety and risk analysis SE0301 Obtained and signed requisition/order documentation as verified by a signature from a SE0302 Signed job card as verified by a signature from a SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person Contextualised Workplace Knowledge Company specific policies and p

TOPIC WA0207: MAKE OFF CABLES

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

WA0304 Checking worksite for safety and risk analysis SE0301 Obtained and signed requisition/order documentation as verified by a signature from a SE0302 Signed job card as verified by a signature from a SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person Contextualised Workplace Knowledge Company specific policies and p

TOPIC WA0208: TERMINATE CABLES AND CONDUCTORS

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

WA0304 Checking worksite for safety and risk analysis SE0301 Obtained and signed requisition/order documentation as verified by a signature from a SE0302 Signed job card as verified by a signature from a

SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person Contextualised Workplace Knowledge Company specific policies and p

TOPIC WA0209: LABEL AND NUMBER CABLES AND CONDUCTORS

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

WA0304 Checking worksite for safety and risk analysis SE0301 Obtained and signed requisition/order documentation as verified by a signature from a SE0302 Signed job card as verified by a signature from a SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person Contextualised Workplace Knowledge Company specific policies and p

IN SUMMARY

In this unit we have focussed on:

WA0201 Application of drawing specifications

WA0202 Use access equipment

WA0203 Mark out and do measurements of the site

WA0204 Drill and/ tap and/chase walls and/complete trenches

WA0205 Install and secure wire ways

WA0206 Draw in conductors and cables

WA0207 Make off cables

WA0208 Terminate cables and conductors

WA0209 Label and number cables and conductors

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 57

WM-02-WE03: UNDERTAKE ALL ACTIVITIES WITHOUT ASSISTANCE, BUT UNDER SUPERVISION, TO INSTALL, WIRE AND CONNECT ELECTRICAL EQUIPMENT AND CONTROL SYSTEMS

In this Chapter we will discuss the following Topics:

WA0301 Apply drawing specifications.

WA0302 Select and use access equipment

WA0303 Mark out and do measurements of the site

WA0304 Drill and/ tap and/or chase walls and/or trenching

WA0305 Install and secure wire ways

WA0306 Draw in conductors and cables

WA0307 Make off cables

WA0308 Terminate cables and conductors

WA0309 Label and number cables and conductors

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

SE0301 Completed observation checklist | SE0302 Completed section in logbook | SE0303 List of equipment installed

TOPIC WA0301: APPLY DRAWING SPECIFICATIONS.

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02-WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-02, Processes of installing, wiring and connecting of electrical equipment and control systems, L4, Cr74 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the process of wiring, connection of electrical equipment and control systems in a real life electrical environment

TOPIC WA0302: SELECT AND USE ACCESS EQUIPMENT

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-02, Processes of installing, wiring and connecting of electrical equipment and control systems, L4, Cr74 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the process of wiring, connection of electrical equipment and control systems in a real life electrical environment

TOPIC WA0303: MARK OUT AND DO MEASUREMENTS OF THE SITE

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

WA0303 Mark out and do measurements of the site WA0304 Drill and/ tap and/or chase walls and/or trenching WA0305 Install and secure wire ways WA0305 Mount and fasten equipment WA0306 Draw in conductors and cables WA0307 Make off cables WA0308 Terminate cables and conductors WA0309 Label and number cables and conductors SE0301 Completed observation checklist SE0302 Completed section in logbook SE0303 List of equipment installed Contextualised Workplace Knowledge Company specific policies and procedures (LV operating

TOPIC WA0304: DRILL AND/ TAP AND/OR CHASE WALLS AND/OR TRENCHING

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

WA0303 Mark out and do measurements of the site WA0304 Drill and/ tap and/or chase walls and/or trenching WA0305 Install and secure wire ways WA0305 Mount and fasten equipment WA0306 Draw in conductors and cables WA0307 Make off cables WA0308 Terminate cables and conductors WA0309 Label and number cables and conductors SE0301 Completed observation checklist SE0302 Completed section in logbook SE0303 List of equipment installed Contextualised Workplace Knowledge Company specific policies and procedures (LV operating

TOPIC WA0305: INSTALL AND SECURE WIRE WAYS

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

WA0303 Mark out and do measurements of the site WA0304 Drill and/ tap and/or chase walls and/or trenching WA0305 Install and secure wire ways WA0305 Mount and fasten equipment WA0306 Draw in conductors and cables WA0307 Make off cables WA0308 Terminate cables and conductors WA0309 Label and number cables and conductors SE0301 Completed observation checklist SE0302 Completed section in logbook SE0303 List of equipment installed Contextualised Workplace Knowledge Company specific policies and procedures (LV operating

TOPIC WA0306: DRAW IN CONDUCTORS AND CABLES

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

WA0303 Mark out and do measurements of the site WA0304 Drill and/ tap and/or chase walls and/or trenching WA0305 Install and secure wire ways WA0305 Mount and fasten equipment WA0306 Draw in conductors and cables WA0307 Make off cables WA0308 Terminate cables and conductors WA0309 Label and number cables and conductors SE0301 Completed observation checklist SE0302 Completed section in logbook SE0303 List of equipment installed Contextualised Workplace Knowledge Company specific policies and procedures (LV operating

TOPIC WA0307: MAKE OFF CABLES

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

WA0303 Mark out and do measurements of the site WA0304 Drill and/ tap and/or chase walls and/or trenching WA0305 Install and secure wire ways WA0305 Mount and fasten equipment WA0306 Draw in conductors and cables WA0307 Make off cables WA0308 Terminate cables and conductors WA0309 Label and number cables and conductors SE0301 Completed observation checklist SE0302 Completed section in logbook SE0303 List of equipment installed Contextualised Workplace Knowledge Company specific policies and procedures (LV operating

TOPIC WA0308: TERMINATE CABLES AND CONDUCTORS

SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- Install, wire and connect electrical equipment and Date Signature WE02 control systems under the direct supervision of a WA0201 Application of drawing specifications WA0202 Use access equipment WA0203 Mark out and do measurements of the site WA0204 Drill and/ tap and/chase walls and/complete WA0205 Install and secure wire ways WA0206 Draw in conductors and cables WA0207 Make off cables WA0208 Terminate cables and conductors WA0209 Label a

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

WA0303 Mark out and do measurements of the site WA0304 Drill and/ tap and/or chase walls and/or trenching WA0305 Install and secure wire ways WA0305 Mount and fasten equipment WA0306 Draw in conductors and cables WA0307 Make off cables WA0308 Terminate cables and conductors WA0309 Label and number cables and conductors SE0301 Completed observation checklist SE0302 Completed section in logbook SE0303 List of equipment installed Contextualised Workplace Knowledge Company specific policies and procedures (LV operating

TOPIC WA0309: LABEL AND NUMBER CABLES AND CONDUCTORS

WA0104 Drill and/ tap and/or chase walls and/or trenching WA0105 Install and secure wire ways WA0106 Mount and fasten equipment WA0107 Draw in conductors and cables WA0108 Make off cables WA0109 Terminate cables and conductors WA0110 Label and number cables and conductors SE0101 Completed observation checklist SE0102 Completed section in logbook SE0103 List of equipment installed WM-02- WE02: Install, wire and connect electrical equipment and control systems under the direct supervision of a qualified electrician Sc

WA0303 Mark out and do measurements of the site WA0304 Drill and/ tap and/or chase walls and/or trenching WA0305 Install and secure wire ways WA0305 Mount and fasten equipment WA0306 Draw in conductors and cables WA0307 Make off cables WA0308 Terminate cables and conductors WA0309 Label and number cables and conductors SE0301 Completed observation checklist SE0302 Completed section in logbook SE0303 List of equipment installed Contextualised Workplace Knowledge Company specific policies and procedures (LV operating

WA0304 Checking worksite for safety and risk analysis SE0301 Obtained and signed requisition/order documentation as verified by a signature from a SE0302 Signed job card as verified by a signature from a

SE0303 Signed workplace log or other relevant documentation to indicate number of hours spent as verified by a signature from a qualified person SE0304 Signed off completed risk assessment checklist as verified by a signature from a qualified person Contextualised Workplace Knowledge Company specific policies and p

IN SUMMARY

In this unit we have focussed on:

WA0301 Apply drawing specifications.

WA0302 Select and use access equipment

WA0303 Mark out and do measurements of the site

WA0304 Drill and/ tap and/or chase walls and/or trenching

WA0305 Install and secure wire ways

WA0306 Draw in conductors and cables

WA0307 Make off cables

WA0308 Terminate cables and conductors

WA0309 Label and number cables and conductors

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 58

WM-03-WE01: OBSERVE AND ASSIST A QUALIFIED ELECTRICIAN IN THE PROCESSES OF TESTING AND INSPECTING OF ELECTRICAL EQUIPMENT, CONTROL SYSTEMS AND INSTALLATIONS

In this Chapter we will discuss the following Topics:

WA0101 Visually and physically inspect electrical equipment

WA0102 Inspect protection devices selection and settings

WA0103 Test electrical equipment

WA0104 Test main and control circuits

WA0105 Visually and physically inspect installations

WA0106 Complete electrical tests

WA0107 Inspect the installation method used.

WA0108 Inspect the selection of cables and conductors.

WA0109 Document findings and test results

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

SE0101 Completed observation checklist | SE0102 Completed section in logbook | SE0103 Completed electrical test report

TOPIC WA0101: VISUALLY AND PHYSICALLY INSPECT ELECTRICAL EQUIPMENT

WA0106 - Complete electrical tests

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of

installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0107 - Inspect the installation method used

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0108 - Inspect the selection of cables and conductors

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

TOPIC WA0102: INSPECT PROTECTION DEVICES SELECTION AND SETTINGS

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

Company specific quality management systems Relevant statutory requirements: Not limited to OHS Act; Construction Regulations; Mine Health and Safety Act, Municipal by-laws, environmental standards, traffic regulations, operating regulations for high voltage systems

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

TOPIC WA0103: TEST ELECTRICAL EQUIPMENT

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

Company specific quality management systems Relevant statutory requirements: Not limited to OHS Act; Construction Regulations; Mine Health and Safety Act, Municipal by-laws, environmental standards, traffic regulations, operating regulations for high voltage systems

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

TOPIC WA0104: TEST MAIN AND CONTROL CIRCUITS

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

Company specific quality management systems Relevant statutory requirements: Not limited to OHS Act; Construction Regulations; Mine Health and Safety Act, Municipal by-laws, environmental standards, traffic regulations, operating regulations for high voltage systems

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

TOPIC WA0105: VISUALLY AND PHYSICALLY INSPECT INSTALLATIONS

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

Company specific quality management systems Relevant statutory requirements: Not limited to OHS Act; Construction Regulations; Mine Health and Safety Act, Municipal by-laws, environmental standards, traffic regulations, operating regulations for high voltage systems

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the processes of commissioning electrical installation. and control systems in a real life electrical environment

TOPIC WA0106: COMPLETE ELECTRICAL TESTS

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

Company specific quality management systems Relevant statutory requirements: Not limited to OHS Act; Construction Regulations; Mine Health and Safety Act, Municipal by-laws, environmental standards, traffic regulations, operating regulations for high voltage systems

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an

opportunity to: Gain exposure to work as part of a team in the processes of commissioning electrical installation. and control systems in a real life electrical environment

TOPIC WA0107: INSPECT THE INSTALLATION METHOD USED.

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

Company specific quality management systems Relevant statutory requirements: Not limited to OHS Act; Construction Regulations; Mine Health and Safety Act, Municipal by-laws, environmental standards, traffic regulations, operating regulations for high voltage systems

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment. WA0202 Inspect protection devises used in electrical equipment and settings verified. WA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations. WA0206 Complete and record electrical tests

TOPIC WA0108: INSPECT THE SELECTION OF CABLES AND CONDUCTORS.

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

Company specific quality management systems Relevant statutory requirements: Not limited to OHS Act; Construction Regulations; Mine Health and Safety Act, Municipal by-laws, environmental standards, traffic regulations, operating regulations for high voltage systems

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment. WA0202 Inspect protection devises used in electrical equipment and settings verified. WA0203 Test electrical equipment and document results

under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations. WA0206 Complete and record electrical tests

TOPIC WA0109: DOCUMENT FINDINGS AND TEST RESULTS

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

Company specific quality management systems Relevant statutory requirements: Not limited to OHS Act; Construction Regulations; Mine Health and Safety Act, Municipal by-laws, environmental standards, traffic regulations, operating regulations for high voltage systems

IN SUMMARY

In this unit we have focussed on:

WA0101 Visually and physically inspect electrical equipment

WA0102 Inspect protection devices selection and settings

WA0103 Test electrical equipment

WA0104 Test main and control circuits

WA0105 Visually and physically inspect installations

WA0106 Complete electrical tests

WA0107 Inspect the installation method used.

WA0108 Inspect the selection of cables and conductors.

WA0109 Document findings and test results

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 59

WM-03-WE02: TEST AND INSPECT ELECTRICAL EQUIPMENT, CONTROL SYSTEMS AND INSTALLATIONS UNDER THE DIRECT SUPERVISION OF A QUALIFIED ELECTRICIAN

In this Chapter we will discuss the following Topics:

WA0109 document findings and test results

WA0201 Visually inspect electrical equipment

WA0202 Protection devises used in electrical equipment are inspected and settings verified

WA0203 Test electrical equipment

WA0204 Test main and control circuits

WA0205 Visually inspect installations

WA0206 Electrical tests are done

WA0207 Installation method used is inspected

WA0208 The selection of cables and conductors are inspected

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

SE0201 Completed observation checklist | SE0202 Completed section in logbook | SE0203 Completed electrical test report

TOPIC WA0109: DOCUMENT FINDINGS AND TEST RESULTS

WA0109 - document findings and test results

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and

document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0201 - Visually inspect electrical equipment

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0202 - Protection devises used in electrical equipment are inspected and settings verified

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0203 - Test electrical equipment

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0204 - Test main and control circuits

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0205 - Visually inspect installations

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0206 - Electrical tests are done

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0207 - Installation method used is inspected

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the

following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0208 - The selection of cables and conductors are inspected

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

TOPIC WA0201: VISUALLY INSPECT ELECTRICAL EQUIPMENT

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

Company specific quality management systems Relevant statutory requirements: Not limited to OHS Act; Construction Regulations; Mine Health and Safety Act, Municipal by-laws, environmental standards, traffic regulations, operating regulations for high voltage systems

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

TOPIC WA0202: PROTECTION DEVISES USED IN ELECTRICAL EQUIPMENT ARE INSPECTED AND SETTINGS VERIFIED

Company specific quality management systems Relevant statutory requirements: Not limited to OHS Act; Construction Regulations; Mine Health and Safety Act, Municipal by-laws, environmental standards, traffic regulations, operating regulations for high voltage systems

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment. WA0202 Inspect protection devises used in electrical equipment and settings verified. WA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations. WA0206 Complete and record electrical tests

TOPIC WA0203: TEST ELECTRICAL EQUIPMENT

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment. WA0202 Inspect protection devises used in electrical equipment and settings verified. WA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations. WA0206 Complete and record electrical tests

TOPIC WA0204: TEST MAIN AND CONTROL CIRCUITS

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment. WA0202 Inspect protection devises used in electrical equipment and settings verified. WA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations. WA0206 Complete and record electrical tests

TOPIC WA0205: VISUALLY INSPECT INSTALLATIONS

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

Company specific quality management systems Relevant statutory requirements: Not limited to OHS Act; Construction Regulations; Mine Health and Safety Act, Municipal by-laws, environmental standards, traffic regulations, operating regulations for high voltage systems

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

TOPIC WA0206: ELECTRICAL TESTS ARE DONE

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

Company specific quality management systems Relevant statutory requirements: Not limited to OHS Act; Construction Regulations; Mine Health and Safety Act, Municipal by-laws, environmental standards, traffic regulations, operating regulations for high voltage systems

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

TOPIC WA0207: INSTALLATION METHOD USED IS INSPECTED

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

Company specific quality management systems Relevant statutory requirements: Not limited to OHS Act; Construction Regulations; Mine Health and Safety Act, Municipal by-laws, environmental standards, traffic regulations, operating regulations for high voltage systems

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

TOPIC WA0208: THE SELECTION OF CABLES AND CONDUCTORS ARE INSPECTED

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

Company specific quality management systems Relevant statutory requirements: Not limited to OHS Act; Construction Regulations; Mine Health and Safety Act, Municipal by-laws, environmental standards, traffic regulations, operating regulations for high voltage systems

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test Guidelines for Work Experiences WM-03-WE01: Observe and assist a qualified electrician in the processes of testing and inspecting of electrical equipment, control systems and installations Scope of Work Experience The person will be expected to engage in the following work activities: WA0101 Vis

IN SUMMARY

In this unit we have focussed on:

WA0109 document findings and test results

WA0201 Visually inspect electrical equipment

WA0202 Protection devises used in electrical equipment are inspected and settings verified

WA0203 Test electrical equipment

WA0204 Test main and control circuits

WA0205 Visually inspect installations

WA0206 Electrical tests are done

WA0207 Installation method used is inspected

WA0208 The selection of cables and conductors are inspected

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 60

WM-03-WE03: UNDERTAKE ALL ACTIVITIES WITHOUT ASSISTANCE, BUT UNDER SUPERVISION, IN THE PROCESSES OF TESTING AND INSPECTING OF ELECTRICAL EQUIPMENT, CONTROL SYSTEMS AND INSTALLATIONS

In this Chapter we will discuss the following Topics:

WA0201 Complete a visual inspection of electrical equipment.

WA0202 Inspect protection devises used in electrical equipment and settings verified.

WA0203 Test electrical equipment and document results under the direct supervision of a qualified artisan

WA0204 Test main and control circuits

WA0205 Complete a visual inspection of installations.

WA0206 Complete and record electrical tests.

WA0207 Inspect the installation method used.

WA0208 Inspect the selection of cables and conductors.

WA0209 Document findings and test results

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

SE0201 Completed observation checklist | SE0202 Completed section in logbook | SE0203 Completed electrical test report

TOPIC WA0201: COMPLETE A VISUAL INSPECTION OF ELECTRICAL EQUIPMENT.

WA0201 - Complete a visual inspection of electrical equipment

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of

installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0202 - Inspect protection devises used in electrical equipment and settings verified

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0203 - Test electrical equipment and document results under the direct supervision of a qualified artisan

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0204 - Test main and control circuits

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and

document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0205 - Complete a visual inspection of installations

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0206 - Complete and record electrical tests

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0207 - Inspect the installation method used

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0208 - Inspect the selection of cables and conductors

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

WA0209 - Document findings and test results

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment wA0202 Inspect protection devises used in electrical equipment and settings verified wA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations

TOPIC WA0202: INSPECT PROTECTION DEVISES USED IN ELECTRICAL EQUIPMENT AND SETTINGS VERIFIED.

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment. WA0202 Inspect protection devises used in electrical equipment and settings verified. WA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations. WA0206 Complete and record electrical tests

WA0209 document findings and test results SE0201 Completed observation checklist SE0202 Completed section in logbook SE0203 Completed electrical test report WM-03- Undertake all activities without assistance, but Date Signature WE03 under supervision, in the processes of testing and inspecting of electrical equipment, control systems WA0201 Complete a visual inspection of electrical WA0202 Inspect protection devises used in electrical equipment and settings verified

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test

The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

TOPIC WA0203: TEST ELECTRICAL EQUIPMENT AND DOCUMENT RESULTS UNDER THE DIRECT SUPERVISION OF A QUALIFIED ARTISAN

WA0209 document findings and test results SE0201 Completed observation checklist SE0202 Completed section in logbook SE0203 Completed electrical test report WM-03- Undertake all activities without assistance, but Date Signature WE03 under supervision, in the processes of testing and inspecting of electrical equipment, control systems WA0201 Complete a visual inspection of electrical WA0202 Inspect protection devises used in electrical equipment and settings verified

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment. WA0202 Inspect protection devises used in electrical equipment and settings verified. WA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations. WA0206 Complete and record electrical tests

TOPIC WA0204: TEST MAIN AND CONTROL CIRCUITS

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment. WA0202 Inspect protection devises used in electrical equipment and settings verified. WA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations. WA0206 Complete and record electrical tests

Company specific quality management systems Relevant statutory requirements: Not limited to OHS Act; Construction Regulations; Mine Health and Safety Act, Municipal by-laws, environmental standards, traffic regulations, operating regulations for high voltage systems

TOPIC WA0205: COMPLETE A VISUAL INSPECTION OF INSTALLATIONS.

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment. WA0202 Inspect protection devises used in electrical equipment and settings verified. WA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations. WA0206 Complete and record electrical tests

WA0209 document findings and test results SE0201 Completed observation checklist SE0202 Completed section in logbook SE0203 Completed electrical test report WM-03- Undertake all activities without assistance, but Date Signature WE03 under supervision, in the processes of testing and inspecting of electrical equipment, control systems WA0201 Complete a visual inspection of electrical WA0202 Inspect protection devises used in electrical equipment and settings verified

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

TOPIC WA0206: COMPLETE AND RECORD ELECTRICAL TESTS.

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment. WA0202 Inspect protection devises used in electrical equipment and settings verified. WA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations. WA0206 Complete and record electrical tests

WA0209 document findings and test results SE0201 Completed observation checklist SE0202 Completed section in logbook SE0203 Completed electrical test report WM-03- Undertake all activities without assistance, but Date Signature WE03 under supervision, in the processes of testing and inspecting of electrical equipment, control systems WA0201 Complete a visual inspection of electrical WA0202 Inspect protection devises used in electrical equipment and settings verified

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

TOPIC WA0207: INSPECT THE INSTALLATION METHOD USED.

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment. WA0202 Inspect protection devises used in electrical equipment and settings verified. WA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations. WA0206 Complete and record electrical tests

Company specific quality management systems Relevant statutory requirements: Not limited to OHS Act; Construction Regulations; Mine Health and Safety Act, Municipal by-laws, environmental standards, traffic regulations, operating regulations for high voltage systems

WA0209 document findings and test results SE0201 Completed observation checklist SE0202 Completed section in logbook SE0203 Completed electrical test report WM-03- Undertake all activities without assistance, but Date Signature WE03 under supervision, in the processes of testing and inspecting of electrical equipment, control systems WA0201 Complete a visual inspection of electrical WA0202 Inspect protection devises used in electrical equipment and settings verified

TOPIC WA0208: INSPECT THE SELECTION OF CABLES AND CONDUCTORS.

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment. WA0202 Inspect protection devises used in electrical equipment and settings verified. WA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations. WA0206 Complete and record electrical tests

Company specific quality management systems Relevant statutory requirements: Not limited to OHS Act; Construction Regulations; Mine Health and Safety Act, Municipal by-laws, environmental standards, traffic regulations, operating regulations for high voltage systems

WA0209 document findings and test results SE0201 Completed observation checklist SE0202 Completed section in logbook SE0203 Completed electrical test report WM-03- Undertake all activities without assistance, but Date Signature WE03 under supervision, in the processes of testing and inspecting of electrical equipment, control systems WA0201 Complete a visual inspection of electrical WA0202 Inspect protection devises used in electrical equipment and settings verified

TOPIC WA0209: DOCUMENT FINDINGS AND TEST RESULTS

earth leakage sensitivity test. The following tests may be included, Earth fault Loop impedance, Elevated voltage on neutral, On load and no load voltage, polarity, phase rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Visually inspect electrical equipment WA0202 Protection devises used in electrical equipment are inspected and settings WA0203 Test electrical equipment WA0204 Test main and control circuits WA0205 Visually inspect in

rotation, fault location test, earth resistance test The person will be expected to engage in the following work activities: WA0201 Complete a visual inspection of electrical equipment. WA0202 Inspect protection devises used in electrical equipment and settings verified. WA0203 Test electrical equipment and document results under the direct supervision of a WA0204 Test main and control circuits WA0205 Complete a visual inspection of installations. WA0206 Complete and record electrical tests

WA0209 document findings and test results SE0201 Completed observation checklist SE0202 Completed section in logbook SE0203 Completed electrical test report WM-03- Undertake all activities without assistance, but Date Signature WE03 under supervision, in the processes of testing and inspecting of electrical equipment, control systems WA0201 Complete a visual inspection of electrical WA0202 Inspect protection devises used in electrical equipment and settings verified

IN SUMMARY

In this unit we have focussed on:

WA0201 Complete a visual inspection of electrical equipment.

WA0202 Inspect protection devises used in electrical equipment and settings verified.

WA0203 Test electrical equipment and document results under the direct supervision of a qualified artisan

WA0204 Test main and control circuits

WA0205 Complete a visual inspection of installations.

WA0206 Complete and record electrical tests.

WA0207 Inspect the installation method used.

WA0208 Inspect the selection of cables and conductors.

WA0209 Document findings and test results

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 61

WM-04-WE01: OBSERVE AND ASSIST A QUALIFIED PERSON IN THE PROCESSES OF COMMISSIONING ELECTRICAL INSTALLATION. AND CONTROL SYSTEMS

In this Chapter we will discuss the following Topics:

WA0101 Implement safety procedures prior to commissioning

WA0102 Perform functionality testing

WA0103 Monitor instrument readings

WA0104 Rectify defects found

WA0105 Complete a commissioning report

WA0106 Hand over the installation

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

SE0101 Completed checklists | SE0102 Completed section in Logbook

TOPIC WA0101: IMPLEMENT SAFETY PROCEDURES PRIOR TO COMMISSIONING

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the processes of commissioning electrical installation. and control systems in a real life electrical environment

SE0102 Completed section in Logbook WM-04-WE02: Commission electrical installation. and control systems under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Implement safety procedures prior to commissioning WA0202 Perform functionality testing WA0203 Monitor instrument readings WA0204 Rectify defects found WA0205 Complete a commissioning report WA0206 Hand over the installation SE0201 Completed checklists SE0

Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 WM-04- Observe and assist and experienced bricklayer Date Signature WE01 with the brick laying processes to construct, repair and/or make alterations to superstructures within the construction environment WA0101 implement safety procedures prior to WA0102 perform functionality testing WA0103 monitor instrument readings WA0104 rectify defects found WA0105 complete a comm

TOPIC WA0102: PERFORM FUNCTIONALITY TESTING

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the processes of commissioning electrical installation. and control systems in a real life electrical environment

Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 WM-04- Observe and assist and experienced bricklayer Date Signature WE01 with the brick laying processes to construct, repair and/or make alterations to superstructures within the construction environment WA0101 implement safety procedures prior to WA0102 perform functionality testing WA0103 monitor instrument readings WA0104 rectify defects found WA0105 complete a comm

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

TOPIC WA0103: MONITOR INSTRUMENT READINGS

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the processes of commissioning electrical installation. and control systems in a real life electrical environment

Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 WM-04- Observe and assist and experienced bricklayer Date Signature WE01 with the brick laying processes to construct, repair and/or make alterations to superstructures within the construction environment WA0101 implement safety procedures prior to WA0102 perform functionality testing WA0103 monitor instrument readings WA0104 rectify defects found WA0105 complete a comm

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

TOPIC WA0104: RECTIFY DEFECTS FOUND

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the processes of commissioning electrical installation. and control systems in a real life electrical environment

Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 WM-04- Observe and assist and experienced bricklayer Date Signature WE01 with the brick laying processes to construct, repair and/or make alterations to superstructures within the construction environment WA0101 implement safety procedures prior to WA0102 perform functionality testing WA0103 monitor instrument readings WA0104 rectify defects found WA0105 complete a comm

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

TOPIC WA0105: COMPLETE A COMMISSIONING REPORT

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the processes of commissioning electrical installation. and control systems in a real life electrical environment

Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 WM-04- Observe and assist and experienced bricklayer Date Signature WE01 with the brick laying processes to construct, repair and/or make alterations to superstructures within the construction environment WA0101 implement safety procedures prior to WA0102 perform functionality testing WA0103 monitor instrument readings WA0104 rectify defects found WA0105 complete a comm

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

TOPIC WA0106: HAND OVER THE INSTALLATION

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the processes of commissioning electrical installation. and control systems in a real life electrical environment

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 WM-04- Observe and assist and experienced bricklayer Date Signature WE01 with the brick laying processes to construct, repair and/or make alterations to superstructures within the construction environment WA0101 implement safety procedures prior to WA0102 perform functionality testing WA0103 monitor instrument readings WA0104 rectify defects found WA0105 complete a comm

IN SUMMARY

In this unit we have focussed on:

WA0101 Implement safety procedures prior to commissioning

WA0102 Perform functionality testing

WA0103 Monitor instrument readings

WA0104 Rectify defects found

WA0105 Complete a commissioning report

WA0106 Hand over the installation

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 62

WM-04-WE02: COMMISSION ELECTRICAL INSTALLATION. AND CONTROL SYSTEMS UNDER THE DIRECT SUPERVISION OF A QUALIFIED ELECTRICIAN

In this Chapter we will discuss the following Topics:

WA0201 Implement safety procedures prior to commissioning

WA0202 Perform functionality testing

WA0203 Monitor instrument readings

WA0204 Rectify defects found

WA0205 Complete a commissioning report

WA0206 Hand over the installation

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

SE0201 Completed checklists | SE0202 Completed section in Logbook

TOPIC WA0201: IMPLEMENT SAFETY PROCEDURES PRIOR TO COMMISSIONING

SE0102 Completed section in Logbook WM-04-WE02: Commission electrical installation. and control systems under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Implement safety procedures prior to commissioning WA0202 Perform functionality testing WA0203 Monitor instrument readings WA0204 Rectify defects found WA0205 Complete a commissioning report WA0206 Hand over the installation SE0201 Completed checklists SE0

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the processes of commissioning electrical installation. and control systems in a real life electrical environment

Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 WM-04- Observe and assist and experienced bricklayer Date Signature WE01 with the brick laying processes to construct, repair and/or make alterations to superstructures within the construction environment WA0101 implement safety procedures prior to WA0102 perform functionality testing WA0103 monitor instrument readings WA0104 rectify defects found WA0105 complete a comm

TOPIC WA0202: PERFORM FUNCTIONALITY TESTING

SE0102 Completed section in Logbook WM-04-WE02: Commission electrical installation. and control systems under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Implement safety procedures prior to commissioning WA0202 Perform functionality testing WA0203 Monitor instrument readings WA0204 Rectify defects found WA0205 Complete a commissioning report WA0206 Hand over the installation SE0201 Completed checklists SE0

Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 WM-04- Observe and assist and experienced bricklayer Date Signature WE01 with the brick laying processes to construct, repair and/or make alterations to superstructures within the construction environment WA0101 implement safety procedures prior to WA0102 perform functionality testing WA0103 monitor instrument readings WA0104 rectify defects found WA0105 complete a comm

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

TOPIC WA0203: MONITOR INSTRUMENT READINGS

SE0102 Completed section in Logbook WM-04-WE02: Commission electrical installation. and control systems under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Implement safety procedures prior to commissioning WA0202 Perform functionality testing WA0203 Monitor instrument readings WA0204 Rectify defects found WA0205 Complete a commissioning report WA0206 Hand over the installation SE0201 Completed checklists SE0

Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 WM-04- Observe and assist and experienced bricklayer Date Signature WE01 with the brick laying processes to construct, repair and/or make alterations to superstructures within the construction environment WA0101 implement safety procedures prior to WA0102 perform functionality testing WA0103 monitor instrument readings WA0104 rectify defects found WA0105 complete a comm

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

TOPIC WA0204: RECTIFY DEFECTS FOUND

SE0102 Completed section in Logbook WM-04-WE02: Commission electrical installation. and control systems under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Implement safety procedures prior to commissioning WA0202 Perform functionality testing WA0203 Monitor instrument readings WA0204 Rectify defects found WA0205 Complete a commissioning report WA0206 Hand over the installation SE0201 Completed checklists SE0

Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 WM-04- Observe and assist and experienced bricklayer Date Signature WE01 with the brick laying processes to construct, repair and/or make alterations to superstructures within the construction environment WA0101 implement safety procedures prior to WA0102 perform functionality testing WA0103 monitor instrument readings WA0104 rectify defects found WA0105 complete a comm

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

TOPIC WA0205: COMPLETE A COMMISSIONING REPORT

SE0102 Completed section in Logbook WM-04-WE02: Commission electrical installation. and control systems under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Implement safety procedures prior to commissioning WA0202 Perform functionality testing WA0203 Monitor instrument readings WA0204 Rectify defects found WA0205 Complete a commissioning report WA0206 Hand over the installation SE0201 Completed checklists SE0

Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 WM-04- Observe and assist and experienced bricklayer Date Signature WE01 with the brick laying processes to construct, repair and/or make alterations to superstructures within the construction environment WA0101 implement safety procedures prior to WA0102 perform functionality testing WA0103 monitor instrument readings WA0104 rectify defects found WA0105 complete a comm

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

TOPIC WA0206: HAND OVER THE INSTALLATION

SE0102 Completed section in Logbook WM-04-WE02: Commission electrical installation. and control systems under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Implement safety procedures prior to commissioning WA0202 Perform functionality testing WA0203 Monitor instrument readings WA0204 Rectify defects found WA0205 Complete a commissioning report WA0206 Hand over the installation SE0201 Completed checklists SE0

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault-finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 WM-04- Observe and assist and experienced bricklayer Date Signature WE01 with the brick laying processes to construct, repair and/or make alterations to superstructures

within the construction environment WA0101 implement safety procedures prior to WA0102 perform functionality testing WA0103 monitor instrument readings WA0104 rectify defects found WA0105 complete a comm

IN SUMMARY

In this unit we have focussed on:

WA0201 Implement safety procedures prior to commissioning

WA0202 Perform functionality testing

WA0203 Monitor instrument readings

WA0204 Rectify defects found

WA0205 Complete a commissioning report

WA0206 Hand over the installation

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 63

WM-04-WE03: UNDERTAKE ALL ACTIVITIES WITHOUT ASSISTANCE, BUT UNDER SUPERVISION, TO COMMISSION ELECTRICAL INSTALLATION. AND CONTROL SYSTEMS

In this Chapter we will discuss the following Topics:

WA0201 Implement safety procedures prior to commissioning

WA0202 Perform functionality testing

WA0203 Monitor instrument readings

WA0204 Rectify defects found

WA0205 Complete a commissioning report

WA0206 Hand over the installation

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

SE0201 Completed checklists | SE0202 Completed section in Logbook

TOPIC WA0201: IMPLEMENT SAFETY PROCEDURES PRIOR TO COMMISSIONING

SE0102 Completed section in Logbook WM-04-WE02: Commission electrical installation. and control systems under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Implement safety procedures prior to commissioning WA0202 Perform functionality testing WA0203 Monitor instrument readings WA0204 Rectify defects found WA0205 Complete a commissioning report WA0206 Hand over the installation SE0201 Completed checklists SE0

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the processes of commissioning electrical installation. and control systems in a real life electrical environment

Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 WM-04- Observe and assist and experienced bricklayer Date Signature WE01 with the brick laying processes to construct, repair and/or make alterations to superstructures within the construction environment WA0101 implement safety procedures prior to WA0102 perform functionality testing WA0103 monitor instrument readings WA0104 rectify defects found WA0105 complete a comm

TOPIC WA0202: PERFORM FUNCTIONALITY TESTING

SE0102 Completed section in Logbook WM-04-WE02: Commission electrical installation. and control systems under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Implement safety procedures prior to commissioning WA0202 Perform functionality testing WA0203 Monitor instrument readings WA0204 Rectify defects found WA0205 Complete a commissioning report WA0206 Hand over the installation SE0201 Completed checklists SE0

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the processes of commissioning electrical installation. and control systems in a real life electrical environment

Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 WM-04- Observe and assist and experienced bricklayer Date Signature WE01 with the brick laying processes to construct, repair and/or make alterations to superstructures within the construction environment WA0101 implement safety procedures prior to WA0102 perform functionality testing WA0103 monitor instrument readings WA0104 rectify defects found WA0105 complete a comm

TOPIC WA0203: MONITOR INSTRUMENT READINGS

SE0102 Completed section in Logbook WM-04-WE02: Commission electrical installation. and control systems under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Implement safety procedures prior to commissioning WA0202 Perform functionality testing WA0203 Monitor instrument readings WA0204 Rectify defects found WA0205 Complete a commissioning report WA0206 Hand over the installation SE0201 Completed checklists SE0

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the processes of commissioning electrical installation. and control systems in a real life electrical environment

Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 WM-04- Observe and assist and experienced bricklayer Date Signature WE01 with the brick laying processes to construct, repair and/or make alterations to superstructures within the construction environment WA0101 implement safety procedures prior to WA0102 perform functionality testing WA0103 monitor instrument readings WA0104 rectify defects found WA0105 complete a comm

TOPIC WA0204: RECTIFY DEFECTS FOUND

SE0102 Completed section in Logbook WM-04-WE02: Commission electrical installation. and control systems under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Implement safety procedures prior to commissioning WA0202 Perform functionality testing WA0203 Monitor instrument readings WA0204 Rectify defects found WA0205 Complete a commissioning report WA0206 Hand over the installation SE0201 Completed checklists SE0

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the processes of commissioning electrical installation. and control systems in a real life electrical environment

Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 WM-04- Observe and assist and experienced bricklayer Date Signature WE01 with the brick laying processes to construct, repair and/or make alterations to superstructures within the construction environment WA0101 implement safety procedures prior to WA0102 perform functionality testing WA0103 monitor instrument readings WA0104 rectify defects found WA0105 complete a comm

TOPIC WA0205: COMPLETE A COMMISSIONING REPORT

SE0102 Completed section in Logbook WM-04-WE02: Commission electrical installation. and control systems under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Implement safety procedures prior to commissioning WA0202 Perform functionality testing WA0203 Monitor instrument readings WA0204 Rectify defects found WA0205 Complete a commissioning report WA0206 Hand over the installation SE0201 Completed checklists SE0

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the processes of commissioning electrical installation. and control systems in a real life electrical environment

Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 WM-04- Observe and assist and experienced bricklayer Date Signature WE01 with the brick laying processes to construct, repair and/or make alterations to superstructures within the construction environment WA0101 implement safety procedures prior to WA0102 perform functionality testing WA0103 monitor instrument readings WA0104 rectify defects found WA0105 complete a comm

TOPIC WA0206: HAND OVER THE INSTALLATION

SE0102 Completed section in Logbook WM-04-WE02: Commission electrical installation. and control systems under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 Implement safety procedures prior to commissioning WA0202 Perform functionality testing WA0203 Monitor instrument readings WA0204 Rectify defects found WA0205 Complete a commissioning report WA0206 Hand over the installation SE0201 Completed checklists SE0

Any other statutory requirements relevant to the specific context Additional Assignments to be Assessed Externally 671101000-WM-04, Processes of commissioning electrical installation. and control systems, L4, Cr15 Purpose of the Work Experience Module The focus of the work experience is on providing the learner an opportunity to: Gain exposure to work as part of a team in the processes of commissioning electrical installation. and control systems in a real life electrical environment

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare

equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

IN SUMMARY

In this unit we have focussed on:

WA0201 Implement safety procedures prior to commissioning

WA0202 Perform functionality testing

WA0203 Monitor instrument readings

WA0204 Rectify defects found

WA0205 Complete a commissioning report

WA0206 Hand over the installation

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 64

WM-05-WE01: OBSERVE AND ASSIST A QUALIFIED ELECTRICIAN IN THE MAINTENANCE PROCESSES FOR EQUIPMENT, CONTROL SYSTEMS AND ELECTRICAL INSTALLATION.

In this Chapter we will discuss the following Topics:

WA0101 isolate and lockout equipment, control systems and installations

WA0102 clean and prepare equipment for maintenance

WA0103 implement maintenance schedules

WA0104 perform fault-finding on installations

WA0105 repair defects

WA0106 complete a maintenance report

WA0107 re-commission and hand over the installation

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

SE0101 Completed observation checklist | SE0102 Completed section in logbook

TOPIC WA0101: ISOLATE AND LOCKOUT EQUIPMENT, CONTROL SYSTEMS AND INSTALLATIONS

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

TOPIC WA0102: CLEAN AND PREPARE EQUIPMENT FOR MAINTENANCE

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault-finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

TOPIC WA0103: IMPLEMENT MAINTENANCE SCHEDULES

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

TOPIC WA0104: PERFORM FAULTFINDING ON INSTALLATIONS

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault-finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

TOPIC WA0105: REPAIR DEFECTS

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

TOPIC WA0106: COMPLETE A MAINTENANCE REPORT

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault-finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

TOPIC WA0107: RE-COMMISSION AND HAND OVER THE INSTALLATION

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

IN SUMMARY

In this unit we have focussed on:

WA0101 isolate and lockout equipment, control systems and installations

WA0102 clean and prepare equipment for maintenance

WA0103 implement maintenance schedules

WA0104 perform fault-finding on installations

WA0105 repair defects

WA0106 complete a maintenance report

WA0107 re-commission and hand over the installation

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 65

WM-05-WE02: MAINTAIN EQUIPMENT, CONTROL SYSTEMS AND ELECTRICAL INSTALLATION. UNDER THE DIRECT SUPERVISION OF A QUALIFIED ELECTRICIAN

In this Chapter we will discuss the following Topics:

WA0201 isolate and lockout equipment, control systems and installations

WA0202 clean and prepare equipment for maintenance

WA0203 implement maintenance schedules

WA0204 perform fault finding on installations

WA0205 repair defects

WA0206 complete a maintenance report

WA0207 re-commission and hand over the installation

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

SE0201 Signed off work orders | SE0202 Completed checklists | SE0203 Work permits/workers register

TOPIC WA0201: ISOLATE AND LOCKOUT EQUIPMENT, CONTROL SYSTEMS AND INSTALLATIONS

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault-finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

TOPIC WA0202: CLEAN AND PREPARE EQUIPMENT FOR MAINTENANCE

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

TOPIC WA0203: IMPLEMENT MAINTENANCE SCHEDULES

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault-finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company

specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

TOPIC WA0204: PERFORM FAULTFINDING ON INSTALLATIONS

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

TOPIC WA0205: REPAIR DEFECTS

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault-finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company

specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

TOPIC WA0206: COMPLETE A MAINTENANCE REPORT

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

TOPIC WA0207: RE-COMMISSION AND HAND OVER THE INSTALLATION

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault-finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

installations and control systems WA0201 implement safety procedures prior to WA0202 perform functionality testing WA0203 monitor instrument readings WA0204 rectify defects found WA0205 complete a commissioning report WA0206 hand over the installation SE0201 Completed checklists SE0202 Completed section in Logbook Contextualised Workplace Knowledge Company specific policies and procedures Company

specific quality management systems Relevant sections of the Occupational Health and Additional Assignments to be Assess

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

IN SUMMARY

In this unit we have focussed on:

WA0201 isolate and lockout equipment, control systems and installations

WA0202 clean and prepare equipment for maintenance

WA0203 implement maintenance schedules

WA0204 perform fault finding on installations

WA0205 repair defects

WA0206 complete a maintenance report

WA0207 re-commission and hand over the installation

Complete the workbook activity linked to this chapter, which is activity 1.

CHAPTER 66

WM-05-WE03: UNDERTAKE ALL ACTIVITIES WITHOUT ASSISTANCE, BUT UNDER SUPERVISION, IN MAINTENANCE PROCESSES FOR EQUIPMENT, CONTROL SYSTEMS AND ELECTRICAL INSTALLATION.

In this Chapter we will discuss the following Topics:

WA0301 isolate and lockout equipment, control systems and installations

WA0302 clean and prepare equipment for maintenance

WA0303 implement maintenance schedules

WA0304 perform fault-finding on installations

WA0305 repair defects

WA0306 complete a maintenance report

WA0307 re-commission and hand over the installation

INTERNAL ASSESSMENT CRITERIA AND WEIGHT

By the end of this Chapter the learner will be able to:

SE0301 Signed off work orders | SE0302 Completed checklists | SE0303 Re-commission and hand over the installation

TOPIC WA0301: ISOLATE AND LOCKOUT EQUIPMENT, CONTROL SYSTEMS AND INSTALLATIONS

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

SE0203 Work permits/workers register WM-05- Undertake all activities without assistance, but Date Signature WE03 under supervision, in maintenance processes for equipment, control systems and electrical WA0301 isolate and lockout equipment, control systems and WA0302 clean and prepare equipment for maintenance WA0303 implement maintenance schedules WA0304 perform fault-finding on installations WA0305 repair defects WA0306 complete a maintenance report WA0307 re-commission and hand over the installation SE0301 Signed

TOPIC WA0302: CLEAN AND PREPARE EQUIPMENT FOR MAINTENANCE

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

SE0203 Work permits/workers register WM-05- Undertake all activities without assistance, but Date Signature WE03 under supervision, in maintenance processes for equipment, control systems and electrical WA0301 isolate and lockout equipment, control systems and WA0302 clean and prepare equipment for maintenance WA0303 implement maintenance schedules WA0304 perform fault-finding on installations WA0305 repair defects WA0306 complete a maintenance report WA0307 re-commission and hand over the installation SE0301 Signed

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

TOPIC WA0303: IMPLEMENT MAINTENANCE SCHEDULES

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

SE0203 Work permits/workers register WM-05- Undertake all activities without assistance, but Date Signature WE03 under supervision, in maintenance processes for equipment, control systems and electrical WA0301 isolate and lockout equipment, control systems and WA0302 clean and prepare equipment for maintenance WA0303 implement maintenance schedules WA0304 perform fault-finding on installations WA0305 repair defects WA0306 complete a maintenance report WA0307 re-commission and hand over the installation SE0301 Signed

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

TOPIC WA0304: PERFORM FAULTFINDING ON INSTALLATIONS

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

SE0203 Work permits/workers register WM-05- Undertake all activities without assistance, but Date Signature WE03 under supervision, in maintenance processes for equipment, control systems and electrical WA0301 isolate and lockout equipment, control systems and WA0302 clean and prepare equipment for maintenance WA0303 implement maintenance schedules WA0304 perform fault-finding on installations WA0305 repair defects WA0306 complete a maintenance report WA0307 re-commission and hand over the installation SE0301 Signed

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

TOPIC WA0305: REPAIR DEFECTS

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

SE0203 Work permits/workers register WM-05- Undertake all activities without assistance, but Date Signature WE03 under supervision, in maintenance processes for equipment, control systems and electrical WA0301 isolate and lockout equipment, control systems and WA0302 clean and prepare equipment for maintenance WA0303 implement maintenance schedules WA0304 perform fault-finding on installations WA0305 repair defects WA0306 complete a maintenance report WA0307 re-commission and hand over the installation SE0301 Signed

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

TOPIC WA0306: COMPLETE A MAINTENANCE REPORT

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

SE0203 Work permits/workers register WM-05- Undertake all activities without assistance, but Date Signature WE03 under supervision, in maintenance processes for equipment, control systems and electrical WA0301 isolate and lockout equipment, control systems and WA0302 clean and prepare equipment for maintenance WA0303 implement maintenance schedules WA0304 perform fault-finding on installations WA0305 repair defects WA0306 complete a maintenance report WA0307 re-commission and hand over the installation SE0301 Signed

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

TOPIC WA0307: RE-COMMISSION AND HAND OVER THE INSTALLATION

WM-05-WE02: Maintain equipment, control systems and electrical installation. under the direct supervision of a qualified electrician Scope of Work Experience The person will be expected to engage in the following work activities: WA0201 isolate and lockout equipment, control systems and installations WA0202 clean and prepare equipment for maintenance WA0203 implement maintenance schedules WA0204 perform fault finding on installations WA0205 repair defects WA0206 complete a maintenance report WA0207 re-commission and

SE0203 Work permits/workers register WM-05- Undertake all activities without assistance, but Date Signature WE03 under supervision, in maintenance processes for equipment, control systems and electrical WA0301 isolate and lockout equipment, control systems and WA0302 clean and prepare equipment for maintenance WA0303 implement maintenance schedules WA0304 perform fault-finding on installations WA0305 repair defects WA0306 complete a maintenance report WA0307 re-commission and hand over the installation SE0301 Signed

WM-05-WE02: Maintenance processes for equipment, control systems and electrical installation. under the direct supervision of a qualified person WM-05-WE03: Undertake all activities without assistance, but under supervision, in maintenance processes for equipment, control systems and electrical installation.a5

IN SUMMARY

In this unit we have focussed on:

WA0301 isolate and lockout equipment, control systems and installations

WA0302 clean and prepare equipment for maintenance

WA0303 implement maintenance schedules

WA0304 perform fault finding on installations

WA0305 repair defects

WA0306 complete a maintenance report

WA0307 re-commission and hand over the installation

Complete the workbook activity linked to this chapter, which is activity 1.